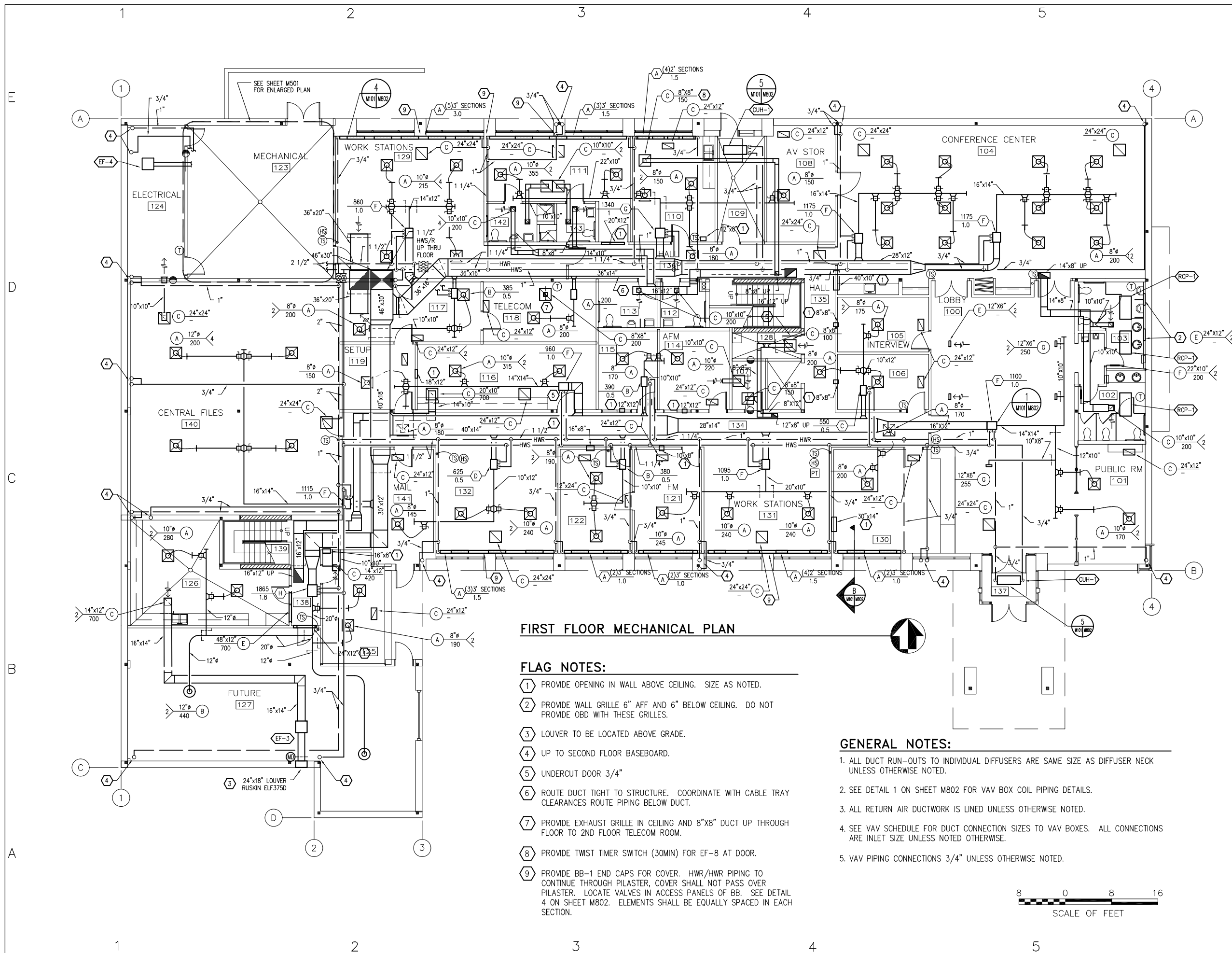


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MECHANICAL LEGEND (NOTE: NOT ALL SYMBOLS SHOWN ARE USED ON THESE DRAWINGS)											
SYMBOL			DESCRIPTION			SYMBOL			DESCRIPTION		
SA		POSITIVE PRESSURE DUCT UP		2-WAY CONTROL VALVE	HWS		HEATING HOT WATER SUPPLY		ELECTRIC MOTOR		
SA		POSITIVE PRESSURE DUCT DOWN		3-WAY CONTROL VALVE	HWR		HEATING HOT WATER RETURN		DIFFERENTIAL PRESSURE TRANSMITTER		
EA/RA		NEGATIVE PRESSURE DUCT UP		BALANCE VALVE	D		CONDENSATE OR EQUIPMENT DRAIN		INDICATING DIFFERENTIAL PRESSURE TRANSMITTER		
EA/RA		NEGATIVE PRESSURE DUCT DOWN		BALL VALVE	CW		DOMESTIC COLD WATER		FLOW TRANSMITTER		
OA		OUTSIDE AIR INTAKE DUCT UP		BUTTERFLY VALVE	HW		DOMESTIC HOT WATER		ELECTRIC\PNEUMATIC TRANSDUCER		
OA		OUTSIDE AIR INTAKE DUCT DOWN		CHECK VALVE	HWC		DOMESTIC HOT WATER CIRCULATING		FIRE ALARM CONTROL PANEL		
		ROUND DUCT UP		GATE VALVE	V		DOMESTIC VENT PIPING		CURRENT SWITCH/TRANSMITTER		
		CONICAL FITTING WITH DAMPER		GLOBE VALVE	W		BURIED WASTE				
		BRANCH FITTING WITH DAMPER		PRESSURE REDUCING VALVE	W		SUSPENDED WASTE				
		CONICAL FITTING WITHOUT DAMPER		PRESSURE RELIEF VALVE	ST		SUSPENDED STORM SEWER		FLAG NOTE		
		BRANCH FITTING WITHOUT DAMPER		HOSE END DRAIN VALVE	ST		BURIED STORM SEWER		REVISION NUMBER		
		ELBOW WITH TURNING VANES		STRAINER	IW		INDIRECT WASTE		MECHANICAL NOTE		
		ELBOW WITHOUT TURNING VANES		STRAINER WITH BLOW OFF VALVE			THERMOSTAT W/ZONE CALLOUT		CONNECT TO EXSITING		
		LONG RADIUS ELBOW		PLUG VALVE			HUMIDITY SENSOR W/ZONE CALLOUT		WORK POINT		
		SHORT RADIUS ELBOW		GAS COCK			TEMP SENSOR W/ZONE CALLOUT		DIFFUSER, REGISTER, GRILL CFM CALLOUT		
		SUPPLY DIFFUSER		SOLENOID VALVE			DIGITAL INPUT		DIFFUSER, REGISTER, GRILL CFM CALLOUT		
		SUPPLY DIFFUSER HIDDEN		PRESSURE REGULATING VALVE			ANALOG INPUT		BASEBOARD/FINTUBE RADIATION HEATING		
		SUPPLY DIFFUSER ROUND NECK		THERMOWELL			DIGITAL OUTPUT		VAV TERMINAL BOX CALLOUT		
		SUPPLY DIFFUSER SQUARE NECK		THERMOMETER WITH THERMOWELL			ANALOG OUTPUT				
		EXHAUST/RETURN GRILLE SQUARE NECK		UNION			CONTROL WIRE		DETAIL BUBBLE DETAIL NUMBER SHEET NUMBER - WHERE DETAIL IS SHOWN SHEET NUMBER - WHERE DETAIL IS CALLED OUT		
		EXHAUST/RETURN GRILLE ROUND NECK		AUTOMATIC AIR VENT			POWER WIRE		SECTION CUT SECTION LETTER SHEET NUMBER - WHERE SECTION IS SHOWN SHEET NUMBER - WHERE SECTION IS CALLED OUT		
		RETURN AIR GRILLE		MANUAL AIR VENT			FLOW ARROW				
		RETURN AIR GRILLE HIDDEN		THERMOMETER			FLOW SWITCH				
		ROUND DIFFUSER		GAUGE			DUCT TEMPERATURE SENSOR				
		SLOT DIFFUSER		GAUGE WITH BALL VALVE			WELL MOUNTED TEMPERATURE SENSOR				
		FLEXDUCT		PRESSURE TEMPERATURE/TEST POINT			ELECTRIC ACTUATOR CONTROL DAMPER				
		DUCTWORK		FLEX CONNECTION			LOW TEMP. LIMIT THERMOSTAT				
		LINED DUCTWORK		PIPE CAP			DUCT SMOKE DETECTOR				
MVD		MANUAL VOLUME DAMPER		FLOW ARROW			DAMPER\VALVE END SWITCH				
		FIRE DAMPER 1 1/2 HOUR UNLESS NOTED		FLOW MEASURING STATION			DIFFERENTIAL PRESSURE SWITCH				
		FIRE/SMOKE DAMPER 1 1/2 HOUR UNLESS NOTED		PIPE BREAK			HIGH PRESSURE SWITCH				
		MOTORIZED DAMPER		PIPE DOWN			PRESSURE SWITCH MANUAL RESET				
		ACCESS PANEL (SIZE)		PIPE UP			AQUATSTAT				
		DUCT BREAK		TEE OUTLET UP			PRESSURE ELECTRIC SWITCH				
				TEE OUTLET DOWN			MOTOR STARTER				
							CONTROL RELAY				
							VARIABLE FREQUENCY DRIVE				
							CONTROL ADDRESSABLE MODULE				

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FIRST FLOOR MECHANICAL PLAN

FLAG NOTES:

- 1 PROVIDE OPENING IN WALL ABOVE CEILING. SIZE AS NOTED.
- 2 PROVIDE WALL GRILLE 6" AFF AND 6" BELOW CEILING. DO NOT PROVIDE OBD WITH THESE GRILLES.
- 3 LOUVER TO BE LOCATED ABOVE GRADE.
- 4 UP TO SECOND FLOOR BASEBOARD.
- 5 UNDERCUT DOOR 3/4"
- 6 ROUTE DUCT TIGHT TO STRUCTURE. COORDINATE WITH CABLE TRAY CLEARANCES ROUTE PIPING BELOW DUCT.
- 7 PROVIDE EXHAUST GRILLE IN CEILING AND 8"x8" DUCT UP THROUGH FLOOR TO 2ND FLOOR TELECOM ROOM.
- 8 PROVIDE TWIST TIMER SWITCH (30MIN) FOR EF-8 AT DOOR.
- 9 PROVIDE BB-1 END CAPS FOR COVER. HWR/HWR PIPING TO CONTINUE THROUGH PILASTER, COVER SHALL NOT PASS OVER PILASTER. LOCATE VALVES IN ACCESS PANELS OF BB. SEE DETAIL 4 ON SHEET M802. ELEMENTS SHALL BE EQUALLY SPACED IN EACH SECTION.

GENERAL NOTES:

1. ALL DUCT RUN-OUTS TO INDIVIDUAL DIFFUSERS ARE SAME SIZE AS DIFFUSER NECK UNLESS OTHERWISE NOTED.
2. SEE DETAIL 1 ON SHEET M802 FOR VAV BOX COIL PIPING DETAILS.
3. ALL RETURN AIR DUCTWORK IS LINED UNLESS OTHERWISE NOTED.
4. SEE VAV SCHEDULE FOR DUCT CONNECTION SIZES TO VAV BOXES. ALL CONNECTIONS ARE INLET SIZE UNLESS OTHERWISE NOTED.
5. VAV PIPING CONNECTIONS 3/4" UNLESS OTHERWISE NOTED.

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SCALE OF FEET

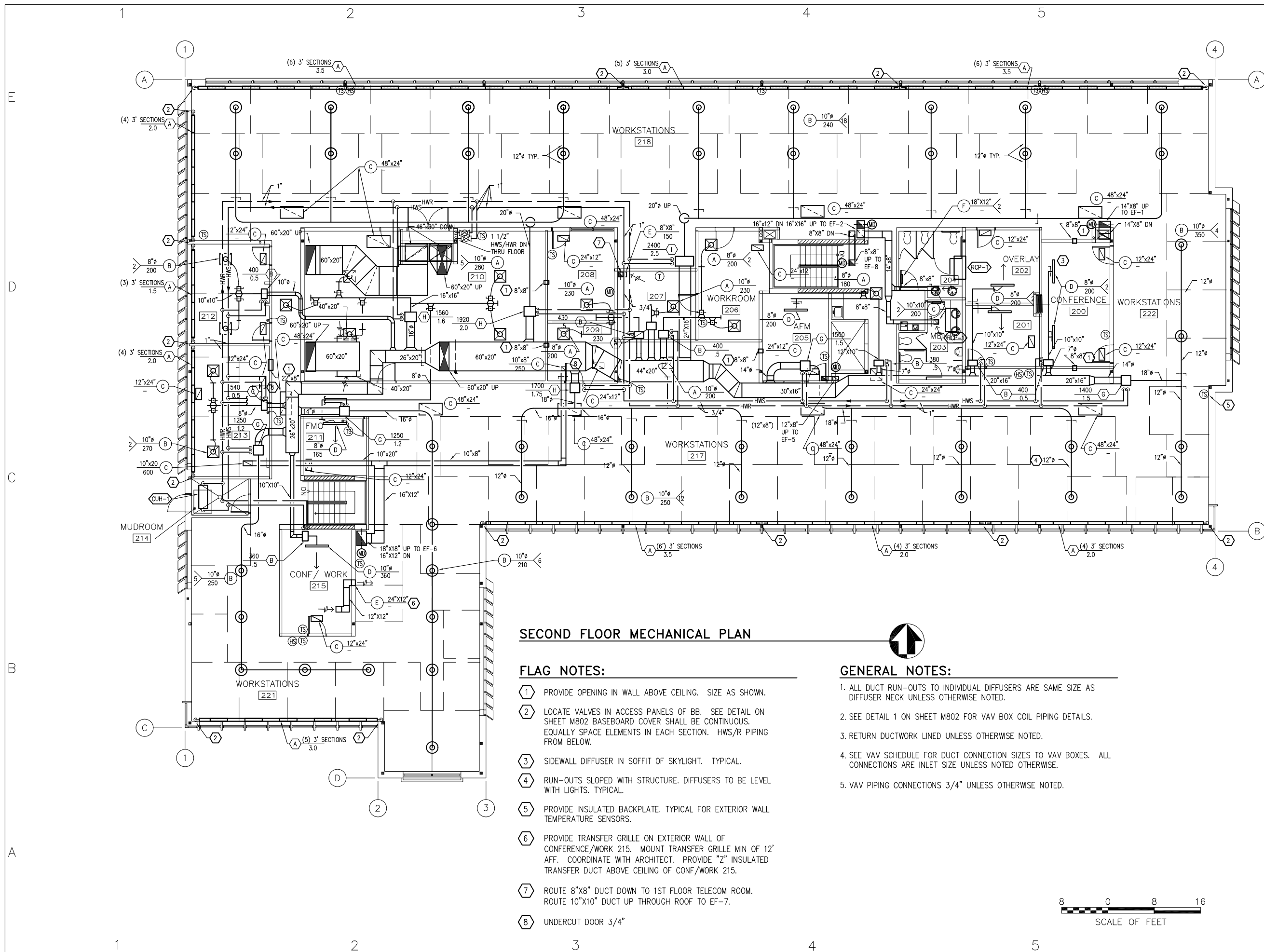


UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
FIRST FLOOR MECHANICAL PLAN
RAWLINS FIELD OFFICE BUILDING

PROJECT NO:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	DESCRIPTION	MARK
16235	RMH	RA	APB	TC	SB		
02-28-2003							

M101
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SECOND FLOOR MECHANICAL PLAN

FLAG NOTES:

- 1 PROVIDE OPENING IN WALL ABOVE CEILING. SIZE AS SHOWN.
- 2 LOCATE VALVES IN ACCESS PANELS OF BB. SEE DETAIL ON SHEET M802 BASEBOARD COVER SHALL BE CONTINUOUS. EQUALLY SPACE ELEMENTS IN EACH SECTION. HWS/R PIPING FROM BELOW.
- 3 SIDEWALL DIFFUSER IN SOFFIT OF SKYLIGHT. TYPICAL.
- 4 RUN-OUTS SLOPED WITH STRUCTURE. DIFFUSERS TO BE LEVEL WITH LIGHTS. TYPICAL.
- 5 PROVIDE INSULATED BACKPLATE. TYPICAL FOR EXTERIOR WALL TEMPERATURE SENSORS.
- 6 PROVIDE TRANSFER GRILLE ON EXTERIOR WALL OF CONFERENCE/WORK 215. MOUNT TRANSFER GRILLE MIN. OF 12' AFF. COORDINATE WITH ARCHITECT. PROVIDE "Z" INSULATED TRANSFER DUCT ABOVE CEILING OF CONF/WORK 215.
- 7 ROUTE 8"x8" DUCT DOWN TO 1ST FLOOR TELECOM ROOM. ROUTE 10"x10" DUCT UP THROUGH ROOF TO EF-7.
- 8 UNDERCUT DOOR 3/4"

GENERAL NOTES:

1. ALL DUCT RUN-OUTS TO INDIVIDUAL DIFFUSERS ARE SAME SIZE AS DIFFUSER NECK UNLESS OTHERWISE NOTED.
2. SEE DETAIL 1 ON SHEET M802 FOR VAV BOX COIL PIPING DETAILS.
3. RETURN DUCTWORK LINED UNLESS OTHERWISE NOTED.
4. SEE VAV SCHEDULE FOR DUCT CONNECTION SIZES TO VAV BOXES. ALL CONNECTIONS ARE INLET SIZE UNLESS NOTED OTHERWISE.
5. VAV PIPING CONNECTIONS 3/4" UNLESS OTHERWISE NOTED.

8 0 8 16
SCALE OF FEET



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

SECOND FLOOR MECHANICAL PLAN

RAWLINS FIELD OFFICE BUILDING

WYOMING

PROJECT NO:	DESIGN OFFICE:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	DESCRIPTION	MARK	APPROVED	DATE
16235	RMH	RA	APB	SB	02-28-2003				

M102
SHEET OF



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

ROOF PLAN

RAWLINS FIELD OFFICE BUILDING

WYOMING

PROJECT NO: 16235			
DESIGN OFFICE: RMH			
DESIGNED BY: RA			
DRAWN BY: APB			
CHK'D BY: TG			
APPROVED BY: SB			
DATE: 02-28-2003	MARK	DESCRIPTION	DATE
			APPROVED

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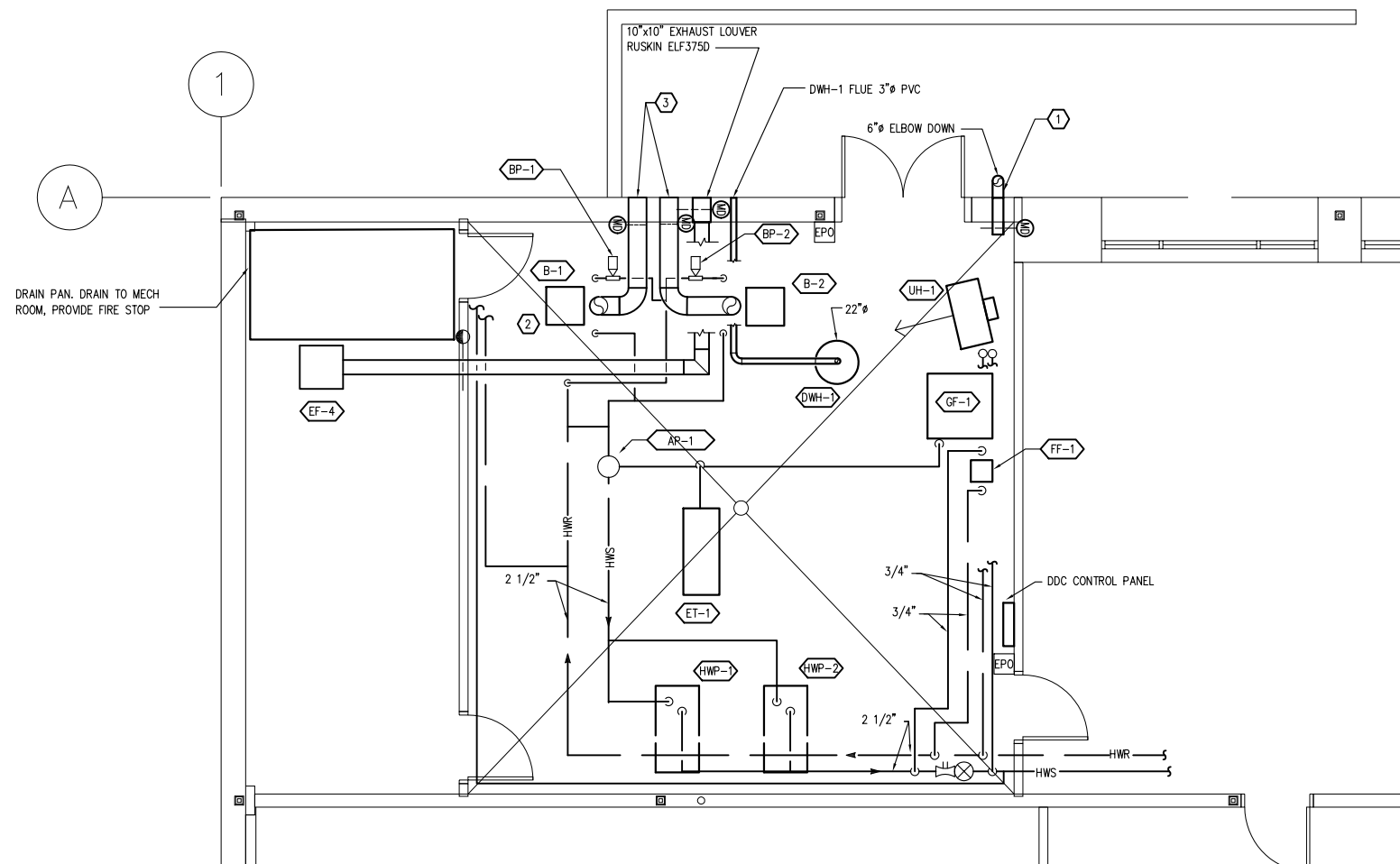
UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

ENLARGED MECHANICAL ROOM PLAN

RAWLINS FIELD OFFICE BUILDING

PROJECT NO: 16235			
DESIGN OFFICE: RMH			
DESIGNED BY: RA			
DRAWN BY: APB			
CHK'D BY: TG			
APPROVED BY: SB			
DATE: 02-28-2003	MARK	DESCRIPTION	DATE
			APPROVED

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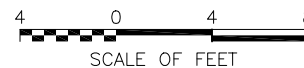


ENLARGED MECHANICAL ROOM PLAN



FLAG NOTES:

- 1 PROVIDE 6"Ø COMBUSTION AIR INTAKE THROUGH WALL FOR DWH-1. PROVIDE MOTORIZED DAMPER INTERLOCKED WITH DWH-1. DWH-1 WILL FIRE WHEN COMBUSTION AIR DAMPER IS PROVED OPEN.
- 2 SEE DETAIL 3 ON SHEET M803 FOR BOILER PIPING AND CONTROL DETAIL.
- 3 COMBUSTION AIR DUCT AND FLUE ARE STACKED. PROVIDE 12"Ø COMBUSTION AIR INTAKE DUCT THROUGH WALL AT 2' AFF AND 10"Ø POSITIVE PRESSURE DOUBLE WALL FLUE MINIMUM 5' ABOVE COMBUSTION AIR INTAKE. CONNECT COMBUSTION AIR AND FLUE TO BOILER. BOILER MANUFACTURER PROVIDED COMBUSTION AIR DUCT DAMPER INSTALLED IN 12"Ø COMBUSTION AIR DUCT.



CUSTOM ROOF TOP AIR HANDLING UNIT SCHEDULE																																	
TAG	PURPOSE	TYPE	SUPPLY FAN SECTION							RETURN FAN SECTION							V/PH/HZ	INDIR EVAP SECTION			DIR EVAP SECTION			HEATING				FILTER	MIN. O.A. %	SIZE	WEIGHT LBS.	MANUFACTURER & MODEL	NOTES
			CFM	TSP	ESP	RPM	HP	DIA.	ARR.	CFM	TSP	ESP	RPM	HP	DIA.	ARR.		EAT	LAT	EAT	LAT	EAT	LAT	MBH	GPM	SIZE							
RTU-1	1ST FLOOR HVAC	IND/DIR EVAP HOOD	13,245	4.5"	2"	1468	20	----	PLEN.	12,800	1.5	.75"	727	7.5	----	PLEN.	460/3/60	85/57	66/50	66/50	55/50	46	60	80	5.5	33"x60"	4"	17%	464x78x100	20,500	ENERGY LABS	①②③④	
RTU-2	2ND FLOOR HVAC	IND/DIR EVAP HOOD	16,235	4.5"	2"	1340	20	----	PLEN.	15,500	1.5	.75"	786	7.5	----	PLEN.	460/3/60	85/57	66/50	66/50	55/50	46	60	98	6.5	39"x60"	4"	15%	453x78x88	21,500	ENERGY LABS	①②③④	
GENERAL NOTES: A. INCLUDE PACKAGED DUCT SMOKE DETECTOR BY MANUFACTURER. B. CAPACITIES ARE AT JOB SITE CONDITIONS. C. HEATING COIL AREA BASED ON 180° EWT AND 160° LWT.			NOTES: ① TWO AIR TO AIR INDIRECT EVAPORATIVE COOLING HEAT EXCHANGERS WITH .5HP FANS AND 3/4HP PUMPS. ② DRAW THRU FAN ARRANGEMENT ASSUMES 3F° FAN HEAT, LAT OF UNITS SHALL BE 55DB. ③ DIRECT EVAPORATIVE TO HAVE TWO SECTIONS WITH DEDICATED PUMPS FOR EACH. STAGED AS NEEDED. ④ PROVIDE SLOPED ROOF CURB TO MATCH ROOF SLOPE.																														

VARIABLE AIR VOLUME UNIT SCHEDULE										
TAG	MAX. CFM COOLING	INLET SIZE	(COIL) OUTLET SIZE	MBH	GPM	MAX NC @ 1" P.D.		NO. OF COIL ROWS	MAX. WPD'	MANUFACTURER & MODEL
						DISCHARGE	RADIATED			
VAV-A	200-300	6"	12"X8"	3-4.7	.5	<20	23	2	0.24	TITUS DESV
VAV-B	400-450	7"	12"X10"	6.2-7	.5	20	23	2	0.33	TITUS DESV
VAV-C	500-560	8"	12"X10"	7.7-8.7	.5	<20	23	2	0.33	TITUS DESV
VAV-D	600-650	9"	14"X12.5"	9.3-10	.75-1.0	<20	25	2	0.65	TITUS DESV
VAV-E	700-800	10"	14"X12.5"	10.8-12.4	.75-1.0	<20	27	2	0.65	TITUS DESV
VAV-F	900-1200	12"	16"X15"	14-18.6	1-1.5	<20	25	2	0.84	TITUS DESV
VAV-G	1300-1600	14"	20"X17.5"	20.1-24.8	1.3-2	<20	32	2	1.25	TITUS DESV
VAV-H	1700-2000	16"	24"X18"	26.3-31	1.5-2.5	<20	23	2	0.90	TITUS DESV
VAV-I	2500-4000	24"X16"	38"X18"	38.7-62	2.5-5	29	35	3	2.2	TITUS DESV
GENERAL NOTES:										
A. CAPACITIES BASED ON 180°F EWT, 150°F LWT, 55°F EAT AND 95°F LAT.										
B. MINIMUM CFM SHALL BE 30% UNLESS NOTED OTHERWISE ON DRAWINGS.										
C. NC LEVELS LISTED ARE WITH ROOM ATTENUATION.										
D. FURNISH ALL UNITS WITH INTEGRAL SOUND ATTENUATOR AND REHEAT COIL.										
E. REQUIRED GPM FOR EACH UNIT IS NOTED ON THE DRAWING.										

FAN SCHEDULE															
TAG	SERVICE	TYPE	CFM	ESP	TSP	RPM	VFD	HP	V/PH/HZ	CURB	VIB. ISO.	WEIGHT LBS.	DRIVE TYPE	MANUFACTURER & MODEL	NOTES
EF-1	REST ROOM EXHAUST	ROOF CENTRIFUGAL	400	1.0	1.0	1705	NA	1/8	120/1/60	Y	INTERNAL	30	BELT	GREENHECK GB-090	① ⑨ ⑦
EF-2	REST ROOM EXHAUST	ROOF CENTRIFUGAL	1200	1.0	1.0	1621	NA	1/3	120/1/60	Y	INTERNAL	50	BELT	GREENHECK GB-121	② ⑦
EF-3	LAB EXHAUST	IN-LINE	700	0.75	0.75	1852	NA	1/3	120/1/60	NA	SPRING D	75	BELT	GREENHECK B50-90-3	③ ⑦ ⑧
EF-4	ELECTRICAL ROOM	IN-LINE	200	0.75	0.75	1584	NA	1/4	120/1/60	NA	SPRING D	50	BELT	GREENHECK B50-80-4	④ ⑦
EF-5	ELEVATOR/JANITOR	ROOF CENTRIFUGAL	300	0.75	0.75	1260	NA	1/20	120/1/60	Y	INTERNAL	30	BELT	GREENHECK GB-080	⑤ ⑦
EF-6	COPY ROOMS	ROOF CENTRIFUGAL	2050	0.75	0.75	1070	NA	1/2	120/1/60	Y	INTERNAL	90	BELT	GREENHECK GB-161	⑥ ⑦
EF-7	TELECOM	ROOF CENTRIFUGAL	350	.75	.75	1200	NA	1/6	120/1/60	Y	INTERNAL	30	BELT	GREENHECK GB-080	⑤ ⑦
EF-8	BREAK ROOM	ROOF CENTRIFUGAL	150	.5	.5	1235	NA	1/6	120/1/60	Y	INTERNAL	30	BELT	GREENHECK GB-070	⑤ ⑦
GENERAL NOTES:						NOTES:									
A. ALL WELDED CONSTRUCTION WITH DRAIN PLUG AND VIB. ISOLATION BASE.						① 10.4 SONES ③ 15.4 SONES ⑤ 6.5 SONES ⑦ PROVIDE MOTORIZED DAMPER ⑨ INTERLOCK WITH OCCUPANCY SENSOR									
B. MOTOR HP AT DESIGN CONDITIONS SHALL BE LESS THAN 125% OF BHP.						② 13 SONES ④ 11 SONES ⑥ 12 SONES ⑧ SWITCH CONTROLLED									
C. STATIC PRESSURES SHOWN ARE AT SEA LEVEL.															

GRILLE/REGISTER/DIFFUSER SCHEDULE								
TAG	PURPOSE	TYPE	MOUNTING	PATTERN	NC	ACCESSORIES	MANUFACTURER & MODEL	NOTES
A	SUPPLY	DIFFUSER	CEILING	LAY-IN	25	---	TITUS TMS-AA	①
B	SUPPLY	DIFFUSER	CEILING	EXPOSED	25	---	TITUS TMR-AA	③
C	RETURN	GRILLE	CEILING	EGG CR.	25	---	TITUS 50F	①
D	SUPPLY	GRILLE	CEILING	2-SLOT	25	---	TITUS TBD - 30	①
E	TRANSFER	GRILLE	WALL	LOUVER	25	---	TITUS 350FL	①
F	TRANSFER	GRILLE	DOOR	LOUVER	25	---	TITUS T-700L	④
G	SUPPLY	GRILLE	WALL	LOUVER	25	---	TITUS T-300FL	②
GENERAL NOTES:			NOTES:					
A. ASSUMES 10 DB ROOM ABSORPTION.			① 1" SLOT, 4' LENGTH, 2 SLOTS			③ AG75 DAMPER		
B. ALUMINUM CONSTRUCTION			② 45° DEFLECTION			④ STEEL CONSTRUCTION		

PUMP SCHEDULE													
TAG	PURPOSE	TYPE	GPM	FT. HEAD	RPM	HP	IMP. DIA. "	% EFF.	V/PH/HZ	WEIGHT LBS.	VFD	MANUFACTURER & MODEL	NOTES
HWP-1	HEATING SYSTEMS	END-SUCTION	84	40	1750	1.5	6.5	67	460/3/60	---	N	B & G 1510 1 1/2AC	☐
HWP-2	HEATING SYSTEMS	END-SUCTION	84	40	1750	1.5	6.5	67	460/3/60	---	N	B & G 1510 1 1/2AC	☐
BP-1	BOILER CIRC	IN-LINE	45	10	1750	.5	3.75	51	120/1/60	---	N	B & G SERIES 60 2AA	☐
BP-2	BOILER CIRC	IN-LINE	45	10	1750	.5	3.75	51	120/1/60	---	N	B & G SERIES 60 2AA	☐

NOTES:
 ① see spec.

BOILER SCHEDULE									
TAG	GPM	EWT °F	LWT °F	EFF %	MBH INPUT	V/PH/HZ	WEIGHT LBS.	MANUFACTURER & MODEL	NOTES
B-1	45	150	180	85%	700	120/1/60	455	PK THERMIFIC SN700	① ② ③
B-2	45	150	180	85%	700	120/1/60	455	PK THERMIFIC SN700	① ② ③

NOTES:

① MBH INPUT IS AT SEA LEVEL ② 30% GLYCOL ③ SEALED COMBUSTION

RADIATION SCHEDULE								
TAG	TYPE	CAPACITY BTUH/FT.	WPD*	GPM	ELEMENT	ENCLOSURE	MANUFACTURER & MODEL	NOTES
A	FIN TUBE	690	---	---	1°C-2 3/4"	16 GAUGE	VULCAN	①
GENERAL NOTES: A. CAPACITIES BASED ON 180°F EWT, 160°F LWT AND 62°F EAT. B. CAPACITIES ARE AT SEA LEVEL CONDITIONS.								
					NOTES: ① CUSTOM ENCLOSURE SEE DETAIL.			

CABINET/UNIT HEATER SCHEDULE												
TAG	TYPE	CFM	RPM	HP	V/PH/HZ	HTG. BTUH	VIB. ISO.	GPM	WPD'	FILTER	MANUFACTURER & MODEL	NOTES
CUH-1	RECESSED CUH	100	980	0.03	120/1/60	6300	D	1.0	0.05	---	TRANE FFE0201	☉
UH-1	VERTICAL UH	1200	1550	0.125	120/1/60	35000	D	2.5	0.19	---	TRANE UHSA090S	☉

GENERAL NOTES:

NOTES:

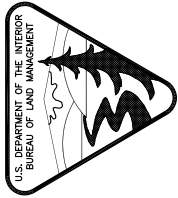
A. HEATING CAPACITIES BASED ON 180°F EWT, 160 LWT AND 60°F EAT.

B. CAPACITIES ARE AT SEA LEVEL CONDITIONS.

C. PROVIDE UNIT WITH WALL MOUNT THERMOSTAT.

D. PROVIDE UNIT WITH DOWN DISCHARGE ADJUSTABLE LOUVER.

EQUIPMENT SCHEDULE	
TAG	PURPOSE
AP-1	HEATING WATER AIR PURGER: AMTROL MODEL 462, 3" SIZE, ASME CONSTRUCTION, OR EQUAL BY B&G OR JOHN WOOD.
ET-1	EXPANSION TANK: B & G D-80, PRESSURIZED ASME HORIZONTAL DIAPHRAM EXPANSION TANK, 21.3 GAL TANK VOLUME, 11.8 GAL ACCEPTANCE.
FF-1	FILTER FEEDER: NEPTUNE MODEL FTF-5 BY-PASS TYPE FILTER FEEDER OF 5 GALLON CAPACITY COMPLETE WITH STAINLESS STEEL BASKET AND PART NO. 106453 5 MICRON FILTERS. SEE SPECIFICATION SECTION 15185 FOR ADDITIONAL DATA.
GF-1	GLYCOL FEEDER: SAGE INDUSTRIES MODEL WC370302, 50 GALLON TANK, 1/3 HP, 120/60/1 15185 FOR ADDITIONAL DATA.
RCP-1	RADIANT CEILING PANEL: AZTEC MODEL CP750, 120/1/60, 750 WATTS, 24"x48"x1" UNIT DIMENSIONS, WALL MOUNTED LINE VOLT THERMOSTAT.



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

MECHANICAL SCHEDULES

RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

PROJECT NO: 16235

DESIGN OFFICE: RM

DESIGNED BY: KC

DRAWN BY: APB

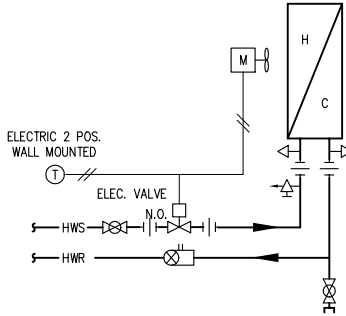
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DATE: 02-28-200

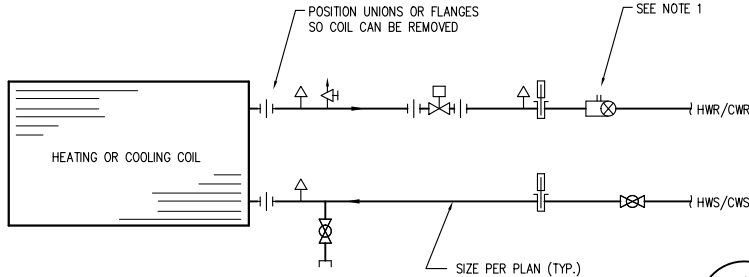
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CUH-1 DETAIL
SCALE: NONE



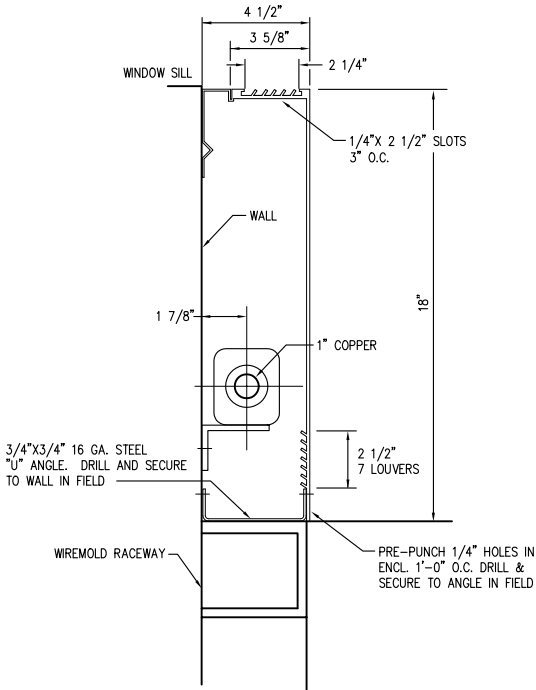
COIL DETAIL
SCALE: NONE



NOTES:

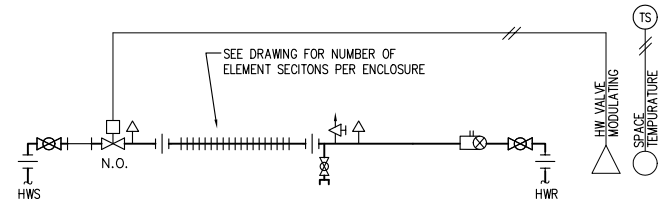
1. INSTALL FLOW MEASURING DEVICES PER MANUFACTURER'S RECOMMENDATIONS
2. REDUCERS SHALL BE PROVIDED AT TEMPERATURE CONTROL VALVE WHERE REQUIRED.

CUSTOM BASEBOARD ENCLOSURE



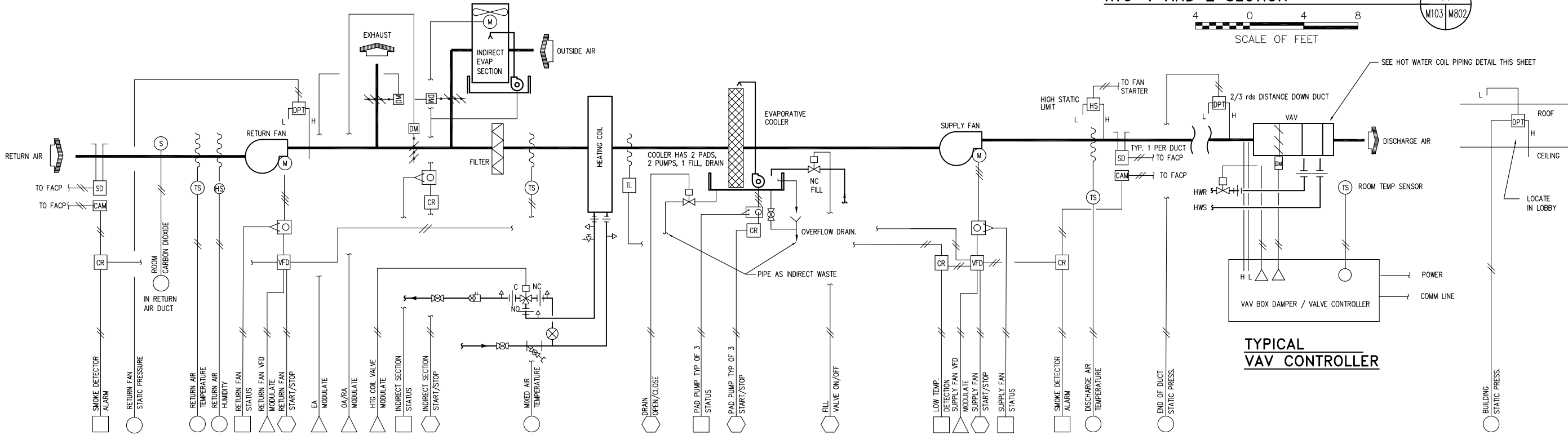
4" 0 4" 8"
SCALE OF INCHES

BASEBOARD HEATING PIPING DETAIL
SCALE: NONE



RTU-1 AND 2 SECTION

4 0 4 8
SCALE OF FEET



RTU-1 AND 2 PIPING AND CONTROL DIAGRAM
SCALE: NO SCALE

TYPICAL
VAV CONTROLLER



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
MECHANICAL DETAILS
RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

PROJECT NO:	DESIGN OFFICE:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	MARK	DESCRIPTION	DATE	APPROVED
16235	RMH	RA	APB	TG	SB	02-28-2003				

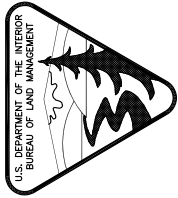
M802
SHEET OF

ELECTRICAL SYMBOLS		
ABBR.	SYMBOL	DESCRIPTION
-SWITCHING DEVICES CONT. -		
TC		LINE VOLTAGE THERMOSTAT
		PHOTOCELL
		TIME CLOCK
		LIGHTING CONTACTOR
-COMMUNICATION DEVICES AND MISCELLANEOUS-		
TELE		PUBLIC TELEPHONE OUTLET
		TELEPHONE OUTLET
		WALL MOUNTED TELEPHONE OUTLET @ +54" A.F.F.
		GIA-TRONICS PHONE OUTLET
		DATA OUTLET
		COMBINATION TELEPHONE/DATA OUTLET
		FLOOR MOUNTED TELEPHONE OUTLET
		FLOOR MOUNTED DATA OUTLET
		FLOOR MOUNTED TELEPHONE/DATA OUTLET
TTB		TELEPHONE TERMINAL BOARD
		CABLE TRAY, AS NOTED
		TELEVISION JACK
		SPEAKER
		MOTORIZED DAMPER
		SPEAKER WITH INTEGRAL VOLUME CONTROL
		PUBLIC ADDRESS SYSTEM AMPLIFIER
		PUBLIC ADDRESS SYSTEM MICROPHONE
		PUBLIC ADDRESS SYSTEM SPEAKER VOLUME CONTROL
		CARD READER
-CIRCUITING-		
(PART)		BRANCH CIRCUIT HOMERUN TO PANELBOARD; NUMBER OF RROWS INDICATE NUMBER OF CIRCUITS; DESIGNATION INDICATES PANEL & CIRCUIT NUMBERS
		(PART) INDICATES PARTIAL CIRCUIT HOMERUN
		LIGHTING OR POWER CONDUIT/ CIRCUITING CONCEALED IN WALLS OR CEILINGS
		LIGHTING OR POWER CONDUIT / CIRCUITING BELOW FLOOR SLAB OR UNDERGROUND
		WIREMOLD
		CONDUIT BREAK SYMBOL
		CONDUIT CAP
		CONDUIT STUB DOWN
		CONDUIT STUB UP
		PARTITION CIRCUIT SPLIT
-POWER-		
PNL		ELECTRICAL PANEL
		EXISTING TO REMAIN ELECTRICAL PANEL
		SWITCHBOARD

ABBR.	SYMBOL	DESCRIPTION
—POWER CONT.—		
DISC SW		CONTROL PANEL
		MOTOR
		NON-FUSED DISCONNECT SWITCH
		FUSED DISCONNECT SWITCH
XFMR		MAGNETIC MOTOR STARTER
		COMBINATION DISCONNECT SWITCH AND MOTOR STARTER
		TRANSFORMER
		VARIABLE FREQUENCY DRIVE
JB		METER
		CEILING MOUNTED JUNCTION BOX
		FLOOR MOUNTED JUNCTION BOX
		WALL MOUNTED JUNCTION BOX
EPO		EMERGENCY POWER OFF SWITCH
GND		ELECTRICAL GROUND
		GROUND ROD WITH INSPECTION TEST WELL
		UNTERMINATED CONDUCTOR/ CABLE, AS NOTED
—FIRE ALARM DEVICES—		
		FIRE ALARM CONTROL PANEL
		FIRE ALARM REMOTE ANNUNCIATOR
		CONTROL ADDRESSABLE MODULE
		MONITOR ADDRESSABLE MODULE
		FIRE SMOKE DAMPER
		MANUAL PULL STATION
		FIRE ALARM HORN
		COMBINATION FIRE ALARM HORN STROBE LIGHT
		FIRE ALARM STROBE LIGHT
		MAGNETIC DOOR HOLD—OPEN
		IONIZATION TYPE SMOKE DETECTOR
		PHOTOELECTRIC SMOKE DETECTOR
		DUCT SMOKE DETECTOR
		THERMAL DETECTOR
		EARLY WARNING SMOKE DETECTOR UNDER RAISED COMPUTER FLOOR
		REMOTE INDICATING LIGHT
		FLOW SWITCH
		TAMPER SWITCH
		PRESSURE SWITCH
		MANUAL ACTIVATION & ABORT BUTTON
		FM 200 SUPPRESSION SYSTEM CONNECTION

BE USED)		
ABBR.	SYMBOL	DESCRIPTION
-ONE LINE SYMBOLS-		
		MULTIPLE BATTERIES
		CIRCUIT BREAKER
		PULL OUT CIRCUIT BREAKER
		CONTACT NORMALLY OPEN
		CONTACT NORMALLY CLOSED
		FUSE
		FUSE WITH SWITCH
		PANEL CIRCUIT BREAKER
XFMR		DRAW-OUT POWER CIRCUIT BREAKER
		TRANSFORMER
ATS		AUTOMATIC TRANSFER SWITCH
		DISCONNECT
-GENERAL-		
		FLAG NOTE
		FEEDER NOTE
		REVISION NUMBER
		DETAIL NOTE
		MECHANICAL EQUIPMENT TAG
		ROOM NUMBER
	DETAIL BUBBLE DETAIL NUMBER SHEET NUMBER - WHERE DETAIL IS SHOWN SHEET NUMBER - WHERE DETAIL IS CALLED OUT SECTION CUT DETAIL NUMBER SHEET NUMBER - WHERE DETAIL IS SHOWN SHEET NUMBER - WHERE DETAIL IS CALLED OUT	
GENERAL NOTES: (APPLIES TO ALL SHEETS) - COORDINATE ALL WORK WITH THE CONSTRUCTION PHASING AS OUTLINED ON THE ARCHITECTURAL DOCUMENTS. COORDINATE WITH OTHER TRADES, MAINTAIN EXISTING SYSTEMS, AND SCHEDULE ALL WORK IN COORDINATION WITH THE CONSTRUCTION PHASING. - ALL ELECTRICAL WORK SHALL BE IN ACCORDANCE WITH LOCAL BUILDING CODES AND THE LATEST EDITION OF THE NATIONAL ELECTRIC CODE (NEC). - CONTRACTOR SHALL REFERENCE SPECIFICATIONS FOR CONSTRUCTION MATERIALS AND INSTALLATION PROCEDURES. SPECIFICATIONS ARE A PART OF THE CONSTRUCTION DOCUMENTS. SHOULD ANY CONFLICT ARISE BETWEEN THE DRAWINGS AND SPECIFICATIONS, SUCH CONFLICT SHALL BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER FOR RESOLUTION. - THESE DRAWINGS ARE DIAGRAMMATIC. THE CONTRACTOR SHALL PROVIDE ALL MATERIALS AND LABOR REQUIRED FOR A COMPLETE AND FULLY OPERATIONAL INSTALLATION. - LOCATIONS AND INFORMATION FOR EXISTING ELECTRICAL DEVICES AND EQUIPMENT SHOWN ON THESE PLANS ARE APPROXIMATE. THIS INFORMATION IS DERIVED FROM NON-INVASIVE FIELD OBSERVATIONS AND DRAWINGS THAT WERE AVAILABLE AT THE TIME THESE DOCUMENTS WERE PREPARED. THE CONTRACTOR IS TO VERIFY ACTUAL FIELD CONDITIONS PRIOR TO STARTING WORK. - ALL PENETRATIONS THROUGH FIRE SEPARATION WALLS ARE TO BE SEALED TO MAINTAIN SEPARATION RATING. - REFER TO MECHANICAL DRAWINGS FOR MECHANICAL EQUIPMENT LOCATIONS AND ELECTRICAL INSTALLATION REQUIREMENTS. - ALL EXPOSED CABLES AND SUPPORTS LOCATED IN PLENUM SPACE SHALL BE PLENUM RATED.		

ABBR.	SYMBOL	DESCRIPTION
-ABBREVIATIONS-		
AC		ABOVE COUNTER
AFF		ABOVE FINISHED FLOOR
AFG		ABOVE FINISHED GRADE
BFG		BELOW FINISHED GRADE
EM		EMERGENCY
GFI		GROUND FAULT INTERRUPTER
NL		NIGHT LIGHT
(E)		EXISTING
(ER)		EXISTING TO REMAIN
ST		SHUNT TRIP
TYP		TYPICAL
UON		UNLESS OTHERWISE NOTED
WP		WEATHERPROOF
WG		WIRE GUARD
OS		OCCUPANCY SENSOR
CPU		COMPUTER



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

LEGEND

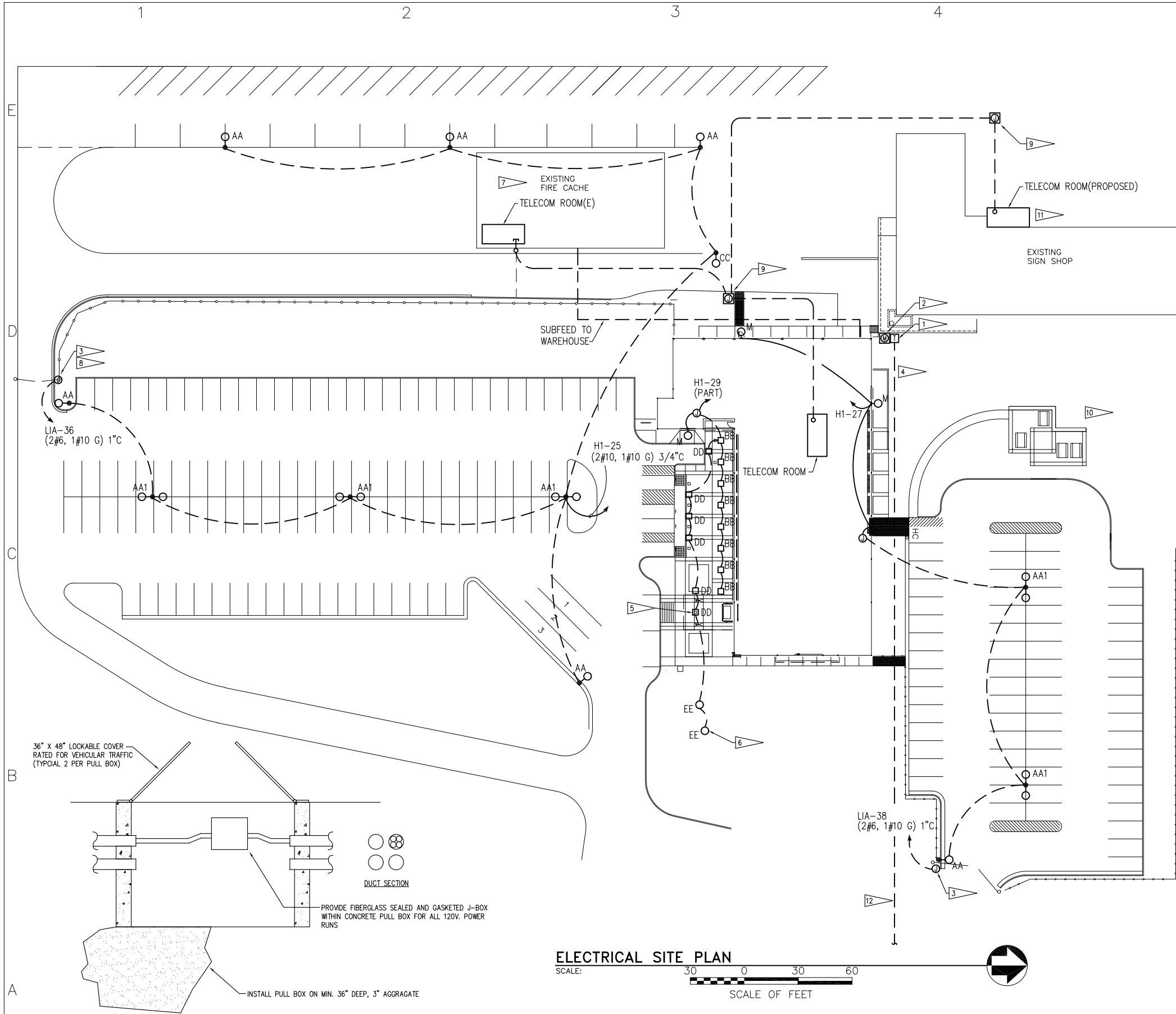
RAWLINS FIELD OFFICE BUILDING

RAWLINS

WYOMING

PROJECT NO: 16235			
DESIGN OFFICE: RMH			
DESIGNED BY: KC			
DRAWN BY: MW			
CHK'D BY: KC			
APPROVED BY: GIR			
DATE: 02-28-2003	MARK	DESCRIPTION	DATE
			APPROVED

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FLAG NOTES:

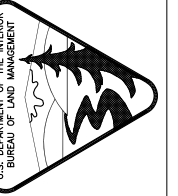
- 1 APPROXIMATE LOCATION OF NEW UTILITY TRANSFORMER. COORDINATE EXACT LOCATION WITH UTILITY COMPANY PRIOR TO ROUGH IN.
- 2 APPROXIMATE LOCATION OF NEW UTILITY METER, CTS, AND CT CAN. COORDINATE EXACT LOCATION WITH UTILITY COMPANY PRIOR TO ROUGH IN. PROVIDE ALL MATERIAL AND LABOR REQUIRED FOR A COMPLETE INSTALLATION INCLUDING SUPPORTING STRUCTURE, IF REQUIRED.
- 3 SECURITY GATE. CONNECT GATE POWER TO CIRCUIT INDICATED. PROVIDE LONG RANGE CARD READER ON PEDESTAL AND GROUND LOOPS FOR GATE ACCESS CONTROL, AND EXITING. (CARD READ AND GROUND LOOPS ON ENTRY, GROUND LOOP EXITING). PROVIDE ALL MATERIALS AND LABOR AS REQUIRED FOR A FULLY OPERATIONAL SYSTEM. PROVIDE CONDUIT AND WIRE AS RECOMMENDED BY THE GATE INSTALLER. VERIFY EXACT LOCATION AND REQUIREMENTS WITH GATE INSTALLER PRIOR TO ROUGH-IN. COORDINATE EXITING EQUIPMENT LOCATIONS WITH PATH OF TRAVEL. SEE GATE ACCESS DETAILS AND DIAGRAMS FOR ADDITIONAL INFORMATION.
- 4 CONNECT IRRIGATION CONTROLS TO L1A-54. VERIFY EXACT LOCATION OF EQUIPMENT WITH INSTALLER PRIOR TO ROUGH-IN.
- 5 CONTRACTOR SHALL VERIFY EXACT MOUNTING LOCATION OF BOLLARDS WITH LANDSCAPE CONTRACTOR PRIOR TO INSTALLATION. (TYPICAL OF 6)
- 6 CONTRACTOR SHALL VERIFY EXACT MOUNTING LOCATION OF FLAG POLE LIGHTS WITH LANDSCAPE CONTRACTOR PRIOR TO INSTALLATION. (TYPICAL OF 2)
- 7 COORDINATE EXACT LOCATION OF MONITOR AND GATE RELEASE PUSH BUTTON IN THIS BUILDING WITH CONTRACTING OFFICER PRIOR TO ROUGH-IN.
- 8 PROVIDE INTERCOM AT GATE CONTROL PEDESTAL, AND CAMERA, LIGHT, AND MOTION DETECTOR ON SEPARATE POLE AT THIS GATE. PROVIDE ALL MATERIALS AND LABOR REQUIRED FOR PUSH BUTTON GATE CONTROL AND VIDEO MONITORING AT THE RECEPTION DESK OF THE NEW BUILDING AND THE FIRE DISPATCH OF THE EXISTING FIRE CACHE BUILDING. VERIFY EXACT LOCATION AND REQUIREMENTS WITH GATE INSTALLER PRIOR TO ROUGH-IN. SEE GATE ACCESS DETAILS AND DIAGRAMS FOR ADDITIONAL INFORMATION.
- 9 PROVIDE 48" X 72" X 36" OPEN BOTTOM CONCRETE PULL BOX RATED FOR VEHICULAR TRAFFIC IN APPROXIMATE LOCATION SHOWN FLUSH WITH FINISHED GRADE. PROVIDE (2) SPLIT HINGED LOCKABLE 48" X 36" TRAFFIC COVERS. PROVIDE (3) 4" CONDUIT AND (1) 4" (4) CONDUIT INTERDUCT WITH PULL STRING TO CONNECT THE NEW BUILDING TELEPHONE TERMINAL BOARD TO THE EXISTING WAREHOUSE AND SIGN SHOP TELEPHONE TERMINAL BOARDS. SEE PULL BOX AND DUCT DIAGRAM THIS SHEET.
- 10 PROVIDE 2" CONDUIT FROM EXISTING SATELLITE DISH LOCATION TO NEW BUILDING. VERIFY EXACT LOCATION AND REQUIREMENTS WITH CIVIL PLANS AND EXISTING EQUIPMENT PRIOR TO ROUGH-IN. ROUTE CABLE FROM SATELLITE DISH TO TELEVISION OUTLETS IN CONFERENCE CENTER PER MANUFACTURER'S RECOMMENDATIONS. THE NEW BUILDING CABLE TRAY MAY BE UTILIZED FOR CABLE ROUTING WITHIN THE BUILDING. SEE SHEET E101 FOR TELEVISION OUTLET LOCATIONS.
- 11 PROVIDE CONCRETE JUNCTION BOXES RATED FOR VEHICULAR TRAFFIC FLUSH WITH GRADE AS REQUIRED TO ROUTE CONDUIT AROUND EXISTING BUILDING TO LOCATION SHOWN OR COORDINATE CONDUIT ROUTING THROUGH EXISTING BUILDING WITH CONTRACTING OFFICER.
- 12 PROVIDE ALL TRENCHING AND BACKFILL FOR UTILITY INSTALLATION. PROVIDE (2) 4" CONDUITS WITH PULLWIRE, TRANSFORMER PAD, ETC. PER THE REQUIREMENTS OF THE SERVING UTILITY. COORDINATE WITH PP&L AND PROCEED ACCORDINGLY.

ELECTRICAL SITE PLAN

SCALE: 30 0 30 60
SCALE OF FEET

PULL BOX AND DUCT DIAGRAM

SCALE: NONE



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

SITE PLAN

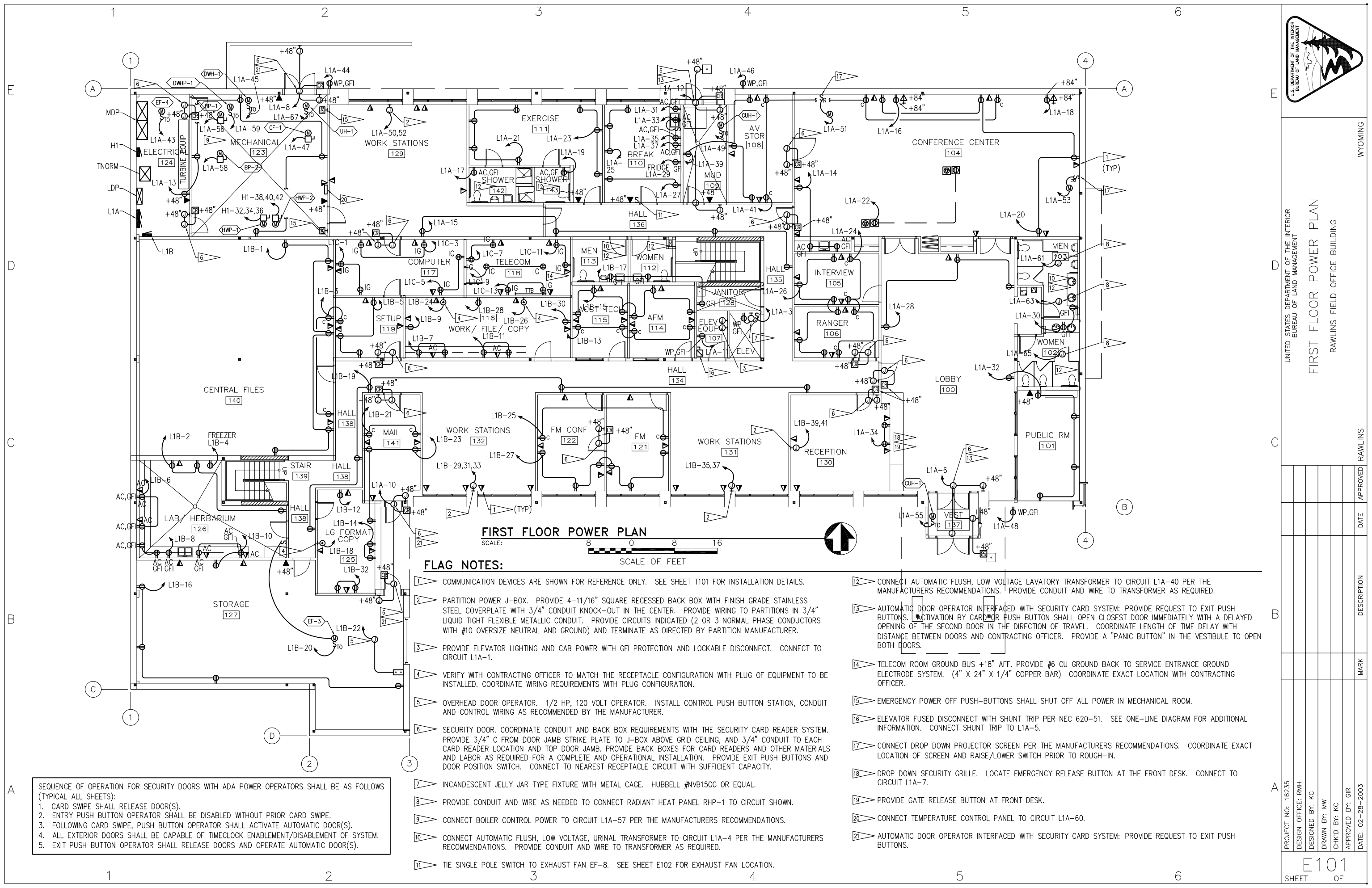
RAWLINS FIELD OFFICE BUILDING

WYOMING

PROJECT NO:	DESIGN OFFICE:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	MARK	DESCRIPTION	DATE	APPROVED	RAWLINS
16235	RMH	KC	MW	KC	GIR	02-28-2003					

E100
SHEET OF

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UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
RAWLINS FIELD OFFICE BUILDING

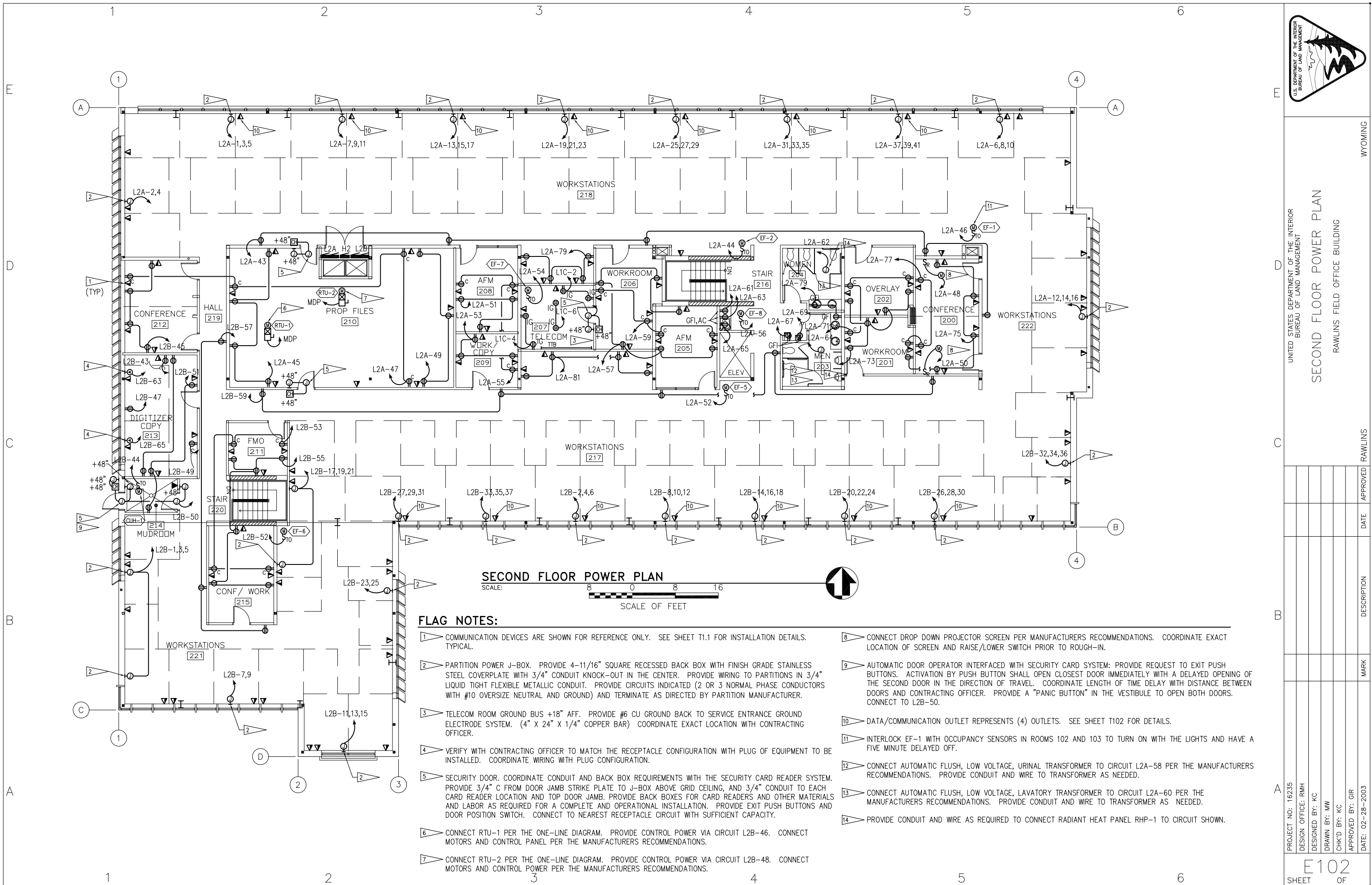
RAWLINS

DATE	APPROVED

DESCRIPTION	MARK

E101
SHEET OF

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UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

SECOND FLOOR POWER PLAN
RAWLINS FIELD OFFICE BUILDING

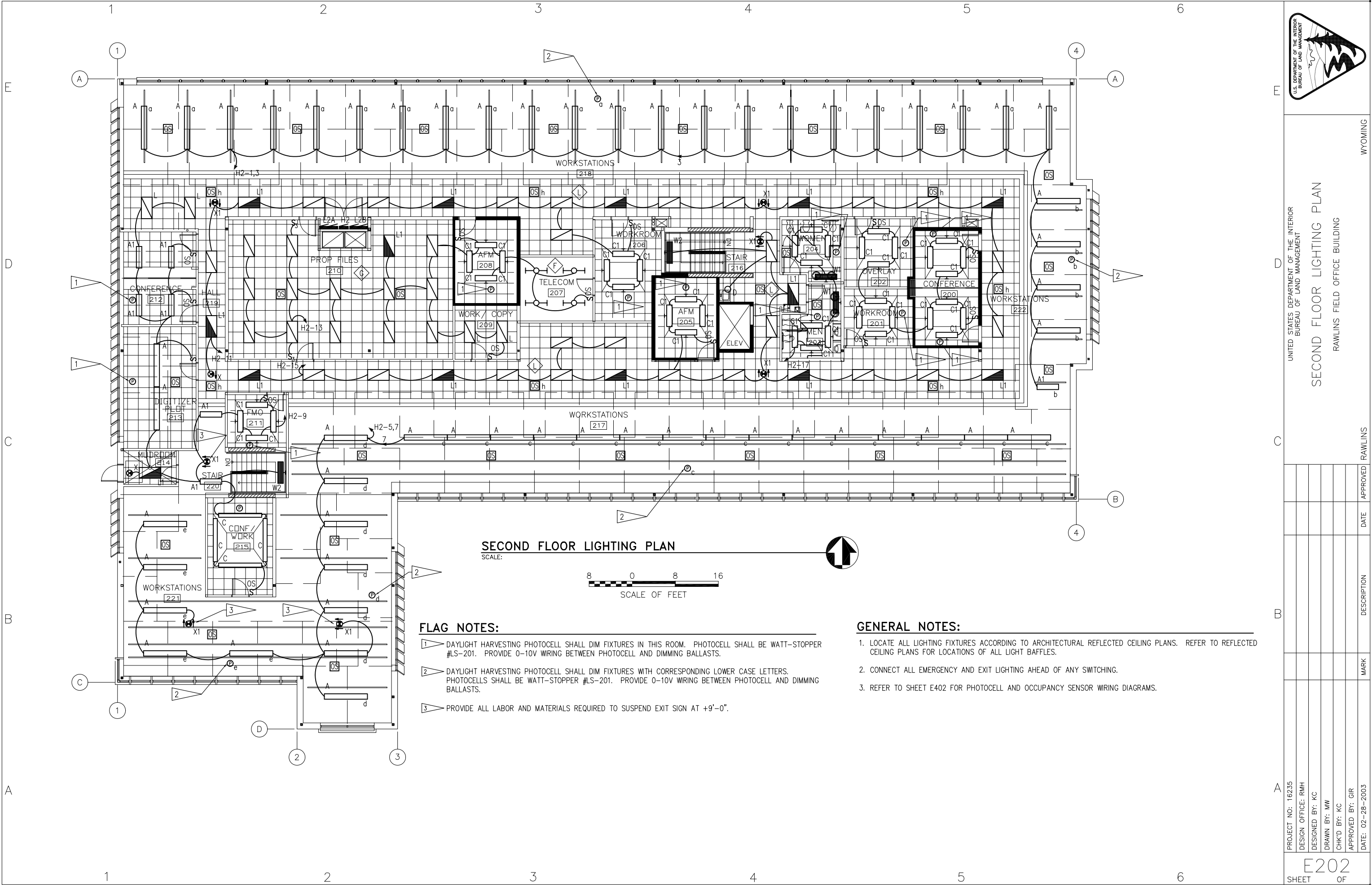
WYOMING

RAWLINS

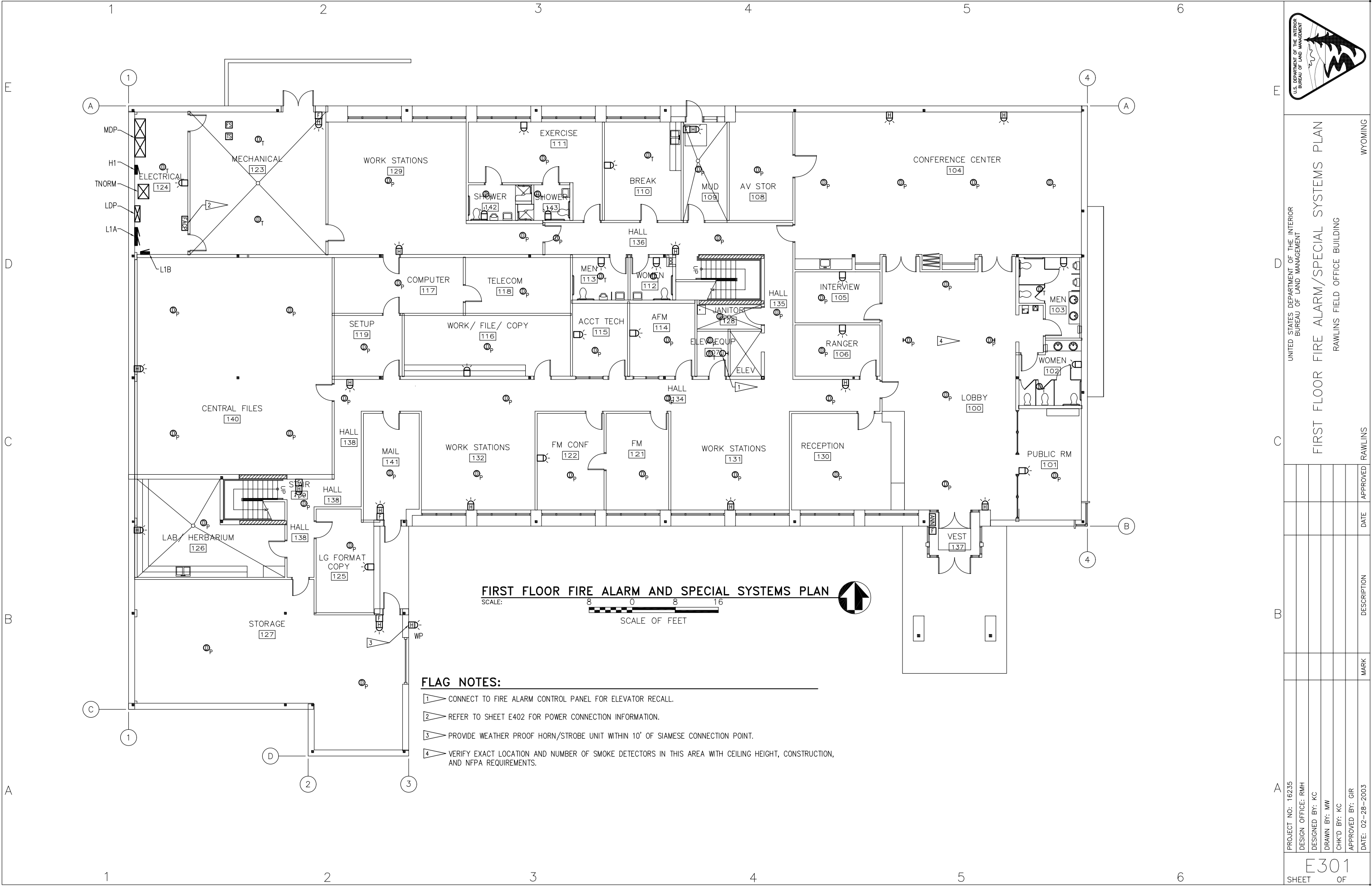
DESCRIPTION	DATE	APPROVED
MARK		
PROJECT NO: 16235		
DESIGN OFFICE: RWH		
DESIGNED BY: KC		
DRAWN BY: MW		
CHK'D BY: KC		
APPROVED BY: GJR		
DATE: 02-28-2003		

E102
SHEET OF

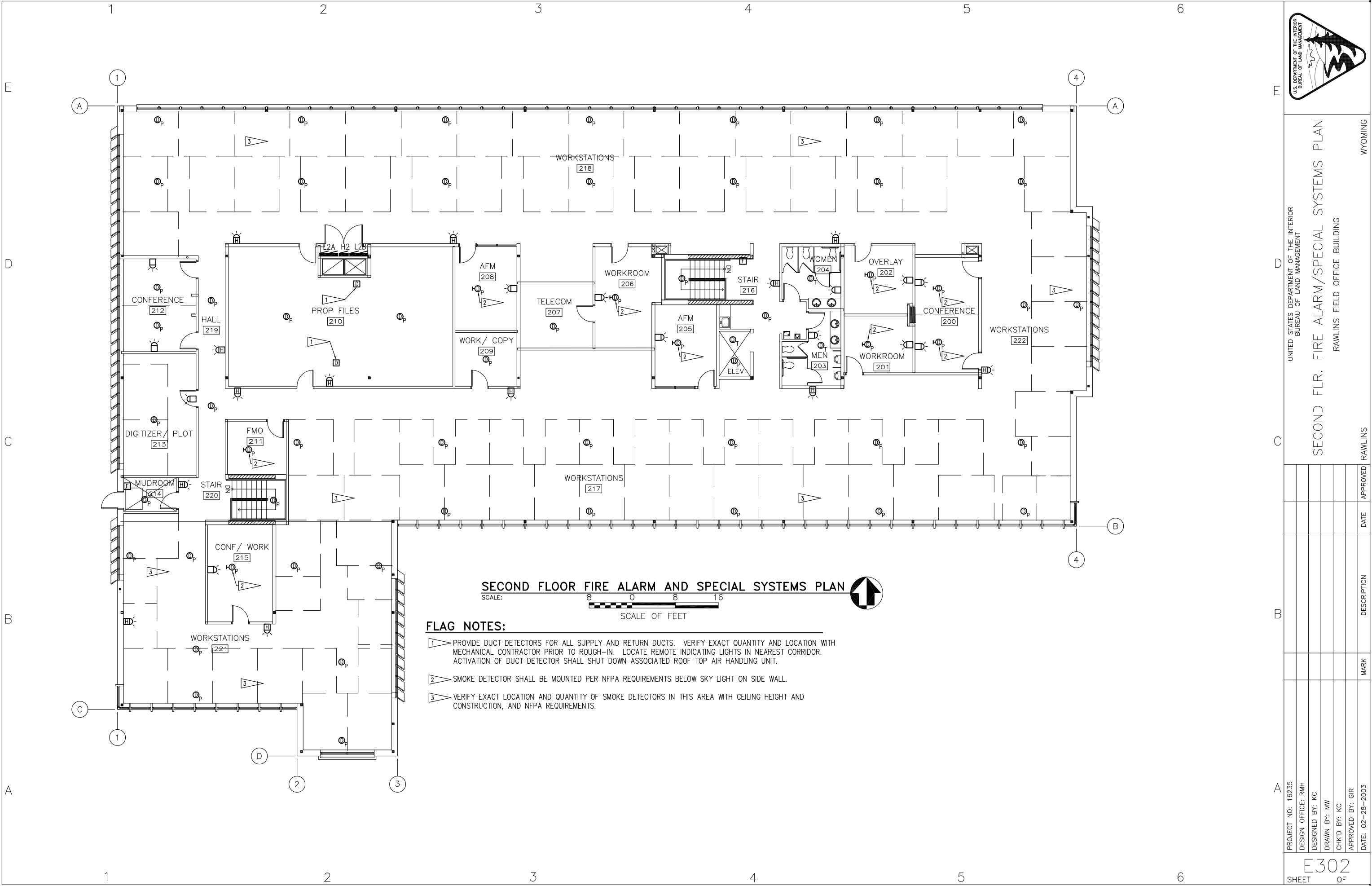
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UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

SECOND FLR. FIRE ALARM/SPECIAL SYSTEMS PLAN

RAWLINS FIELD OFFICE BUILDING

WYOMING

PROJECT NO:	DESIGN OFFICE:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	DESCRIPTION	MARK	DATE	APPROVED	RAWLINS
16235	RMH	KC	MW	KC	GIR	02-28-2003					

E302
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LUMINAIRE SCHEDULE										
LAMPING			DESCRIPTION	FINISH	MOUNT	SPECIFICATIONS				
KEY	QTY	SPEC				MANUFACTURER	CATALOG NUMBER	VOLT	VA	NOTES
A	4	F32 T8 3500K 80+ CRI	10.5" X 3" X 8' DIRECT/INDIRECT FLUORESCENT WITH 0-10V DC DIMMING BALLAST, 70% UPLIGHT, 30% DOWNLIGHT AND 92% EFFICIENCY	WHITE	CABLE 8'-6" AFF	PEERLESS OR EQUAL	10CRM8-2-32-X-R8-277- ADVANCE MARK VII- LP835-X-X	277	130	1
A1	2	F32 T8 3500K 80+ CRI	SAME AS TYPE A EXCEPT 4' LONG	WHITE	CABLE 8'-6" AFF	PEERLESS OR EQUAL	10CRM8-2-32-X-R4-277- ADVANCE MARK VII- LP835-X-X	277	70	1
A2	4	F32 T8 3500K 80+ CRI	SAME AS TYPE A EXCEPT WITH NON-DIM ELECTRONIC BALLASTS	WHITE	CABLE 8'-6" AFF	PEERLESS OR EQUAL	10CRM8-2-32-X-R8-277- LP835-X-X	277	130	1
B	6	F32 T8 3500K 80+ CRI	10.5" X 3" X 8' DIRECT/INDIRECT FLUORESCENT WITH 0-10V DC DIMMING BALLAST, 70% UPLIGHT, 30% DOWNLIGHT AND 92% EFFICIENCY	WHITE	CABLE 8'-6" AFF	PEERLESS OR EQUAL	10CRM8-3-32-X-R8-277- GEB10-LP835-X-X	277	200	1
C	4	F32 T8 3500K 80+ CRI	7.6" X 3" X 8' DIRECT/INDIRECT FLUORESCENT WITH 0-10V DC DIMMING BALLAST, 95% UPLIGHT, 5% DOWNLIGHT AND 62% EFFICIENCY	WHITE	SURFACE WALL	PEERLESS OR EQUAL	CRW7-2-32-WHR-X-X-DM 0-10V DIMMING-SCT- LP835	277	130	2
C1	2	F32 T8 3500K 80+ CRI	7.6" X 3" X 4' DIRECT/INDIRECT FLUORESCENT WITH 0-10V DC DIMMING BALLAST, 95% UPLIGHT, 5% DOWNLIGHT AND 62% EFFICIENCY	WHITE	SURFACE WALL	PEERLESS OR EQUAL	CRW7-2-32-WHR-X-X-DM 0-10V DIMMING-SCT- LP835	277	70	2
C2	2	F32 T8 3500K 80+ CRI	7.6" X 3" X 8' DIRECT/INDIRECT FLUORESCENT WITH LUTRON DIMMING BALLAST, 95% UPLIGHT, 5% DOWNLIGHT AND 62% EFFICIENCY	WHITE	SURFACE WALL	PEERLESS OR EQUAL	CRW7-1-32-WHR-X-X-DM LUTRON ECHO 10 DIM- SCT-LP835	120	130	2
C3	2	F32 T8 3500K 80+ CRI	7.6" X 3" X 4' DIRECT/INDIRECT FLUORESCENT WITH ELECTRONIC BALLAST, 95% UPLIGHT, 5% DOWNLIGHT AND 62% EFFICIENCY	WHITE	SURFACE WALL	PEERLESS OR EQUAL	CRW7-2-32-WHR-X-X-EL- SCT-LP835	277	70	2
D	1	32W PLT 3500K 80+ CRI	7-1/2" APERTURE COMPACT FLUORESCENT DOWNLIGHT WITH CLEAR ALZAK SELF-TRIMMING CONE AND ELECTRONIC BALLAST	MATTE CLEAR	RECESSED 12"	PRESCOLITE OR EQUAL	CFT832EB-STF802-MFC	277/ 120	35	3
D1	1	32W PLT 4100K	SAME AS TYPE D EXCEPT WITH LUTRON 1% DIMMING BALLAST	MATTE CLEAR	RECESSED 12"	PRESCOLITE OR EQUAL	CFT832EB-STF802-MFC LUTRON HI-LUME BALLAST	120	35	
D2	1	32W PLT 4100K	SAME AS TYPE D EXCEPT WITH DAMP LOCATION LISTING	MATTE CLEAR	RECESSED 12"	PRESCOLITE OR EQUAL	CFT832EB-STF802-MFC DAMP LABEL	120	35	
D3	1	32W PLT 4100K	SAME AS TYPE D EXCEPT WITH BODINE EMERGENCY BALLAST	MATTE CLEAR	RECESSED 12"	PRESCOLITE OR EQUAL	CFT832EB-STF802-BODINE EM. LABEL	120	35	
D4	1	32W PLT 3500K	6" APERTURE COMPACT FLUORESCENT DOWNLIGHT WITH PRISMATIC ACRYLIC SHOWER LENS	MATTE CLEAR	RECESSED 8"	PRESCOLITE OR EQUAL	TBX-TBXIC-TL3	120	35	
F	2	F32 T8 3500K	4' FLUORESCENT STRIPLIGHT WITH WIREGUARD AND ELECTRONIC BALLAST	STANDARD	PENDANT 8'-0" AFF	COLUMBIA OR EQUAL	CS-4-2-32-EBBLH-WG	277	70	
F1	1	F32 T8 3500K	2.75" WIDE X 4' FLUORESCENT STRIPLIGHT WITH ELECTRONIC BALLAST	STANDARD	PENDANT 8'-0" AFF	LITHONIA OR EQUAL	S-1-32-277-GEB	277	35	
G	3	F32 T8 3500K	2' X 4' FLUORESCENT WITH PRISMATIC ACRYLIC .125" THICK LENS AND ELECTRONIC BALLAST	STANDARD	RECESSED	COLUMBIA OR EQUAL	4PS-24-3-32-G-FS-A12.125- EBBLH-277	277	105	
G1	2	F32 T8 3500K	SAME AS TYPE G EXCEPT WITH TWO LAMPS	STANDARD	RECESSED	COLUMBIA OR EQUAL	4PS-24-2-32-G-FS-A12.125- EBBLH-277	277	70	
H	2	F32 T8 3500K 80+ CRI	10" X 4' FLUORESCENT WITH WRAP AROUND PRISMATIC ACRYLIC LENS AN ELECTRONIC BALLAST	STANDARD	SURFACE	COLUMBIA OR EQUAL	WC4-2-32-EBBLH-277	277	70	
L	2	F32 T8 3500K 80+ CRI	2' X 4' FLUORESCENT WITH 12 CELL, 3" DEEP, PARABOLIC LOUVER AND ELECTRONIC BALLAST	STANDARD	RECESSED	COLUMBIA OR EQUAL	P4-24-2-32-G-LD-2-6-S- EBBLH-277	277	70	
L1	2	F32 T8 3500K 80+ CRI	SAME AS TYPE L EXCEPT EQUIPPED WITH A BATTERY PACK	STANDARD	RECESSED	COLUMBIA OR EQUAL	P4-24-2-32-G-LD-2-6-S- EBBLH-277-EM BALLAST	277	70	
M	1	70W MH, 3500K	WALL MOUNTED CUTOFF FLOOD LIGHT, CAST ALUMINUM RECTANGULAR HOUSING, WET LOCATION	PANTONE 17-1550 TP STEEPLE GRAY	WALL PER ARCHITECT	MCPHILBEN OR EQUAL	101-F7-70MH-277-PANTONE 17-1500 TP "STEEPLE GRAY"	277	100	4
N	1	Q50MR16/FL 4000 HOUR	4" APERTURE LOW-VOLTAGE DOWNLIGHT WITH CLEAR ALZAK CONE AND MAG. XFMR	CLEAR ALZAK CONE	RECESSED	PRESCOLITE OR EQUAL	LVH4-STLV2C-LV19	120	50	5
P	4	F40 TWIN 3500K	24" X 24" COMPACT FLUORESCENT PENDANT WITH PERFORATED DIFFUSER AND ELECTRONIC BALLAST	MATTE WHITE	PENDANT	DM LIGHTING OR EQUAL	5300-ELECTRONIC BALLAST	277	180	6
P1	4	F40 TWIN 3500K	SAME AS TYPE P EXCEPT WITH LUTRON ECO 10 10% DIMMING BALLASTS, 120V	MATTE WHITE	PENDANT	DM LIGHTING OR EQUAL	5300-LUTRON ECO-10 10% DIMMING BALLASTS	120	180	6
Q	1	F40 TWIN 3500K 80+ CRI	8" X 2' COMPACT FLUORESCENT WALL-WASHER WITH LUTRON ECO-10 10% DIMMING BALLAST AND CROSS-BAFFLE	STANDARD	RECESSED	ALERA OR EQUAL	EN-82-1-40TT-F-LD-B- EBTT-120-LUTRON ECO-10 10% DIMMING BALLAST	120	50	
R	1	12V Q10 HALOGEN	2.72" DIA. X 1.2" TALL HALOGEN LOW-VOLTAGE DOWNLIGHT WITH FROSTED GLASS LENS AND REMOTE ELECTRONIC TRANSFORMER	MATTE WHITE	SEMI- RECESSED	LUCIFER OR EQUAL	FPK-SC-12-10-W-ET60-12	120	10	
T	1	F32 T8 3500K 80+ CRI	4' UNDERCOUNTER FLUORESCENT TASKLIGHT WITH ELECTRONIC BALLAST AND SOLID FRONT	STANDARD	SURFACE	ALOKO OR EQUAL	SF332-ELECTRONIC BALLAST	277	35	
W	1	F32 T8 3500K 80+ CRI	7" X 3" DIRECT/INDIRECT FLUORESCENT WITH PERFORATED BAFFLE, EXTRUDED ALUMINUM HOUSING AND ELECTRONIC BALLAST	MATTE WHITE	SURFACE WALL	LINEAR OR EQUAL	S37P2-B-1-E78-277-MBP- WMT-277-XX-X'	277	35	7
W1	1	F32 T8 3500K 80+ CRI	SAME AS TYPE W EXCEPT WITH EMERGENCY BATTERY BALLAST	MATTE WHITE	SURFACE WALL	LINEAR OR EQUAL	S37P2-B-1-E78-277-MBP- WMT-277-XX-4'-EM. BALLAST	277	35	7
W2	2	F32 T8 3500K 80+ CRI	7" X 3" X 4' DIRECT/INDIRECT FLUORESCENT WITH PERFORATED BAFFLE, EXTRUDED ALUMINUM HOUSING AND EMERGENCY BATTERY BALLAST	MATTE WHITE	SURFACE WALL	LINEAR OR EQUAL	S37P2-B-2-E78-277-MBP- WMT-277-XX-4'-EM. BALLAST	277	70	7
X		LEDS INCLUDED	5/8" THICK CAST ALUMINUM LED EXIT SIGN WITH BATTERY, ARROWS AND MOUNTING AS INDICATED	PER LOCAL CODE	AS INDICATED	EVENLITE OR EQUAL	RZR-EM-X-1-X-X-X-277	277	35	
X1		LEDS INCLUDED	SAME AS TYPE X EXCEPT DOUBLE SIDED	PER LOCAL CODE	AS INDICATED	EVENLITE OR EQUAL	RZR-EM-X-2-X-X-X-277	277	35	
Y	2	PROVIDED WITH UNIT	EMERGENCY BATTERY PACK LIGHT WITH SELF-DIAGNOSTICS	WHITE	SURFACE	DUAL-LITE OR EQUAL	LZ21	277	10	

LUMINAIRE SCHEDULE										
LAMPING			DESCRIPTION	FINISH	MOUNT	SPECIFICATIONS				
KEY	QTY	SPEC				MANUFACTURER	CATALOG NUMBER	VOLT	V-A	NOTES
Z	2	MR16 INCL WITH UNIT	4.5" X 6.5" EMERGENCY LIGHT WITH CONCEALED LAMPS BEHIND FOLD DOWN DOOR. EQUIPPED WITH BATTERY.	WHITE	RECESSED	CONCEALITE OR EQUAL	F5-20-90-NS	120	35	8
Z1	2	LAMPS INCLUDED	EMERGENCY LIGHT, UNIVERSAL MOUNTING, SELF TESTING DIAGNOSTICS, MAINTENANCE FREE LEAD BATTERY.	WHITE	WALL 9' AFF	DUAL LITE OR EQUAL	EZ-21	277	10	
AA	1	175 MH	16" X 23" METAL HALIDE POLE LIGHT WITH DIE CAST ALUMINUM HOUSING, SEGMENTED TYPE 3 OPTICS AND 30' TALL POLE	BLACK	POLE CONCRETE BASE	KIM OR EQUAL	1A-ET3-175MH277-XXX-30' SQUARE STEEL POLE	277	225	9
AA1	2	175 MH	16" X 23" METAL HALIDE POLE LIGHTS (2) WITH DIE CAST ALUMINUM HOUSING, SEGMENTED TYPE 3 OPTICS AND 30' TALL POLE	BLACK	POLE CONCRETE BASE	KIM OR EQUAL	2B-ET3-175MH277-XXX-30' SQUARE STEEL POLE	277	450	9
BB	1	70 MH CERAMIC ARC TUBE	6" APERTUR METAL HALIDE DOWNLIGHT WITH MAXIMUM BRIGHTNESS OF 6008 FOOTLAMBERTS AT 45 DEGREES AND 68% EFFICIENCY	SOFT GLOW CLEAR ALZAK	RECESSED CANOPY 6.5" MAX	KURT VERSEN OR EQUAL	R7344-DL-SC	277	100	
CC	1	70 MH CERAMIC ARC TUBE	12" X 17.5" METAL HALIDE POLE LIGHT WITH DIE CAST ALUMINUM HOUSING, SEGMENTED TYPE 3 OPTICS AND 15' TALL POLE.	BLACK	POLE CONCRETE BASE	KIM OR EQUAL	1A-SET3-70MH277-XXX- 15' SQUARE STEEL POLE	277	100	9
DD	1	70 MH	36" SQUARE BOLLARD WITH LOUVERS					277	100	10
EE	1	70 MH PAR/FL	DIE CAST ALUMINUM ACCENT LIGHT, SWIVEL WITH LOCKING TEETH, WATER PROOF, VANDAL RESISTANT, PRISMATIC LENS.	BLACK	DIRECT	KIM OR EQUAL	67300-70MH277- BLACK	277	100	11

NOTES:

- FIXTURE SHALL HAVE EXTRUDED ALUMINUM SPLINE WITH ONE-PIECE ALUMINUM BAFFLE AND DIE-CAST ALUMINUM ENDCAPS.
- UNDERSIDE OF FIXTURE SHALL BE FULLY PERFORATED WITH PRECISION DIE- PUNCHED HOLES, PERFORATIONS SHALL PROVIDE 7% DOWNLIGHT. TOTAL FIXTURE EFFICIENCY SHALL BE 62% OR HIGHER. VIRGIN ACRYLIC UV-STABILIZED ACRYLIC OPAL DIFFUSER SHALL BE PROVIDED ABOVE PERFORATIONS.
- FIXTURE SHALL HAVE 45 DEGREE VERTICAL CUT-OFF TO LAMP, OR REFLECTED LAMP IMAGE. ELECTRONIC BALLAST SHALL HAVE END OF LIFE CIRCUIT PROTECTION LOW INRUSH CURRENT AND SUITABLE FOR 120 OR 277 VOLT OPERATION. BALLAST SHALL BE CAPABLE OF OPERATING 26, 32 AND 42 WATT TRIPLE TUBE LAMPS.
- FIXTURE SHALL HAVE SINGLE PIECE DIE-CAST ALUMINUM DOOR FRAME. DOOR FRAME SHALL BE HINGED WITH (2) STAINLESS CAPTIVE SCREWS. SINGLE PIECE GASKET AND 1/8" THICK TEMPERED GLASS LENS SHALL BE MECHANICALLY FASTENED TO DOOR FRAME.
- ADJUSTABLE DOWNLIGHT SHALL HAVE SELF-TRIMMING CLEAR ALZAK CONE AND SOFTENING LENS. FIXTURE SHALL HAVE 35 DEGREE VERTICAL AIMING CAPABILITY.
- PENDANT SHALL HAVE CURVED BOTTOM WITH WHITE ACRYLIC LENS OVERLAY ABOVE PERFORATIONS. PERFORATIONS SHALL HAVE 0.25" DIAMETER HOLES ON 3/8" STAGGERED SPACING, (40% OPEN). FIXTURE SHALL HAVE TWO PENDANT ROD SUPPORTS WITH SINGLE MOUNTING CANOPY, 36" OVERALL HEIGHT INCLUDING HOUSING.
- FIXTURE HOUSING SHALL BE TWO PIECE 0.093" THICK EXTRUDED ALUMINUM WITH CURVED FRONT. PERFORATED STEEL UPPER FRONT SHALL HAVE 0.045" DIA. HOLES, STAGGERED 0.90" SPACING (23% OPEN) WITH EXTRUDED 0.04" THICK WHITE ACRYLIC LENS INSERT BEHIND PERFORATIONS. PERFORATED WHITE LOUVER BLADES SHALL HAVE 3/4" HEIGHT X 3-1/4" WIDTH X 3/16" THICKNESS X 1.2" FREQUENCY. LOUVER BLADES SHALL BE MADE WITH 0.0235" PERFORATED ALUMINUM.
- OPENING FOR UNIT SHALL BE 4.5" X 6.5" X 3.5" DEEP. HOUSING SHALL BE FABRICATED FROM COLD ROLLED STEEL. DOOR SHALL BE IN RETRACTED POSITION (CONCEALED LAMPS DURING NORMAL POWER CONDITIONS). UPON LOSS OF UTILITY POWER, THE DOOR WILL OPENED AND THE LAMPS WILL BE ENERGIZED.
- ONE PIECE DIE CAST HOUSING SHALL HAVE INTEGRAL COOLING FINS ON TOP SURFACES ABOVE OPTICAL AND ELECTRICAL CHAMBER. ALL HARDWARE SHALL BE STAINLESS STEEL OR ELECTRO-ZINC PLATED STEEL. FIXTURE LENS FRAME SHALL BE CAST ALUMINUM WITH 1" MIN. THICKNESS AROUND GASKET FLANGE, GASKET SHALL BE ONE PIECE EXTRUDED SILICONE WITH VULCANIZED END CLOSURE. LENS SHALL BE 3/16" THICK TEMPERED FLAT GLASS. REFLECTOR MODULE SHALL BE FIELD ROTATABLE IN 90 DEGREE INCREMENTS AND BE CONSTRUCTED WITH SPECULAR ALZAK OPTICAL SEGMENTS RIGIDLY MOUNTED IN AN ALUMINUM FRAME. HOUSING FINISH SHALL BE SUPER TGIC THERMO SET POLYESTER POWDER-COAT PAINT.
- BOLLARD DESIGNED AND SPECIFIED BY LANDSCAPE ARCHITECT. FINAL CONNECTION SHALL BE BY ELECTRICAL CONTRACTOR.
- HOUSING FINISH SHALL BE SUPER TGIC THERMO SET POLYESTER POWDER-COAT PAINT.



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
LUMINAIRE SCHEDULE
RAWLINS FIELD OFFICE BUILDING

RAWLINS

APPROVED

DATE

DESCRIPTION

MARK

PROJECT NO: 16235
DESIGN OFFICE: RMH

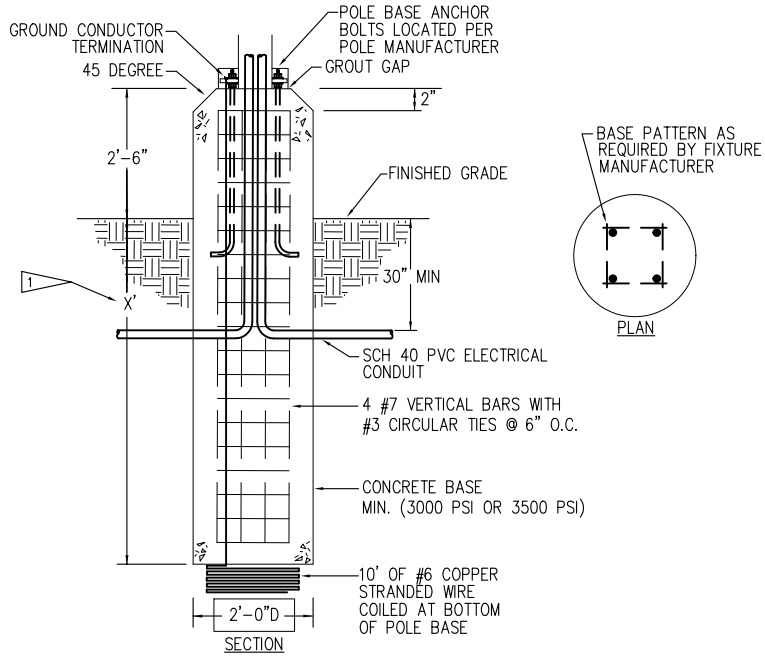
DESIGNED BY: KC
DRAWN BY: WM

CHKD BY: KC
APPROVED BY: GIR

DATE: 02-28-2003

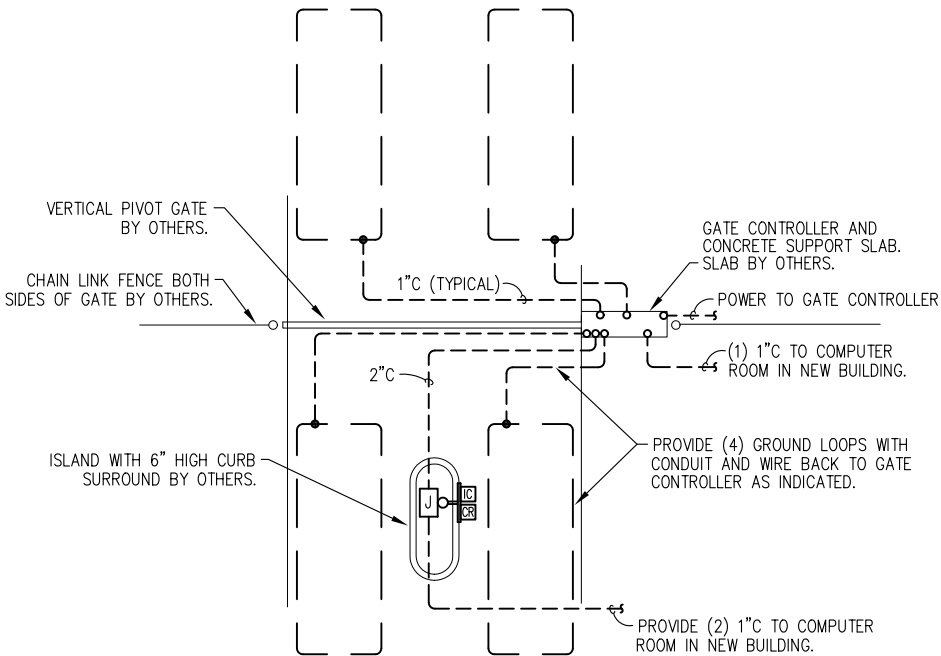
E801
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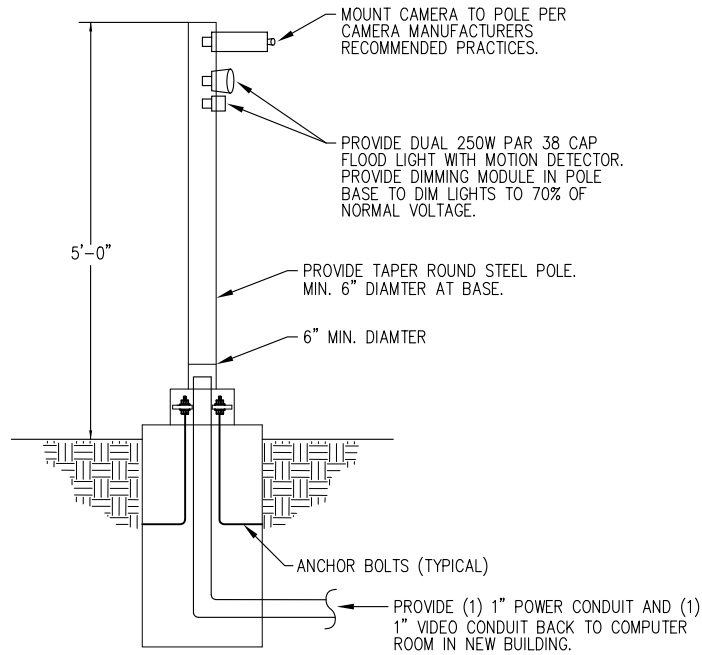
PARKING LOT POLE BASE DETAIL

SCALE: NONE



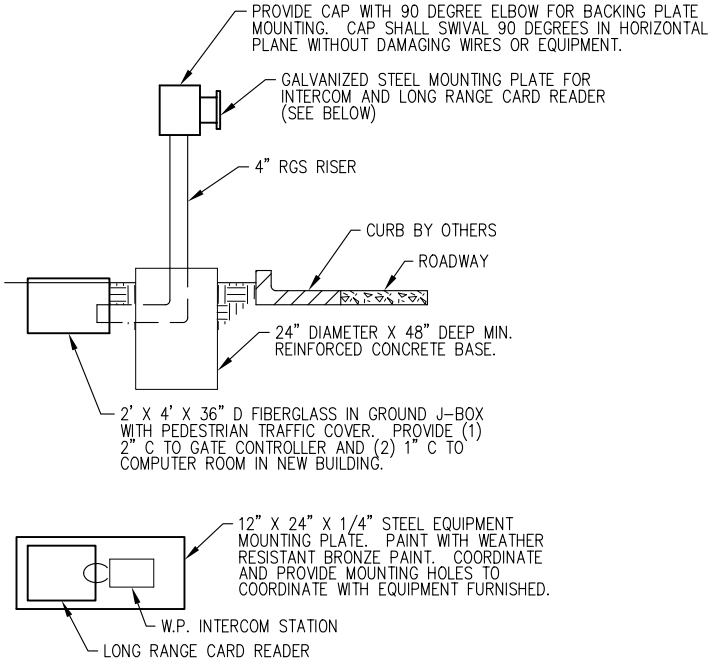
GATE PLAN AND GATE ACCESS EQUIPMENT DETAIL

SCALE: NONE



CAMERA/LIGHT MOUNTING DETAIL

SCALE: NONE



GATE CONTROL STATION DIAGRAM

SCALE: NONE

FLAG NOTES:

- 1 FOR FIXTURES TYPE "AA" AND "AA1" POLE BASE DEPTH BELOW GROUND LEVEL SHALL BE 8'-0". FOR FIXTURE TYPE "CC" POLE BASE DEPTH BELOW GROUND LEVEL SHALL BE 5'-0".



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

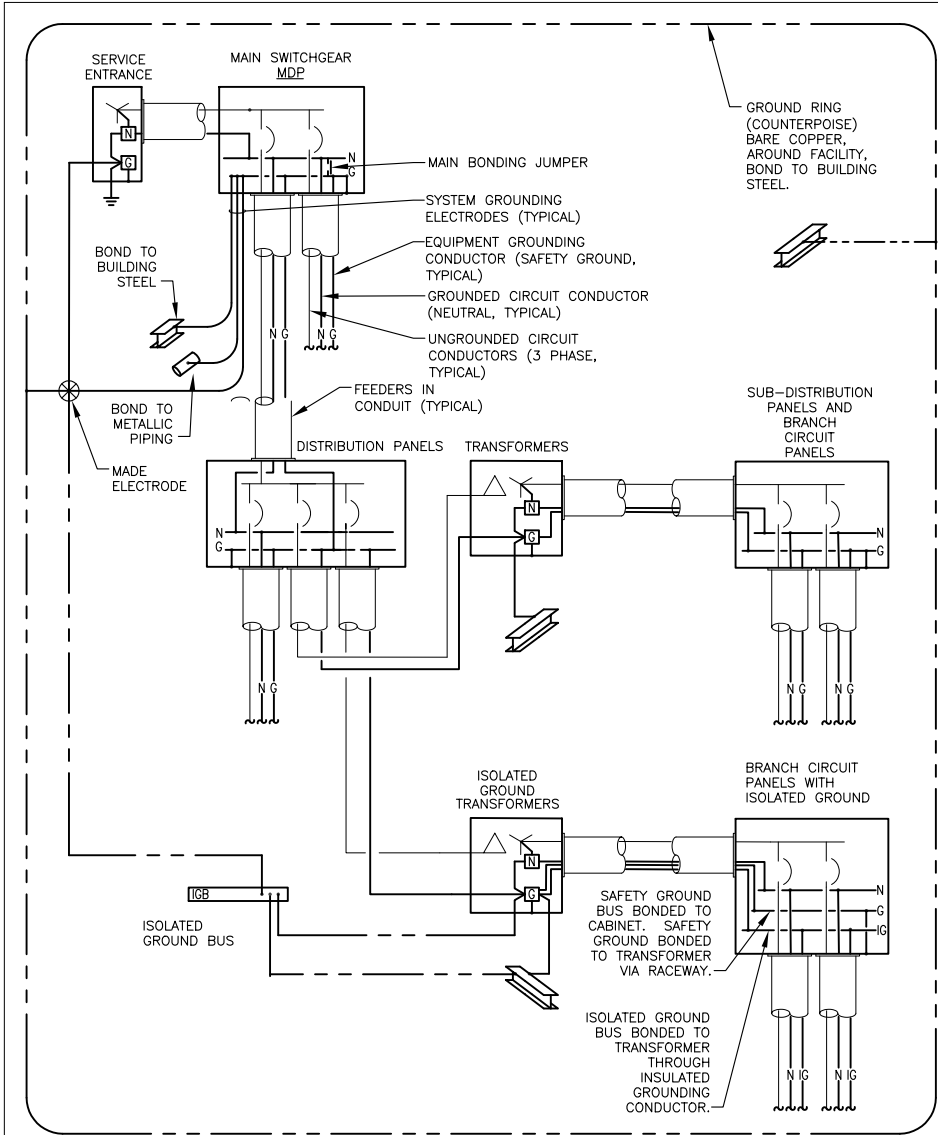
DETAILS

RAWLINS FIELD OFFICE BUILDING

WYOMING

PROJECT NO:	DESIGN OFFICE:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	MARK	DESCRIPTION	DATE	APPROVED
16235	RMH	KC	MW	KC	GIR	02-28-2003				

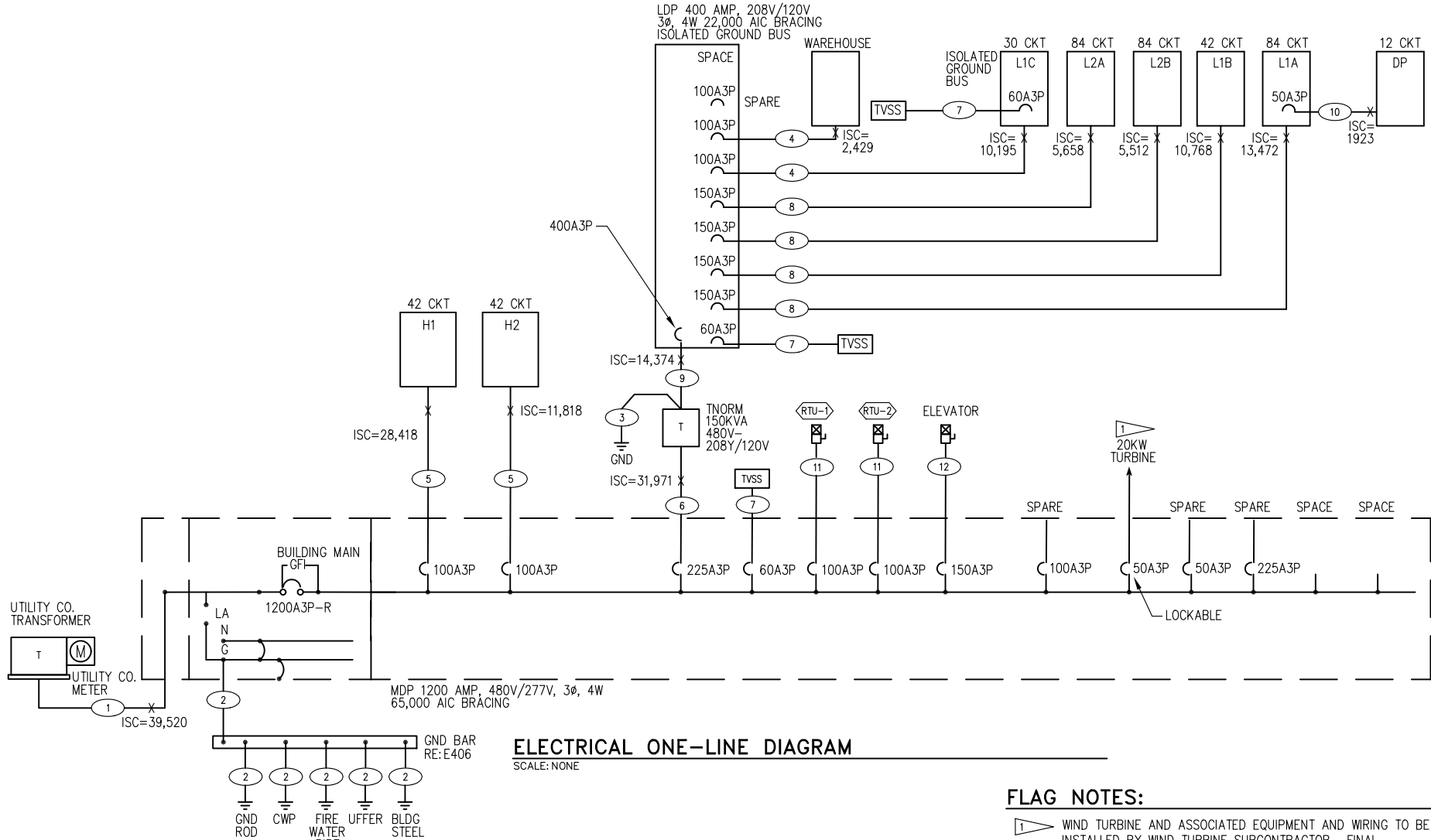
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ELECTRICAL GROUNDING DIAGRAM

SCALE: NONE
NOTE: DIAGRAM IS SCHEMATIC IN NATURE AND SHOWS THE INTENT OF THE GROUNDING SYSTEM. ADJUST AS REQUIRED TO REFLECT ACTUAL EQUIPMENT TO BE INSTALLED AS SHOWN ON THE ONE-LINE DIAGRAM.

MDP LOAD SUMMARY								
Panel Name	CONNECTED LOAD (VA)							Total
	Lighting	Recept	Motor	Heater	Kitchen	Misc	Sub-Pnl	
H1	19765		4144					23909
H2	15685							15685
L1A	365	20240	9476	2250		7000	3060	42391
L1B		30810	2040					32850
L1C		7200				50		7250
L2A		34650	3664	1500		1000		40814
L2B		38140	1368			1500		41008
RTU-1			35706					35706
RTU-2			35706					35706
Elevator			54042					54042
25% Largest motor			13511					13511
Warehouse	14440	14440						28880
CONNECTED TOTAL	50255	145480	159657	3750		9550		
NEC DEMAND	125%	50%	100%	100%	100%	100%		
DEMAND TOTAL	62819	77740	159657	3750		9550		
TOTAL CODE DEMAND							314	KVA
							378	AMPS



ELECTRICAL ONE-LINE DIAGRAM

SCALE: NONE

FLAG NOTES:

1 WIND TURBINE AND ASSOCIATED EQUIPMENT AND WIRING TO BE INSTALLED BY WIND TURBINE SUBCONTRACTOR. FINAL CONNECTION TO MAIN DISTRIBUTION PANEL SHALL BE BY ELECTRICAL CONTRACTOR.

LDP LOAD SUMMARY								
Panel Name	CONNECTED LOAD (VA)							Total
	Lighting	Recept	Motor	Heater	Kitchen	Misc	Sub-Pnl	
L1A	365	20240	9476	2250		7000	3060	42391
L1B		30810	2040					32850
L1C		7200				50		7250
L2A		34650	3664	1500		1000		40814
L2B		38140	1368			1500		41008
25% Largest motor			648					648
Warehouse	14440	14440						28880
CONNECTED TOTAL	14805	145480	17196	3750		9550		
NEC DEMAND	125%	50%	100%	100%	100%	100%		
DEMAND TOTAL	18506	77740	17196	3750		9550		
TOTAL CODE DEMAND						127		KVA
						352		AMPS

FEEDER SCHEDULE	
KEY	DESCRIPTION
1	4[(4#350KCMIL) 2-1/2"C]
2	(1#3/0 GND) 1"C
3	(1#2 GND) 3/4"C
4	(4#1, 1#8 GND, 1#8 IG) 2"C
5	(4#1, 1#8 GND) 1-1/2"C
6	(3#4/0, 1#4 GND) 2"C
7	(4#4, 1#8 GND) 1-1/4"C
8	(4#1/0, 1#6 GND) 1-1/2"C
9	2[(4#3/0, 1#2 GND, 1#2 IG) 2-1/2"C]
10	(4#6, 1#10 GND) 1"C
11	(3#1, 1#8 GND) 1-1/2"C
12	(3#1/0, 1#6 GND) 1-1/2"C



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
ONE-LINE DIAGRAM AND LOAD SUMMARY
RAWLINS FIELD OFFICE BUILDING
WYOMING

RAWLINS
APPROVED
DATE
MARK

PROJECT NO: 16235
DESIGN OFFICE: RMH
DESIGNED BY: KC
DRAWN BY: MW
CHK'D BY: KC
APPROVED BY: GJR
DATE: 02-28-2003

E804
SHEET OF

Name:		LIA										Mnt		Method:		Surface						
Volt:		208/120										Bus		Rating:		225						
Ph:		3										Main		Breaker:		Main Lug Only						
Wire:		4										AIC		Rating:		22,000						
	Code	Description	Load VA	Bkr	P	Ph A	Ph B	Ph C	P	Bkr	Load VA	Description	Code									
1	6	Elev cab power	500	20	1	*			1	20	500	FACP control power	6	2								
3	6	Elev pit	500	20	1		*		1	20	500	Auto flush urinal	6	4								
5	6	Elev shunt trip	500	20	1			*	1	20	500	ADA main entry	6	6								
7	3	Security grille	500	20	1	*			1	20	500	ADA N entrances	6	8								
9	6	Lighting contactor	500	20	1		*		1	20	500	ADA loading entrance	6	10								
11	2	Elev equip rm recept	500	20	1			*	1	20	500	ADA N entrances	6	12								
13	2	Rm 123,124 recept	900	20	1	*			1	20	540	Rm 104 recept	2	14								
15	2	N corridor recept	720	20	1		*		1	20	200	Rm 104 TV recept	2	16								
17	2	Shower recept	180	20	1			*	1	20	200	Rm 104 TV recept	2	18								
19	2	Shower recept	180	20	1	*			1	20	720	Rm 104 recept	2	20								
21	2	Rm 111 recept	720	20	1		*		1	20	360	Rm 104 flr recept	2	22								
23	2	Rm 111 recept	540	20	1			*	1	20	720	CPU recept	2	24								
25	2	Break rm recept	540	20	1	*			1	20	1080	Rm 105,106 recept	2	26								
27	2	Break rm recept	700	20	1		*		1	20	720	Lobby recept	2	28								
29	2	Break rm fridge	830	20	1			*	1	20	360	E rest room recept	2	30								
31	2	Break rm coffee	1500	20	1	*			1	20	900	Rm 101 recept	2	32								
33	2	Break rm disposal	1130	20	1		*		1	20	720	Receptionist recept	2	34								
35	2	Break rm microwave	1500	20	1			*	1	20	1176	Security gate	3	36								
37	2	Break rm dedicated	180	20	1	*			1	20	1176	Security gate	3	38								
39	2	Rm 108,109 recept	720	20	1		*		1	20	500	Auto flush Lavatory	6	40								
41	2	CPU recept	720	20	1			*	1	20		Spare		42								
43	3	EF-4	696	20	1	*			1	20	180	N outdoor recept	2	44								
45	3	DWH-1	500	20	1		*		1	20	180	N outdoor recept	2	46								
47	3	GF-1	864	20	1			*	1	20	180	S outdoor recept	2	48								
49	3	CUH-1 room 109	192	20	1	*			1	20	900	Work stn room 129	2	50								
51	3	Projector	500	20	1		*		1	20	720	Work stn room 129	2	52								
53	3	Projector	500	20	1			*	1	20	500	Irrigation	6	54								
55	3	CUH-1 room 137	192	20	1	*			1	20	1176	BP-1	3	56								
57	6	Boiler control panel	500	20	1		*		1	20	1176	BP-2	3	58								
59	3	DWHP-1	360	20	1			*	1	20	500	Temp control panel	6	60								
61	4	Radiant heat panel	750	20	1	*			1	20		Spare		62								
63	4	Radiant heat panel	750	20	1		*		1	20		Spare		64								
65	4	Radiant heat panel	750	20	1			*	1	20		Spare		66								
67	3	UH-1	468	20	1	*			1	20		Spare		68								
69		Spare		20	1		*		1	20	365	Restroom & Display ltr	1	70								
71		Spare		20	1			*	1	20		Spare		72								
73		Spare		20	1	*			1	20		Spare		74								
75		Spare		20	1		*		1	20		Spare		76								
77		Spare		20	1			*	1	20		Spare		78								
79		Spare		20	1	*			3	50	3060	Panel DP	7	80								
81		Spare		20	1		*		/					82								
83		Spare		20	1			*	/					84								
Code															Description		Load VA		Den		Load Summary	
1		Lighting		365		125%				Per Phase												
2		Rec up to 10,000		10000		100%				Ph A		15290		VA								
		Rec over 10,000		10240		50%				Ph B		14201		VA								
3		Motor		8300		100%				Ph C		12900		VA								
		Largest Motor		1176		125%																
4		Heater		2250		100%				Connected				42391 VA								
5		Kitchen		0		100%				Code Demand				37656 VA								
6		Miscellaneous		7000		100%																
7		Sub Panel		3060		100%				Code Demand				104.65 Amps								

FLAG NOTES:

1 PANEL 'L1C' SHALL BE AN ISOLATED GROUND PANEL. ALL CIRCUITS FED FROM THIS PANEL SHALL BE PROVIDED WITH AN ISCOLATED GROUND.

Name:		L1B				Mnt				Method:		Surface			
Volt:		208/120				Bus				Rating:		225			
Ph:		3				Main				Breaker:		Main lug only			
Wire:		4				AIC				Rating:		22,000			

		Code	Description	Load	Bkr	P	Ph	Ph	Ph	P	Bkr	Load	Description	Code	
				VA			A	B	C			VA			
1	2	Rm 140 recept	900	20	1	*				1	20	540	Rm 126 recept	2	2
3	2	CPU recept	540	20	1		*			1	20	830	Rm 126 freezer recept	2	4
5	2	Rm 119 recept	540	20	1			*		1	20	360	Rm 126 recept	2	6
7	2	Rm 115 recept	1000	20	1	*				1	20	540	Rm 126 recept	2	8
9	2	Rm 115 recept	1500	20	1		*			1	20	540	Rm 126 recept	2	10
11	2	Rm 115 recept	1000	20	1			*		1	20	1000	Rm 125 recept	2	12
13	2	CPU recept	720	20	1	*				1	20	1000	Rm 125 recept	2	14
15	2	Rm 114,115 recept	1080	20	1		*			1	20	720	Rm 127 recept	2	16
17	2	N rest room recept	540	20	1			*		1	30	2000	Rm 125 recept	2	18
19	2	S corridor recept	900	20	1	*				1	20	864	EF-3	3	20
21	2	CPU recept	360	20	1		*			1	20	1176	Overhead Door	3	22
23	2	Rm 125,141 recept	540	20	1			*		1	30	2000	Rm 116 recept	2	24
25	2	CPU recept	360	20	1	*				1	30	2000	Rm 116 recept	2	26
27	2	Rm 121,122 recept	1080	20	1		*			1	20	1000	Rm 116 recept	2	28
29	2	S partition recept	720	20	1			*		1	20	1000	Rm 116 recept	2	30
31	2	S partition recept	720	20	1	*				1	20	1000	Rm 125 recept	2	32
33	2	S partition recept	720	20	1		*			1	20		Spare		34
35	2	S partition recept	720	20	1			*		1	20		Spare		36
37	2	S partition recept	900	20	1	*				1	20		Spare		38
39	2	S partition recept	720	20	1		*			1	20		Spare		40
41	2	S partition recept	720	20	1			*		1	20		Spare		42

Code	Description	Load	VA	Dem	Load Summary
1	Lighting	0	125%		Per Phase
2	Rec up to 10,000	10000	100%		Ph A 11444 VA
	Rec over 10,000	20810	50%		Ph B 10266 VA
3	Motor	864	100%		Ph C 11140 VA
	Largest Motor	1176	125%		
4	Heater	0	100%		Connected 32850 VA
5	Kitchen	0	100%		Code Demand 22739 VA
6	Miscellaneous	0	100%		
7	Sub Panel	0	100%		Code Demand 63.19 Amps

Name:		L1C				Mnt				Method:		Surface			
Volt:		208/120				Bus				Rating:		100			
Ph:		3				Main				Breaker:		Main lug only			
Wire:		4				AIC				Rating:		22,000			

		Code	Description	Load	Bkr	P	Ph	Ph	Ph	P	Bkr	Load	Description	Code	
				VA			A	B	C			VA			
1	2	CPU room recept	720	20	1	*				1	20	720	2nd fl telecom recept	2	2
3	2	CPU room recept	720	20	1		*			1	20	720	2nd fl telecom recept	2	4
5	2	CPU room recept	720	20	1			*		1	20	720	2nd fl telecom recept	2	6
7	2	1st fl telecom recept	720	20	1	*				1	20		Spare		8
9	2	1st fl telecom recept	720	20	1		*			1	20		Spare		10
11	2	1st fl telecom recept	720	20	1			*		1	20		Spare		12
13	2	1st fl telecom recept	720	20	1	*				1	20		Spare		14
15		Spare		20	1		*			1	20		Spare		16
17		Spare		20	1			*		1	20		Spare		18
19		Spare		20	1	*				1	20		Spare		20
21		Spare		20	1		*			1	20		Spare		22
23		Spare		20	1			*		1	20		Spare		24
25		Spare		20	1	*				3	60	50	TVSS	6	26
27		Spare		20	1		*			/					28
29		Spare		20	1			*		/					30
31		NOT USABLE		-		*				-			NOT USABLE		32
33		NOT USABLE		-			*			-			NOT USABLE		34
35		NOT USABLE		-				*		-			NOT USABLE		36
37		NOT USABLE		-		*				-			NOT USABLE		38
39		NOT USABLE		-			*			-			NOT USABLE		40
41		NOT USABLE		-				*		-			NOT USABLE		42

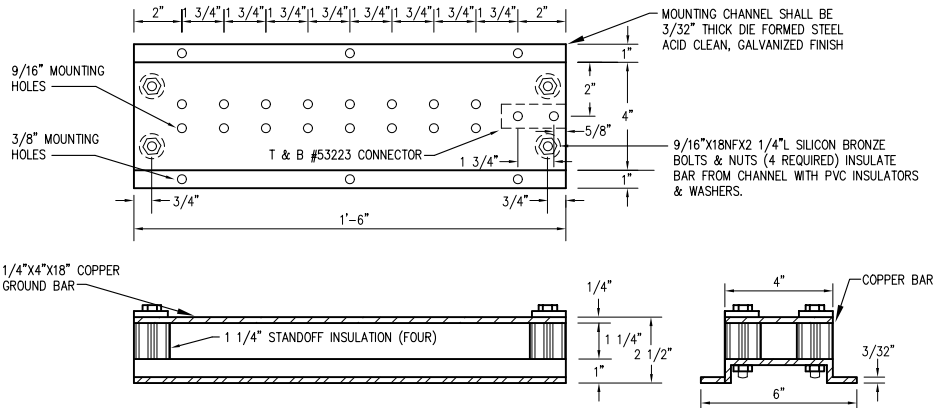
Code	Description	Load	VA	Dem	Load Summary
1	Lighting	0	125%		Per Phase
2	Rec up to 10,000	7200	100%		Ph A 2897 VA
	Rec over 10,000	0	50%		Ph B 2177 VA
3	Motor	0	100%		Ph C 2177 VA
	Largest Motor	0	125%		
4	Heater	0	100%		Connected 7250 VA
5	Kitchen	0	100%		Code Demand 7250 VA
6	Miscellaneous	50	100%		
7	Sub Panel	0	100%		Code Demand 20.15 Amps

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Name:	L2A	Mnt	Method:	Surface
Volt:	208/120	Bus	Rating:	225
Ph:	3	Main	Breaker:	Main Lug Only
Wire:	4	AIC	Rating:	10,000

	Code	Description	Load	Bkr	P	Ph	Ph	Ph	P	Bkr	Load	Description	Code	
		VA				A	B	C			VA			
1	2	N partition recept	720	20	1	*			1	20	900	N partition recept	2	2
3	2	N partition recept	720	20	1	*			1	20	720	N partition recept	2	4
5	2	N partition recept	720	20	1			*	1	20	720	N partition recept	2	6
7	2	N partition recept	720	20	1	*			1	20	720	N partition recept	2	8
9	2	N partition recept	720	20	1	*		*	1	20	540	N partition recept	2	10
11	2	N partition recept	720	20	1			*	1	20	540	N partition recept	2	12
13	2	N partition recept	720	20	1	*			1	20	540	N partition recept	2	14
15	2	N partition recept	720	20	1	*		*	1	20	540	N partition recept	2	16
17	2	N partition recept	720	20	1			*	1	20		Spare		18
19	2	N partition recept	720	20	1	*			1	20		Spare		20
21	2	N partition recept	720	20	1	*		*	1	20		Spare		22
23	2	N partition recept	720	20	1			*	1	20		Spare		24
25	2	N partition recept	720	20	1	*			1	20		Spare		26
27	2	N partition recept	720	20	1	*		*	1	20		Spare		28
29	2	N partition recept	720	20	1			*	1	20		Spare		30
31	2	N partition recept	720	20	1	*			1	20		Spare		32
33	2	N partition recept	720	20	1	*		*	1	20		Spare		34
35	2	N partition recept	720	20	1			*	1	20		Spare		36
37	2	N partition recept	720	20	1	*			1	20		Spare		38
39	2	N partition recept	720	20	1			*	1	20		Spare		40
41	2	N partition recept	720	20	1			*	1	20		Spare		42
43	2	N partition recept	900	20	1	*			1	20	864	EF-2	3	44
45	2	Rm 210 recept	720	20	1			*	1	20	468	EF-1	3	46
47	2	CPU recept	720	20	1			*	1	20	500	Projector	3	48
49	2	Rm 210 recept	540	20	1	*			1	20	500	Projector	3	50
51	2	Rm 208 recept	540	20	1	*		*	1	20	276	EF-5	3	52
53	2	Rm 209 recept	540	20	1			*	1	20	528	EF-7	3	54
55	2	CPU recept	540	20	1	*		*	1	20	528	EF-8	3	56
57	2	Rm 205,206 recept	1080	20	1	*		*	1	20	500	Auto flush urinal	6	58
59	2	CPU recept	720	20	1			*	1	20	500	Auto flush lavatory	6	60
61	2	Break rm microwave	1500	20	1	*		*	1	20	750	RHP-1	4	62
63	2	Break rm coffee	1500	20	1			*	1	20	750	RHP-1	4	64
65	2	Break rm disposal	1130	20	1			*	1	20		Spare		66
67	2	EWC	500	20	1	*		*	1	20		Spare		68
69	2	EWC	500	20	1	*		*	1	20		Spare		70
71	2	CPU recept	540	20	1			*	1	20		Spare		72
73	2	Rm 201,202 recept	900	20	1	*		*	1	20		Spare		74
75	2	Rm 200,201 recept	900	20	1	*		*	1	20		Spare		76
77	2	CPU recept	540	20	1			*	1	20		Spare		78
79	2	Work station recept	720	20	1	*		*	1	20		Spare		80
81	2	Work station recept	720	20	1	*		*	1	20		Spare		82
83		Spare		20	1			*	1	20		Spare		84

Code	Description	Load	VA	Dem	Per	Phase	Load Summary
1	Lighting	0	125%				
2	Rec up to 10,000	10000	100%		Ph A	15442	VA
	Rec over 10,000	26090	50%		Ph B	14794	VA
3	Motor	2800	100%		Ph C	12018	VA
	Largest Motor	864	125%				
4	Heater	1500	100%		Connected		42254 VA
5	Kitchen	0	100%		Code Demand		29425 VA
6	Miscellaneous	1000	100%				
7	Sub Panel	0	100%		Code Demand		81.77 Amps



GROUND BAR DETAIL

SCALE: NONE

Name:	L2B	Mnt	Method:	Surface
Volt:	208/120	Bus	Rating:	225
Ph:	3	Main	Breaker:	Main Lug Only
Wire:	4	AIC	Rating:	10,000

	Code	Description	Load	Bkr	P	Ph	Ph	Ph	P	Bkr	Load	Description	Code	
		VA				A	B	C			VA			
1	2	S partition recept	720	20	1	*			1	20	720	S partition recept	2	2
3	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	4
5	2	S partition recept	720	20	1			*	1	20	720	S partition recept	2	6
7	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	8
9	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	10
11	2	S partition recept	720	20	1			*	1	20	720	S partition recept	2	12
13	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	14
15	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	16
17	2	S partition recept	720	20	1			*	1	20	720	S partition recept	2	18
19	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	20
21	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	22
23	2	S partition recept	720	20	1			*	1	20	720	S partition recept	2	24
25	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	26
27	2	S partition recept	720	20	1			*	1	20	720	S partition recept	2	28
29	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	30
31	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	32
33	2	S partition recept	720	20	1	*		*	1	20	720	S partition recept	2	34
35	2	S partition recept	720	20	1			*	1	20	720	S partition recept	2	36
37	2	S partition recept	720	20	1	*		*	1	20		Spare		38
39		Spare		20	1			*	1	20		Spare		40
41		Spare		20	1			*	1	20		Spare		42
43	2	Rm 213 recept	1000	20	1	*		*	1	20	192	CUH-1 room 214	3	44
45	2	Rm 212 recept	720	20	1	*		*	1	20	500	RTU-1 control pwr	6	46
47	2	Rm 213 recept	1000	20	1			*	1	20	500	RTU-2 control pwr	6	48
49	2	Rm 213 recept	1000	20	1	*		*	1	20	500	ADA W entrance	6	50
51	2	Rm 213 recept	540	20	1			*	1	20	1176	EF-6	3	52
53	2	CPU recept	720	20	1			*	1	20		Spare		54
55	2	Rm 211,215 recept	1080	20	1	*		*	1	20		Spare		56
57	2	W corridor recept	540	20	1	*		*	1	20		Spare		58
59	2	S corridor recept	900	20	1			*	1	20		Spare		60
61		Spare		20	1	*		*	1	20		Spare		62
63	2	Rm 213 recept	2000	30	1	*		*	1	20		Spare		64
65	2	Rm 213 recept	2000	30	1			*	1	20		Spare		66
67		Spare		20	1	*		*	1	20		Spare		68
69		Spare		20	1	*		*	1	20		Spare		70
71		Spare		20	1			*	1	20		Spare		72
73		Spare		20	1	*		*	1	20		Spare		74
75		Spare		20	1	*		*	1	20		Spare		76
77		Spare		20	1			*	1	20		Spare		78
79		Spare		20	1	*		*	1	20		Spare		80
81		Spare		20	1			*	1	20		Spare		82
83		Spare		20	1			*	1	20		Spare		84

Code	Description	Load	VA	Dem	Per	Phase	Load Summary
1	Lighting	0	125%				
2	Rec up to 10,000	10000	100%		Ph A	13132	VA
	Rec over 10,000	28140	50%		Ph B	14116	VA
3	Motor	192	100%		Ph C	13760	VA
	Largest Motor	1176	125%				
4	Heater	0	100%		Connected		41008 VA
5	Kitchen	0	100%		Code Demand		27232 VA
6	Miscellaneous	1500	100%				
7	Sub Panel	0	100%		Code Demand		75.68 Amps

NOTES:

1. VERTICALLY MOUNT OPGPB GROUND BARS 10'-0" A.F.F.
2. HORIZONTALLY MOUNT MAIN GROUND BAR 9'-0" A.F.F.
3. PROVIDE LAMACOID ENGRAVED NAMEPLATE, BLACK WITH 3/8" H. WHITE , LETTERS AT READING "MAIN GROUND BAR".



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

PANEL SCHEDULES
RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

APPROVED

DATE

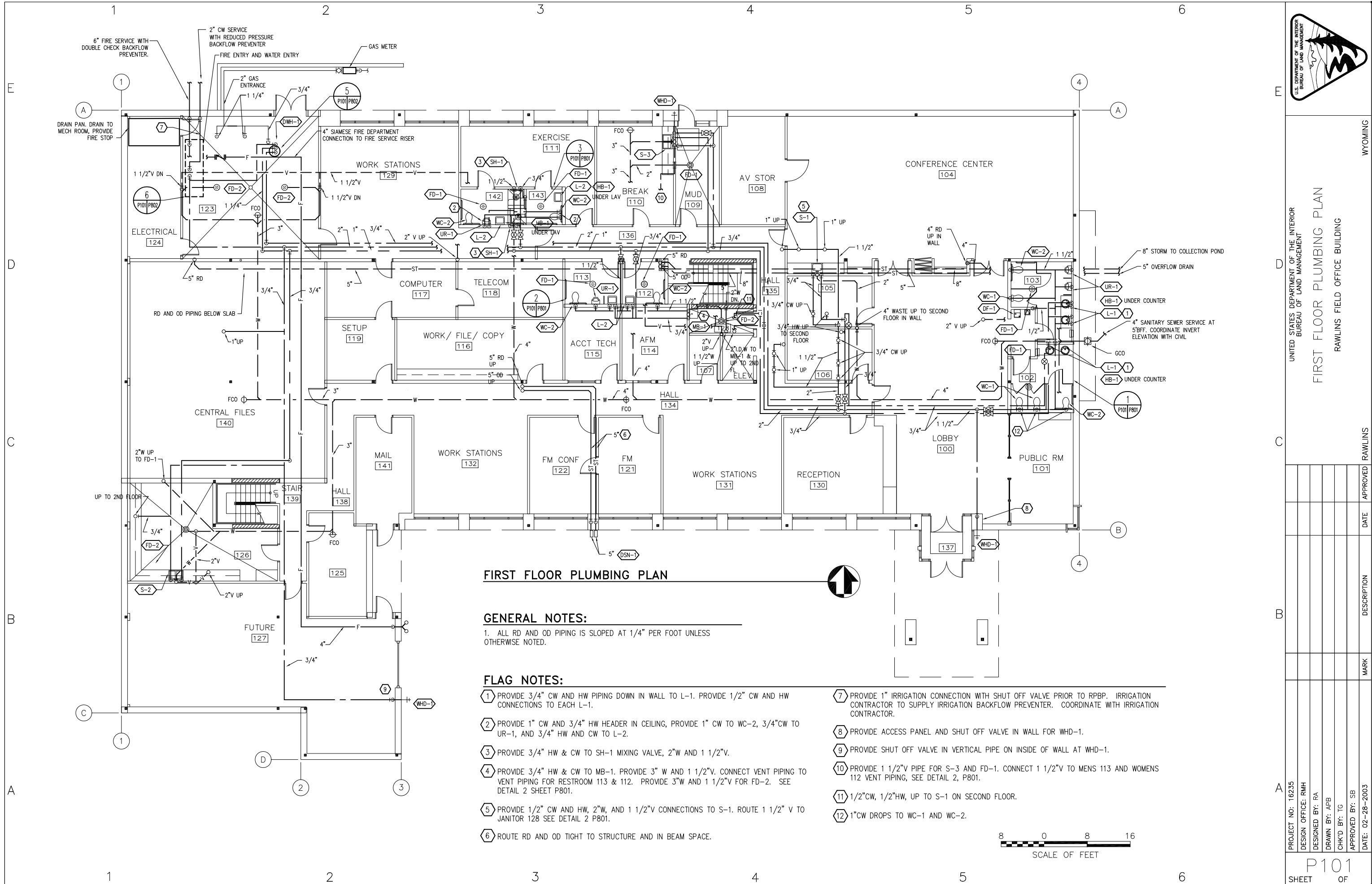
DESCRIPTION

MARK

PROJECT NO: 16235
DESIGN OFFICE: RMH
DESIGNED BY: KC
DRAWN BY: MW
CHK'D BY: KC
APPROVED BY: GIR
DATE: 02-28-2003

E807
SHEET OF

H:\J05956\16235\62559\01.dwg, 4/25/2003 11:32:15 AM, ctidd, INT7, 1,2, The RMH Group, Inc., Inc.



FIRST FLOOR PLUMBING PLAN

GENERAL NOTES:

1. ALL RD AND OD PIPING IS SLOPED AT 1/4" PER FOOT UNLESS OTHERWISE NOTED.

FLAG NOTES:

- 1 PROVIDE 3/4" CW AND HW PIPING DOWN IN WALL TO L-1. PROVIDE 1/2" CW AND HW CONNECTIONS TO EACH L-1.
- 2 PROVIDE 1" CW AND 3/4" HW HEADER IN CEILING, PROVIDE 1" CW TO WC-2, 3/4" CW TO UR-1, AND 3/4" HW AND CW TO L-2.
- 3 PROVIDE 3/4" HW & CW TO SH-1 MIXING VALVE, 2"W AND 1 1/2"V.
- 4 PROVIDE 3/4" HW & CW TO MB-1. PROVIDE 3" W AND 1 1/2"V. CONNECT VENT PIPING TO VENT PIPING FOR RESTROOM 113 & 112. PROVIDE 3"W AND 1 1/2"V FOR FD-2. SEE DETAIL 2 SHEET P801.
- 5 PROVIDE 1/2" CW AND HW, 2"W, AND 1 1/2"V CONNECTIONS TO S-1. ROUTE 1 1/2" V TO JANITOR 128 SEE DETAIL 2 P801.
- 6 ROUTE RD AND OD TIGHT TO STRUCTURE AND IN BEAM SPACE.
- 7 PROVIDE 1" IRRIGATION CONNECTION WITH SHUT OFF VALVE PRIOR TO RBPB. IRRIGATION CONTRACTOR TO SUPPLY IRRIGATION BACKFLOW PREVENTER. COORDINATE WITH IRRIGATION CONTRACTOR.
- 8 PROVIDE ACCESS PANEL AND SHUT OFF VALVE IN WALL FOR WHD-1.
- 9 PROVIDE SHUT OFF VALVE IN VERTICAL PIPE ON INSIDE OF WALL AT WHD-1.
- 10 PROVIDE 1 1/2"V PIPE FOR S-3 AND FD-1. CONNECT 1 1/2"V TO MENS 113 AND WOMENS 112 VENT PIPING, SEE DETAIL 2, P801.
- 11 1/2"CW, 1/2"HW, UP TO S-1 ON SECOND FLOOR.
- 12 1"CW DROPS TO WC-1 AND WC-2.

8 0 8 16
SCALE OF FEET



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
FIRST FLOOR PLUMBING PLAN
RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

APPROVED

DESCRIPTION

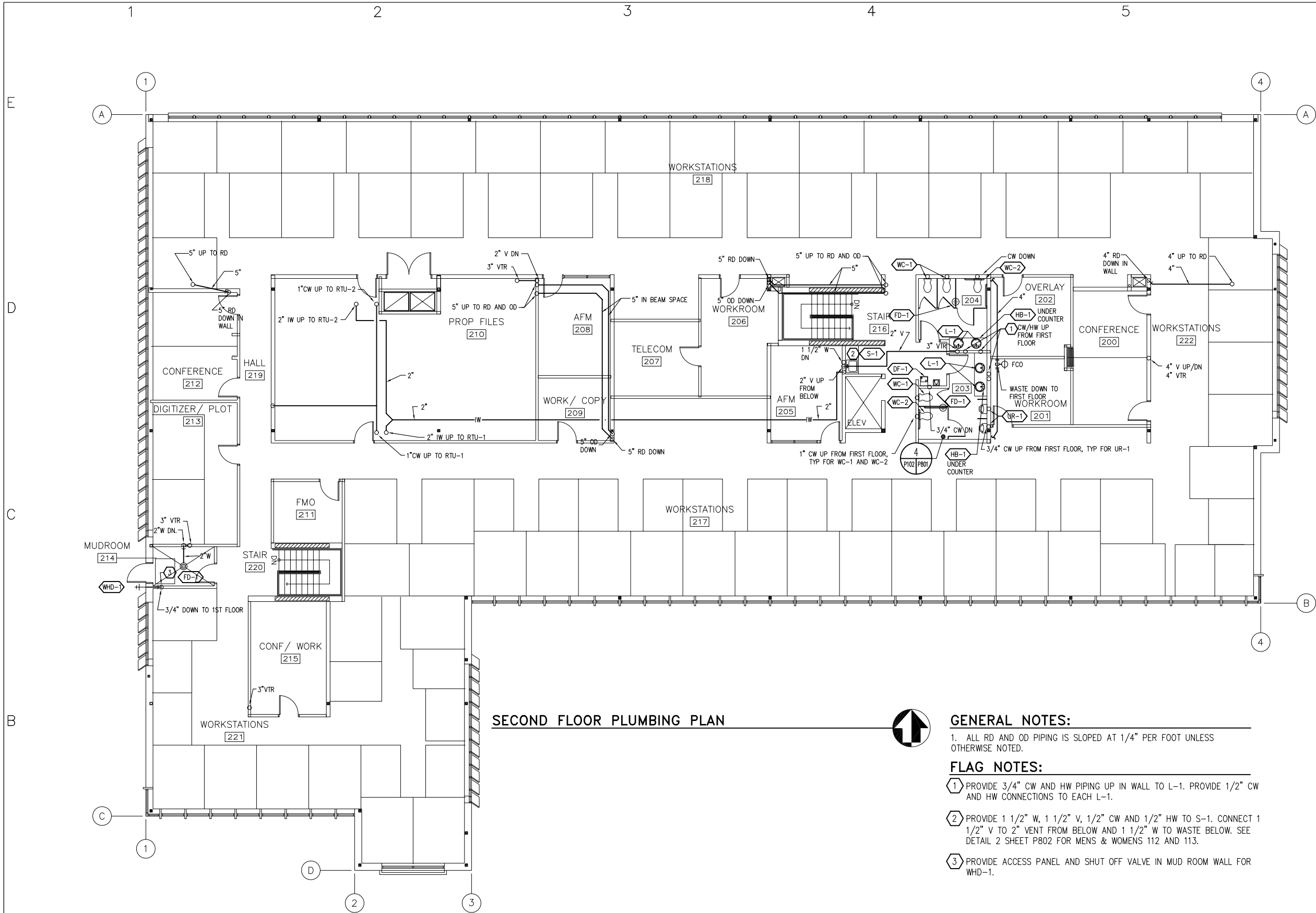
MARK

DATE

PROJECT NO: 16235
DESIGN OFFICE: RMH
DESIGNED BY: RA
DRAWN BY: APB
CHK'D BY: TC
APPROVED BY: SB
DATE: 02-28-2003

P101
SHEET OF

H:\J05956\16235\6255F102.dwg, 4/25/2003 11:33:10 AM, cldd, 11/7, 1:2, The RWH Group, Inc. Inc.



SECOND FLOOR PLUMBING PLAN

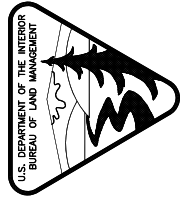


GENERAL NOTES:

1. ALL RD AND OD PIPING IS SLOPED AT 1/4" PER FOOT UNLESS OTHERWISE NOTED.

FLAG NOTES:

- 1 PROVIDE 3/4" CW AND HW PIPING UP IN WALL TO L-1. PROVIDE 1/2" CW AND HW CONNECTIONS TO EACH L-1.
- 2 PROVIDE 1 1/2" W, 1 1/2" V, 1/2" CW AND 1/2" HW TO S-1. CONNECT 1 1/2" V TO 2" VENT FROM BELOW AND 1 1/2" W TO WASTE BELOW. SEE DETAIL 2 SHEET P802 FOR MENS & WOMENS 112 AND 113.
- 3 PROVIDE ACCESS PANEL AND SHUT OFF VALVE IN MUD ROOM WALL FOR WHD-1.

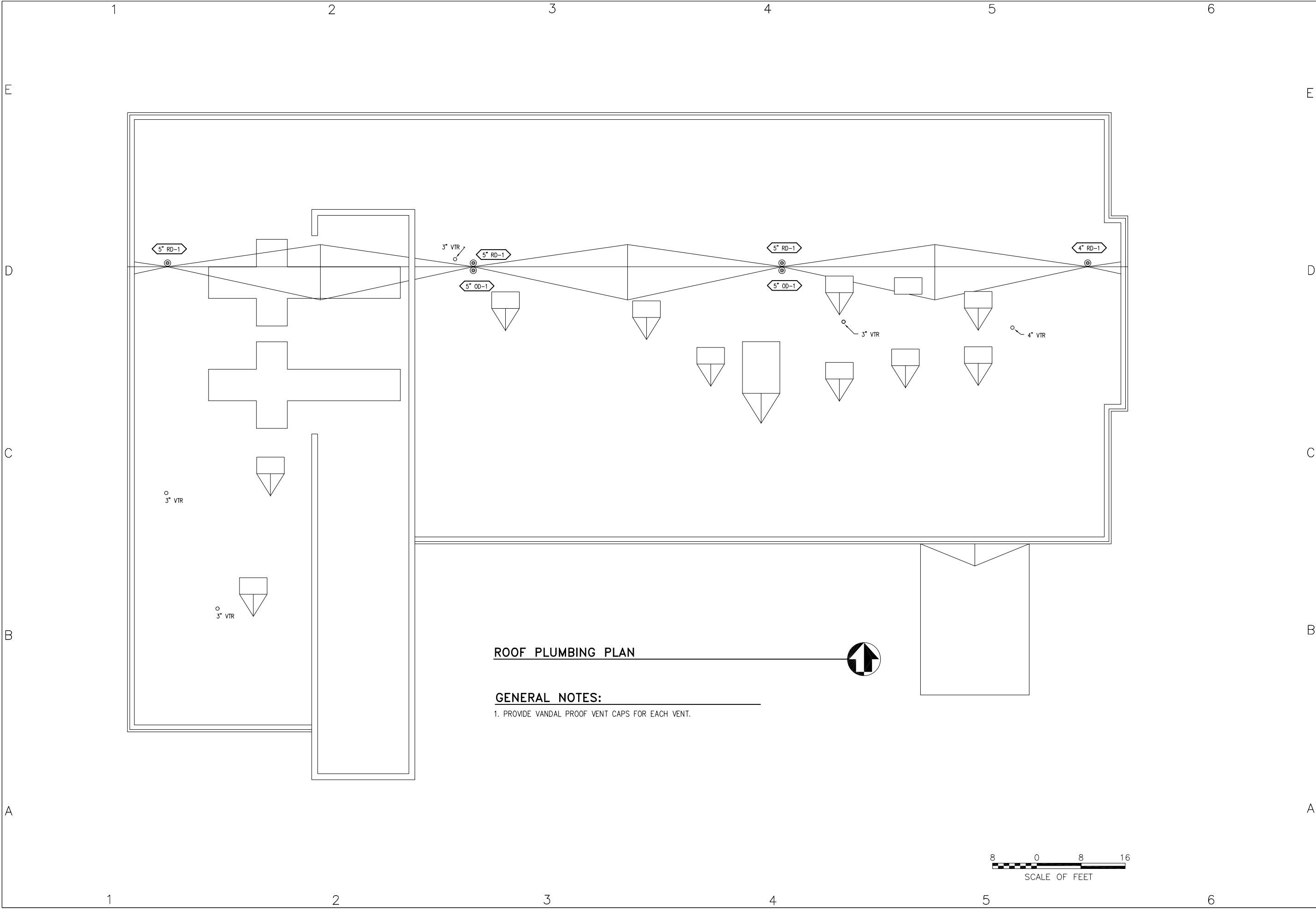


UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
SECOND FLOOR PLUMBING PLAN
RAWLINS FIELD OFFICE BUILDING

PROJECT NO:	DESIGN OFFICE:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	DESCRIPTION	DATE	APPROVED	RAWLINS
16235	RMH	RA	APB	TC	SB	02-28-2003				

P102
SHEET OF

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ROOF PLUMBING PLAN

GENERAL NOTES:

1. PROVIDE VANDAL PROOF VENT CAPS FOR EACH VENT.

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

WYOMING

UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

RAWLINS FIELD OFFICE BUILDING

ROOF PLAN

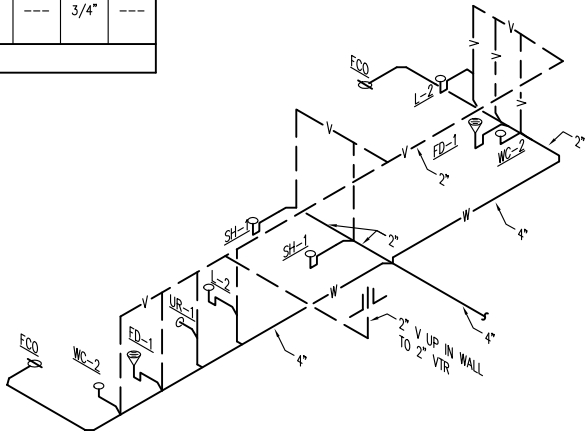
PROJECT NO: 16235							
DESIGN OFFICE: RMH							
DESIGNED BY: RA							
DRAWN BY: APB							
CHK'D BY: TC							
APPROVED BY: SB							
DATE: 02-28-2003							
	MARK	DESCRIPTION	DATE	APPROVED	RAWLINS		

P103

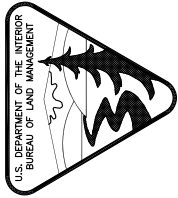
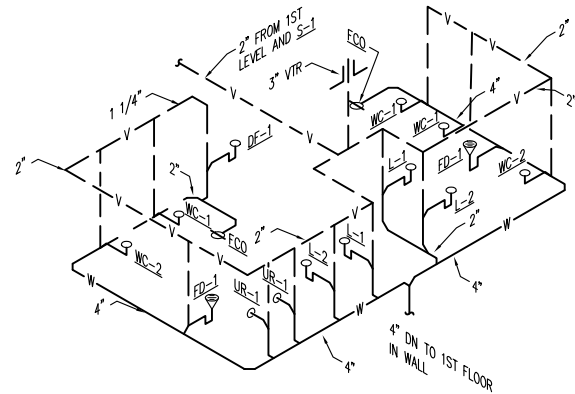
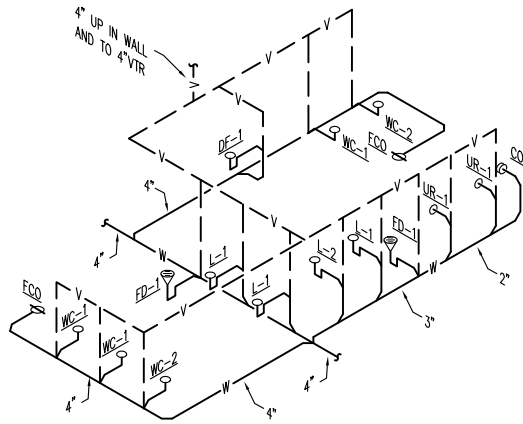
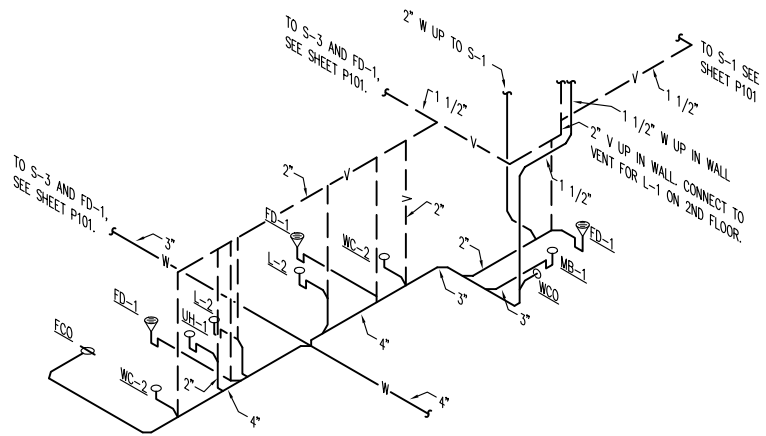
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PLUMBING FIXTURE SCHEDULE						
TAG	DESCRIPTION	CONNECTION SIZES				
		*WASTE	VENT	COLD	HOT	
DSN-1	DOWNSPOUT NOZZLE – JOSAM SERIES 25010 DOWNSPOUT NOZZLE.	AS NOTED				
DF-1	DRINKING FOUNTAIN – ELKAY MODEL EDPF-17-C, HIGH/LOW DESIGN, ADA COMPLIANT, STAINLESS STEEL SATIN FINISH, STAINLESS STEEL BACK PLATE, PERFORATED DOME STRAINERS WITH FLEXI-GUARD SAFETY BUBBLER, FRONT PUSHBARS, VANDAL RESISTANT BOTTOM COVER PLATES AND LK-464 DRAIN AND TRAP.	1 1/4"	1 1/4"	1/2"	---	
FD-1	FLOOR DRAIN (FINISHED AREAS) – JOSAM SUPER-FLO STRAINER SERIES 30000-A COATED CAST IRON FLOOR DRAIN, TWO PIECE BODY WITH DOUBLE DRAINAGE FLANGE, BOTTOM NO-HUB SPOUT OUTLET CONNECTION AND ADJUSTABLE SATIN BRONZE STRAINER AND VANDAL PROOF SCREWS.	2"	1 1/4"	---	---	
FD-2	FLOOR DRAIN (FINISHED AREAS) – JOSAM SUPER-FLO STRAINER SERIES 30000-A COATED CAST IRON FLOOR DRAIN, TWO PIECE BODY WITH DOUBLE DRAINAGE FLANGE, BOTTOM NO-HUB SPOUT OUTLET CONNECTION AND ADJUSTABLE SATIN BRONZE STRAINER AND VANDAL PROOF SCREWS.	3"	1 1/2"	---	---	
HB-1	HOSE BIBB – CHICAGO FAUCET 952 WITH VACUUM BREAKER & LOOSE TEE HANDLE, POLISHED CHROME PLATE.	---	---	1/2"	---	
L-1	LAVATORY – (COUNTERTOP, HANDICAPPED), KOHLER PENNINGTON, K-2196-4, WHITE VITREOUS CHINA, SELF RIMMING 20-1/4"x17-1/2", 4" CENTERS, 1-1/4" TAILPIECE, CHROME PLATED CAST BRASS ADJUSTABLE 1-1/4" P-TRAP WITH C.O. AND GRID STRAINER, 3/8" FLEXIBLE SUPPLIES WITH ANGLE STOPS, LOOSE KEY, SLOAN OPTIMA ETF-600 INFRARED FAUCET, ADA COMPLIANT, 5GPM FLOW, 4" CENTERS, 4 1/2" SPOUT, BOM MECHANICAL MIXING VALVE, SOLENOID WITH Y STRAINER, VANDAL RESISTANT HEAD, 24VAC CONTROL, ADJUSTABLE TIME OUT ON FAUCET RUN TIME, PROVIDE BOX MOUNT 120VAC TRANSFORMER FOR EACH MENS OR WOMENS RESTROOM (ONE PER RESTROOM). CONNECT TRANSFORMER TO FAUCETS. COORDINATE WITH ELECTRICAL CONTRACTOR.	1 1/4"	1 1/4"	1/2"	1/2"	
L-2	LAVATORY – (WALL MOUNTED, HANDICAPPED), KOHLER GREENWICH K-2027, WHITE VITREOUS CHINA, 18 5/8"x16", 4" CENTERS, SOAP DISPENSER, 1-1/4" TAILPIECE, CHROME PLATED CAST BRASS ADJUSTABLE 1-1/4" P-TRAP WITH C.O. K-13885 OFFSET DRAIN, AND GRID STRAINER, 3/8" FLEXIBLE SUPPLIES WITH ANGLE STOPS, LOOSE KEY, PROVIDE FLOOR MOUNTED CARRIER. SLOAN OPTIMA ETF-600 INFRARED FAUCET, ADA COMPLIANT, 5GPM FLOW, 4" CENTERS, 4 1/2" SPOUT, BOM MECHANICAL MIXING WITH LOOSE VALVE, SOLENOID WITH Y STRAINER, VANDAL RESISTANT HEAD, 24VAC CONTROL, ADJUSTABLE TIME OUT ON FAUCET RUN TIME, PROVIDE BOX MOUNT 120VAC TRANSFORMER FOR EACH MENS OR WOMENS RESTROOM (ONE PER RESTROOM). CONNECT TRANSFORMER TO FAUCETS. COORDINATE WITH ELECTRICAL CONTRACTOR. PROVIDE TRUEBORG INSULATION KIT.	1 1/4"	1 1/4"	1/2"	1/2"	
MB-1	MOP BASIN – FIAT, MOLDED STONE MSB-2424, 24"x24"x10" COMBINATION DOME STRAINER AND LINT BASKET, FAUCET CHICAGO FAUCET 540-LD-897S-WXF CHROME PLATED, ADA COMPLIANT, WITH INTEGRAL STOPS, VACUUM BREAKER, WALL BRACE, MOP HANGER, PAIL HOOK AND 3/4" HOSE TREAD ON SPOUT. VINYL RIM GUARDS ALL EXPOSED SIDES. PROVIDE CHECK VALVES AND 1.6 GPM RESTRICTORS IN HOT AND COLD WATER LINES. PROVIDE STAINLESS STEEL WALL GUARDS ON TWO EXPOSED WALLS.	3"	1 1/2"	1/2"	1/2"	
RD-1	ROOF DRAIN – JOSAM SERIES 21000 WITH COATED CAST IRON BODY, NO-HUB BOTTOM OUTLET, ADJUSTABLE TOP, LOCKING VANDAL PROOF POLYPROPYLENE STRAINER, DECK-CLAMP, CLAMP RING, GRAVEL GUARD, AND DRAIN RECEIVER.	AS NOTED	---	---	---	
OD-1	OVERFLOW DRAIN – JOSAM SERIES 26000, INTEGRAL WATERGUARD, 2" INTERNAL PLASTIC STANDPIPE, WITH COATED CAST IRON BODY, NO-HUB BOTTOM OUTLET, ADJUSTABLE TOP, LOCKING VANDAL PROOF POLYPROPYLENE STRAINER, DECK-CLAMP, CLAMP RING, GRAVEL GUARD, AND DRAIN RECEIVER.	AS NOTED	---	---	---	
SA-1	SHOCK ABSORBER – JOSAM SERIES 75000. PROVIDE ACCESS DOOR AND ISOLATION VALVE. SIZE PER PDI STANDARDS.	---	---	---	---	
SH-1	INDIVIDUAL SHOWERS – LAWLER CONCEALED THERMOSTATIC AND PRESSURE BALANCED VALVE MODEL 7500-SC5 WITH SATIN CHROME TRIM, ADA COMPLIANT LEVER HANDLE, BUILT IN VOLUME CONTROL, 1.5 GPM STANDARD FLOW, AND VANDAL RESISTANT, ADJUSTABLE TEMPERATURE LIMIT TO 115F, STRAINER IN ANGLE STOPS, AND CHECK VALVES. PROVIDE SH-2 SHOWER HEAD. SHOWER BASE BY ARCHITECT.	2"	1 1/4"	1/2"	1/2"	
S-1	SINK – ELKAY DLR191910, SELF RIMMING, SINGLE COMPARTMENT, 18 GAUGE, TYPE 304 STAINLESS STEEL SINK, BOWL SIZE 19 1/2" X19"x10" DEEP, 3 HOLES. FAUCET – CHICAGO FAUCETS MODEL 2300-8 SINGLE LEVER FAUCET, ADA COMPLIANT, 4" CENTERS, 10" CHROME PLATED SPOUT, INTEGRAL FLOW CONTROL, ADJUSTABLE LIMIT STOP, E-28 VANDAL PROOF 3/4 GPM AERATOR, LKVR-18 DRAIN FITTING, 3" VANDAL PROOF GRID STRAINER. CHROME PLATED BRASS 1 1/2" TAILPIECE, 17 GAUGE CHROME PLATED BRASS TRAP W/CO AND CONTINUOUS WASTE, ANGLE SUPPLIES WITH LOOSE KEY, AND FLEXIBLE SUPPLIES.	1 1/2"	1 1/4"	1/2"	1/2"	
S-2	SINK – ELKAY DLR432210, SELF RIMMING, DOUBLE COMPARTMENT SINK, 18 GAUGE TYPE 304 STAINLESS STEEL SINK, 43"x22"x10"DEEP, 3 HOLES, 4" CENTERS. FAUCET: CHICAGO FAUCETS MODEL 895-317-L9 ADA COMPLIANT, TYPE 317 BLADE HANDLES, 4" CENTERS, L9 9 1/2" CHROME PLATED SPOUT, INTEGRAL FLOW CONTROL, ADJUSTABLE LIMIT STOP, E3 2 GPM AERATOR. LKVR-18 3" GRID STRAINER. CHROME PLATED BRASS 1 1/2" TAILPIECE, 17 GAUGE CHROME PLATED BRASS TRAP W/CO AND CONTINUOUS WASTE, ANGLE SUPPLIES WITH LOOSE KEY, AND FLEXIBLE SUPPLIES.	1 1/2"	1 1/4"	1/2"	1/2"	
S-3	SINK – ELKAY LR3319, SELF RIMMING, DOUBLE COMPARTMENT SINK, 18 GAUGE TYPE 304 STAINLESS STEEL SINK, 33"x19"x7 1/2" DEEP, 3 HOLES, 4" CENTERS. FAUCET: CHICAGO FAUCETS MODEL 2300-8 SINGLE LEVER FAUCET, ADA COMPLIANT, 4" CENTERS, 10" CHROME PLATED SPOUT, INTEGRAL FLOW CONTROL, ADJUSTABLE LIMIT STOP, E28 VANDAL PROOF 3/4 GPM AERATOR. LKVR-18 DRAIN FITTING, 3" VANDAL PROOF GRID STRAINER. CHROME PLATED BRASS 1 1/2" TAILPIECE, 17 GAUGE CHROME PLATED BRASS TRAP W/CO AND CONTINUOUS WASTE, ANGLE SUPPLIES WITH LOOSE KEY, AND FLEXIBLE SUPPLIES. PROVIDE GARBAGE DISPOSAL – INSINKERATOR 333-SS, 1/2HP, 120V, 4YR WARRANTY, STAINLESS STEEL GRINDING ELEMENTS, AND FULL SOUND ABSORBING.	1 1/2"	1 1/4"	1/2"	1/2"	
UR-1	URINAL (WALL MOUNTED) – KOHLER BARDON MODEL K-4960-ET, WASHOUT, WHITE VITREOUS CHINA, .5 GPF, 3/4" TOP SPOUT, 2" BACK OUTLET. FLUSH VALVE – SLOAN OPTIMA MODEL 186-.5-ES-5 INFRARED 24VAC ELECTRIC EXPOSED FLUSH VALVE, TOP SPOUT, CHROME PLATED ELECTRICAL BOX COVER PLATE WITH SENSOR AND VANDAL RESISTANT SCREWS, OVERRIDE BUTTON, .5GPF, OVERRIDE BUTTON, 3/4" ANGLE STOP, 3YR WARRANTY, PROVIDE ONE TRANSFORMER FOR EACH MENS RESTROOM, COORDINATE WITH ELECTRICAL CONTRACTOR. PROVIDE FIXTURE CARRIER WITH TUBULAR UPRIGHTS BOLTED TO FLOOR SEE ARCHITECTURAL DRAWINGS FOR MOUNTING HEIGHT.	2"	1 1/4"	3/4"	---	
WC-1	WATER CLOSET (HANDICAPPED, FLOOR MOUNTED, FLUSH VALVE) – KOHLER HIGHCREST MODEL K-4302, WHITE VITREOUS CHINA, SIPHON JET ELONGATED RIM BOWL WITH 1 1/2" TOP SPOUT. OPEN FRONT SEAT LESS COVER, 1.6GPF. FLUSH VALVE SLOAN ROYAL 111-1.6-YO, HANDICAPPED, FLUSH VALVE HANDLE. MOUNTING HEIGHT 16 1/2" TO RIM.	4"	2"	1"	---	
WC-2	WATER CLOSET (HANDICAPPED, FLOOR MOUNTED, FLUSH VALVE) – KOHLER HIGHCREST MODEL K-4302, WHITE VITREOUS CHINA, SIPHON JET ELONGATED RIM BOWL WITH 1 1/2" TOP SPOUT. OPEN FRONT SEAT LESS COVER, 1.6GPF. FLUSH VALVE SLOAN ROYAL 111-1.6-YO, HANDICAPPED, FLUSH VALVE HANDLE. MOUNTING HEIGHT 16 1/2" TO RIM.	4"	2"	1"	-	
WHD-1	WALL HYDRANT – JOSAM SERIES 71000, NON-FREEZE, VACUUM BREAKER, SATIN NICKEL BRONZE FINISH, WITH VANDAL PROOF LOCK, WALL CLAMP, UNION ELBOW ASSEMBLY, BRONZE OPERATING PARTS, BRONZE CASING.	---	---	3/4"	---	

*WASTE INDICATES TRAP AND TRAP ARM SIZING UNLESS OTHERWISE SHOWN ON DRAWINGS.



PLUMBING EQUIPMENT SCHEDULE	
TAG	PURPOSE
DHW-1	DOMESTIC WATER HEATER – LOCHINVAR DIRECT VENT DVN052, 48 GAL. DIRECT VENT, 65,000 BTUH INPUT AT S.L., 69 GAL. RECOVERY AT 90°F TEMPERATURE RISE AT SEA LEVEL, 3"ø SIDE WALL VENT, 3/4" CW AND HW CONNECTIONS. 120/1/60. 185 LBS.. 3YR WARRANTY.
DWHP-1	DOMESTIC HOT WATER CIRCULATING PUMP – GRUNDFOS MODEL UP-26-96F, 4GPM, 20FT HD, 1/12HP, 115/1/60.
ET-2	DOMESTIC WATER EXPANSION TANK – B&G PTA-5, VERTICAL DIAPHRAGM, 2 GAL TANK WITH .9 GAL ACCEPTANCE, 3/4" CONNECTION, 10 1/4"ø X9 1/4, 21 LBS.



RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

APPF

DATE _____

DESCRIPTION

MARK

PROJECT NO: 16235

DESIGN OFFICE: RMH

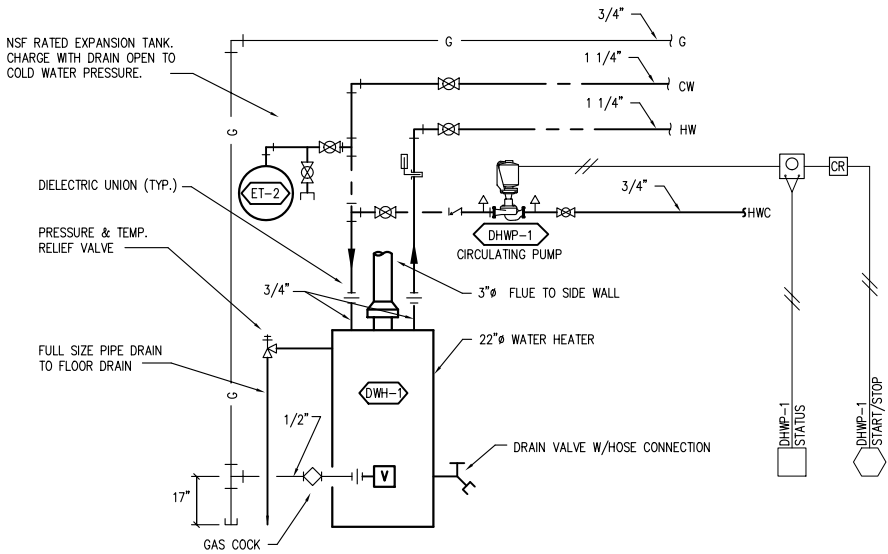
DESIGNED BY: RA

DRAWN BY: APB

CHK'D BY: TG

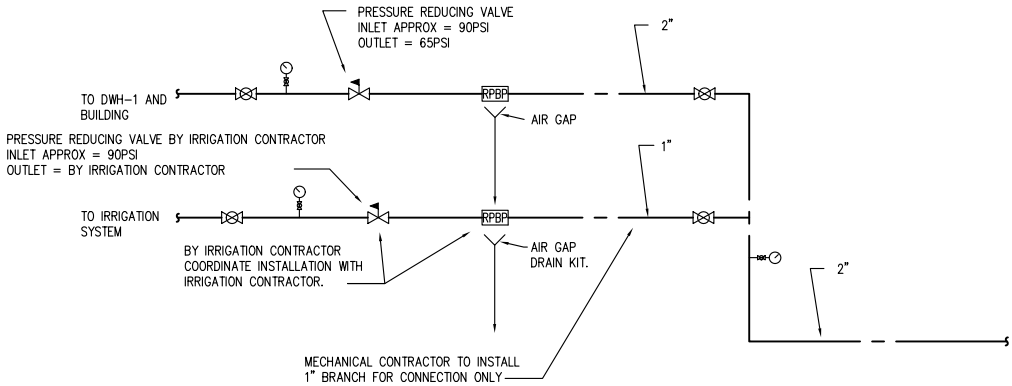
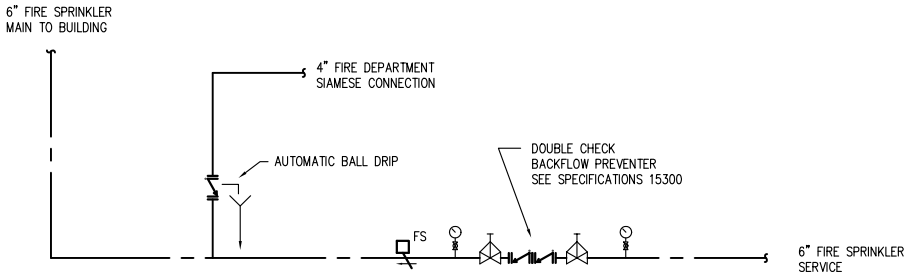
P801
SHEET OF

H:\JOB916\16235\16235-P802.dwg, 4/25/2003 10:23:28 AM, ctdd, 11/17, 1.2, The RWH Group, Inc., Inc.



GAS-FIRED DOMESTIC WATER HEATER DETAIL
SCALE: NTS

5
P101 P802



DOMESTIC WATER AND FIRE ENTRY DETAIL
SCALE: NTS

6
P101 P802



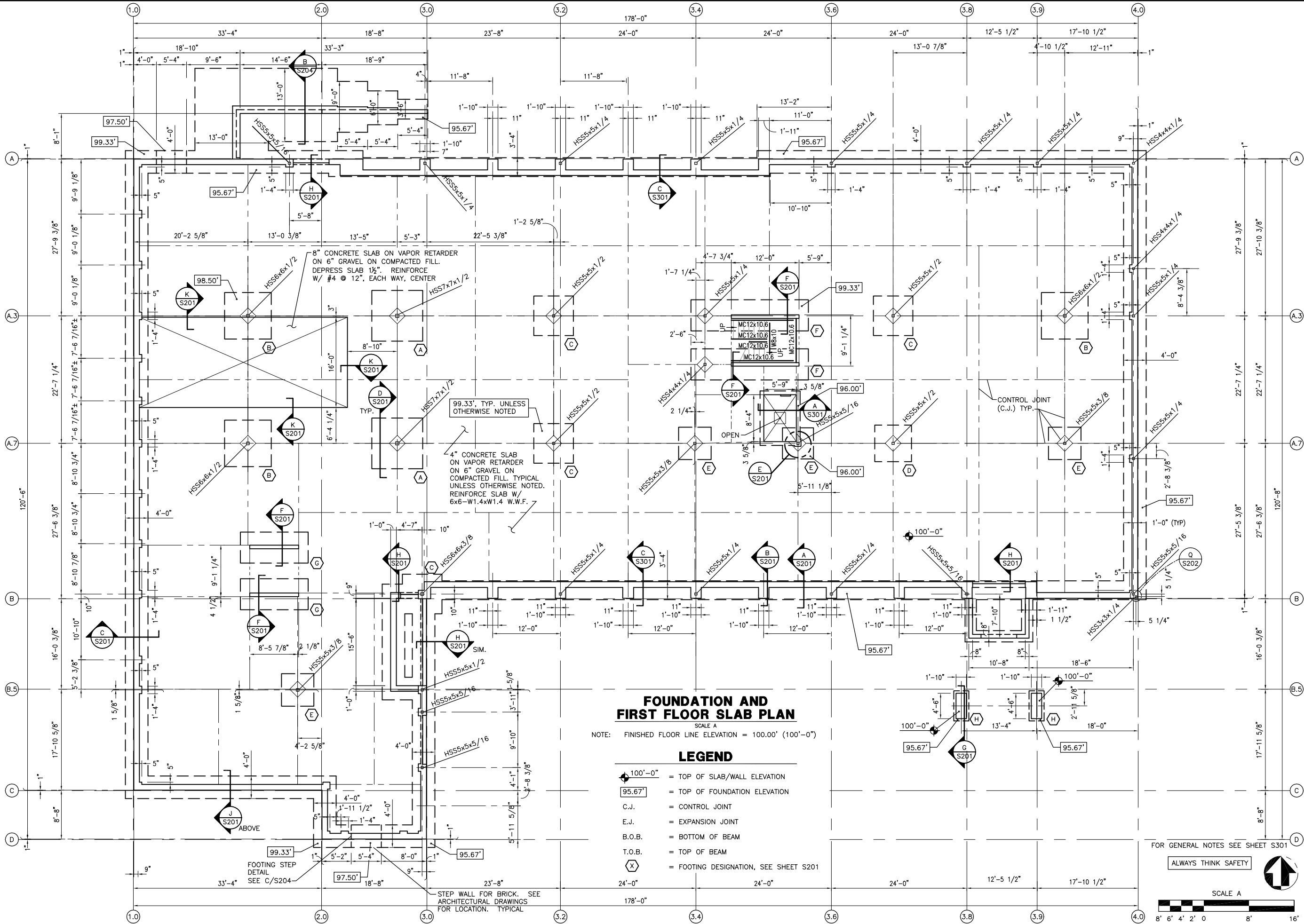
UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT
PLUMBING DETAILS
RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

PROJECT NO:	DESIGN OFFICE:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	DESCRIPTION	MARK
16235	RMH	RA	APB	TG	SB	02-28-2003		

P802
SHEET OF



**FOUNDATION AND
FIRST FLOOR SLAB PLAN**

SCALE A

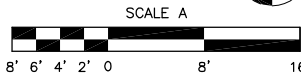
NOTE: FINISHED FLOOR LINE ELEVATION = 100.00' (100'-0")

LEGEND

- 100'-0" = TOP OF SLAB/WALL ELEVATION
- 95.67' = TOP OF FOUNDATION ELEVATION
- C.J. = CONTROL JOINT
- E.J. = EXPANSION JOINT
- B.O.B. = BOTTOM OF BEAM
- T.O.B. = TOP OF BEAM
- X = FOOTING DESIGNATION, SEE SHEET S201

FOR GENERAL NOTES SEE SHEET S301

ALWAYS THINK SAFETY



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

FOUNDATION AND FIRST FLOOR SLAB PLAN

RAWLINS FIELD OFFICE BUILDING

WYOMING

PROJECT NO: P02124
DESIGN OFFICE: LONGO, INC.

DESIGNED BY: JAS

DRAWN BY: EAB

CHK'D BY: JAS

APPROVED BY:

DATE: 2-28-03

S101
SHEET OF

RAWLINS

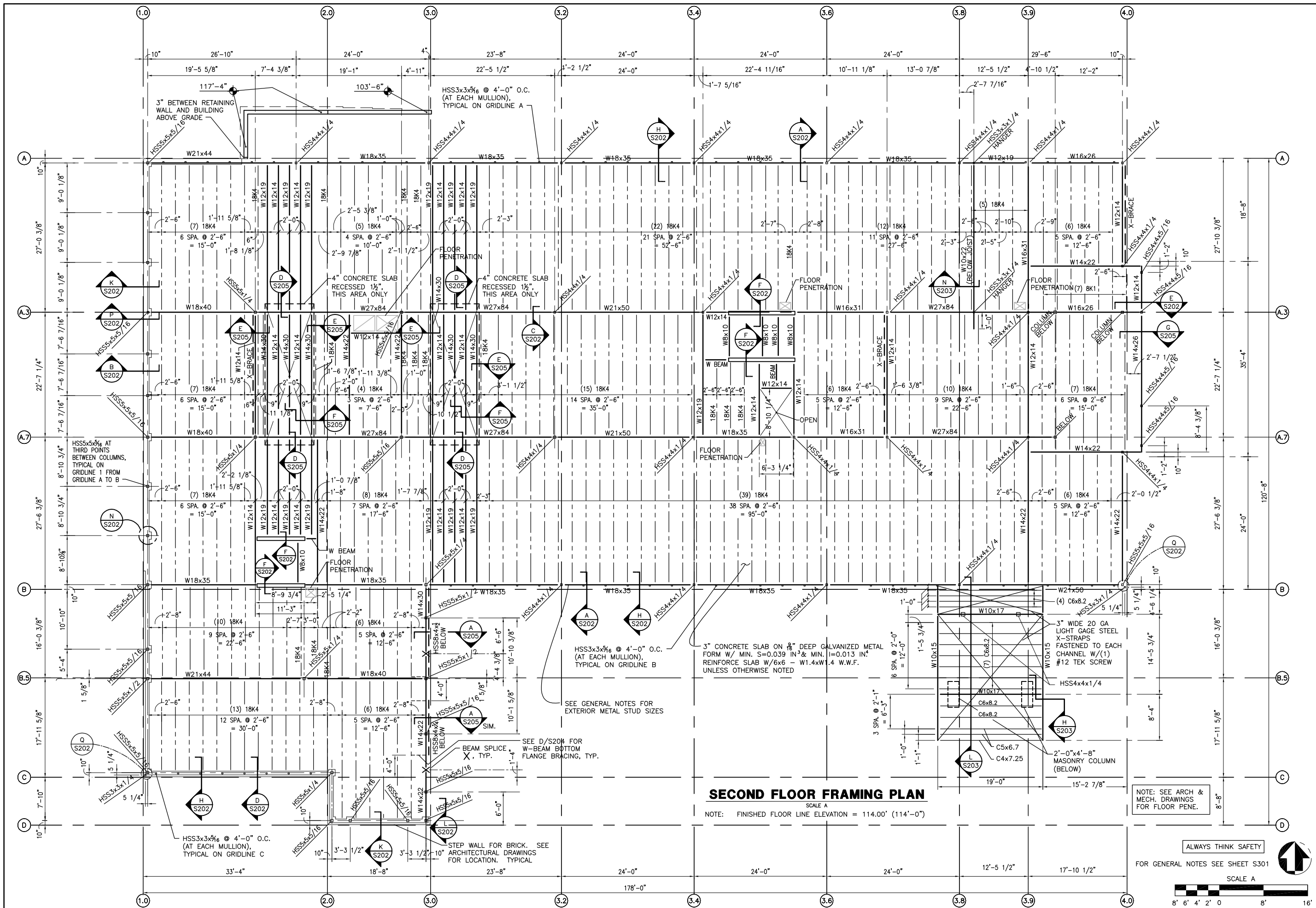
APPROVED

DATE

DESCRIPTION

MARK

Drawing Name: F:\2002\PO2124\Drawings\S102 2nd Floor Plan.DWG - Current tab: S102



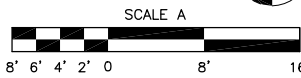
SECOND FLOOR FRAMING PLAN

SCALE A
NOTE: FINISHED FLOOR LINE ELEVATION = 114.00' (114'-0")

NOTE: SEE ARCH & MECH. DRAWINGS FOR FLOOR PENE.

ALWAYS THINK SAFETY

FOR GENERAL NOTES SEE SHEET S301



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

SECOND FLOOR FRAMING PLAN

RAWLINS FIELD OFFICE BUILDING

RAWLINS

WYOMING

PROJECT NO: P02124

DESIGN OFFICE: LONGO, INC.

DESIGNED BY: JAS

DRAWN BY: BFB

CHK'D BY: JAS

APPROVED BY:

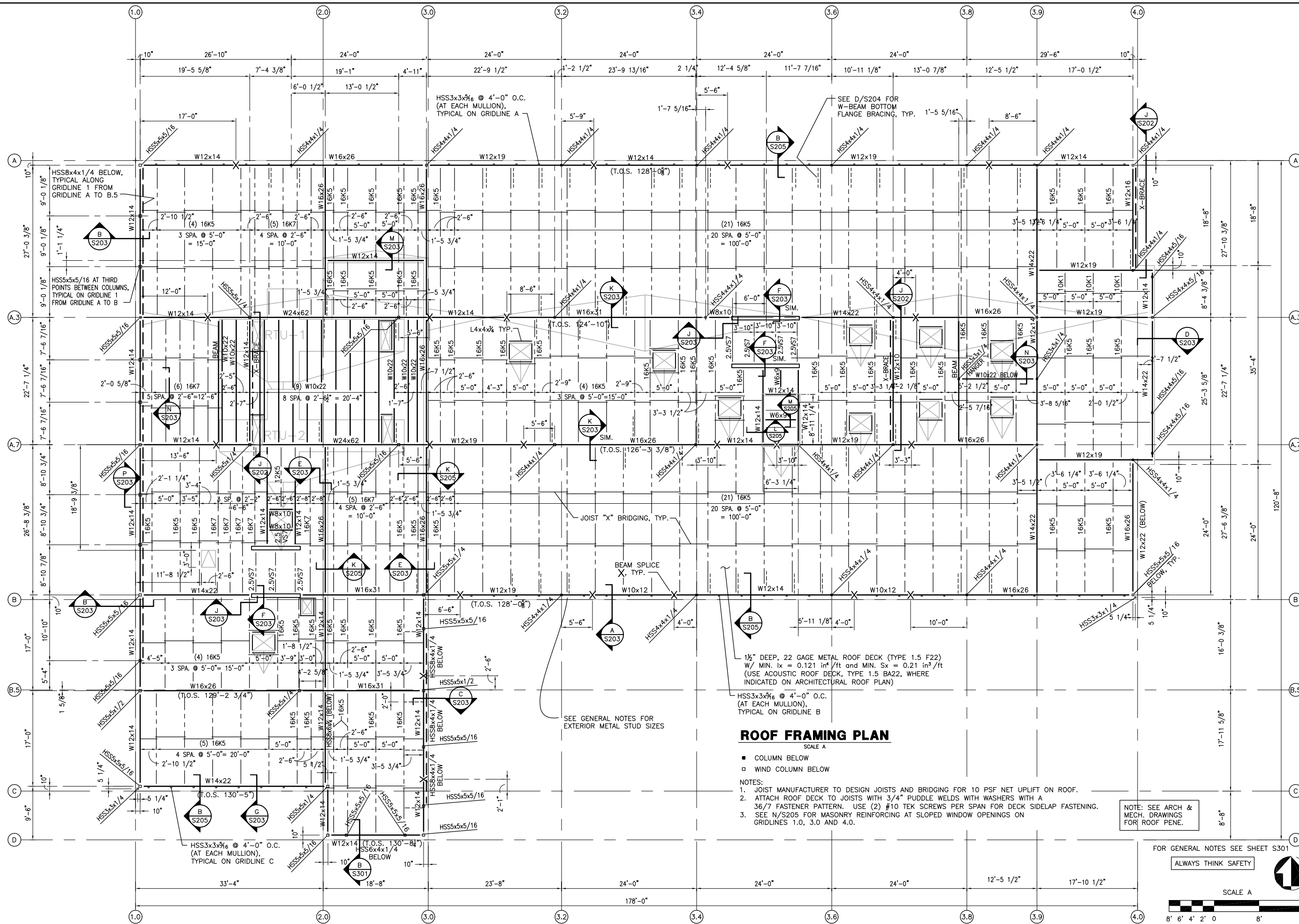
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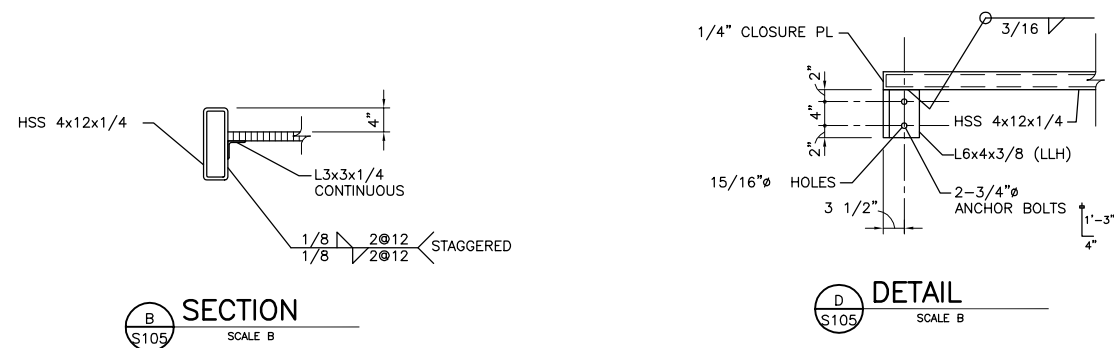
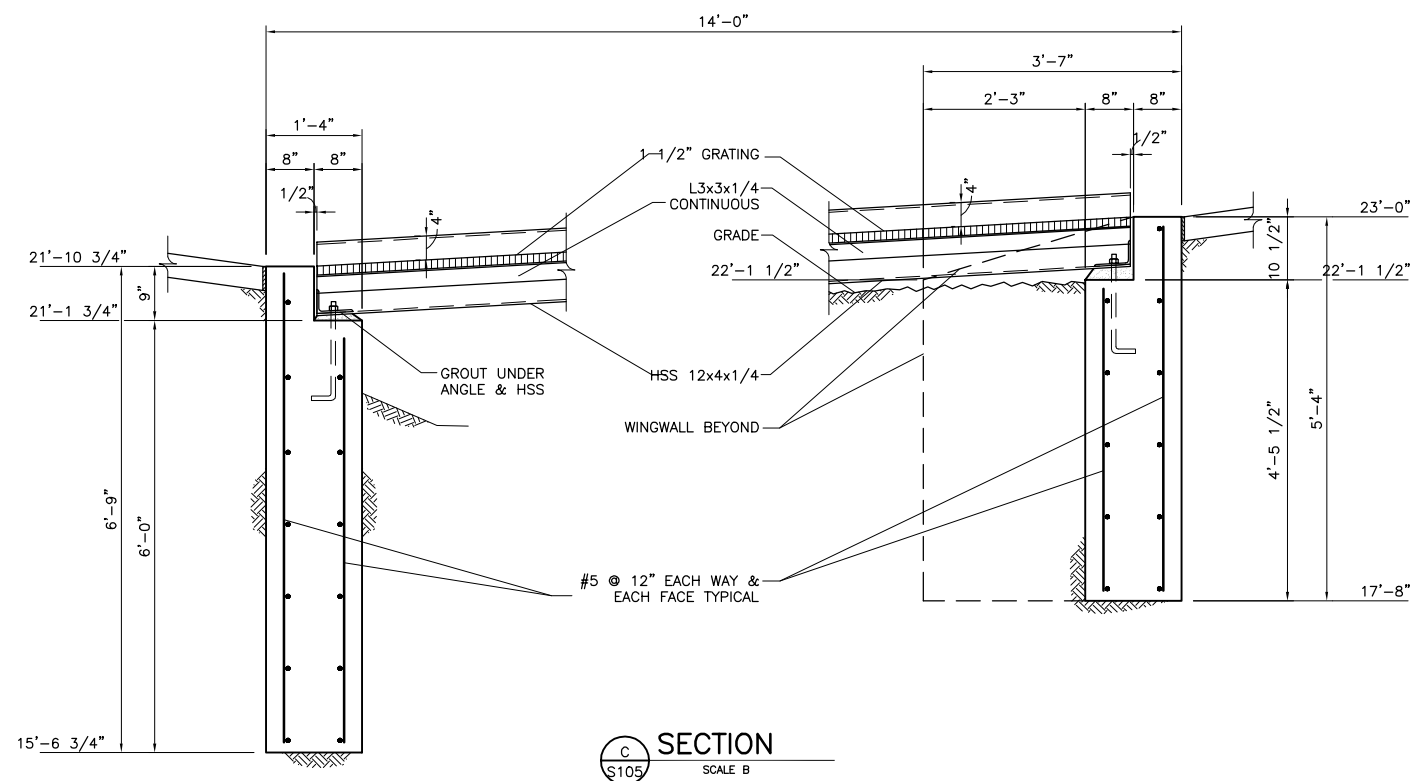
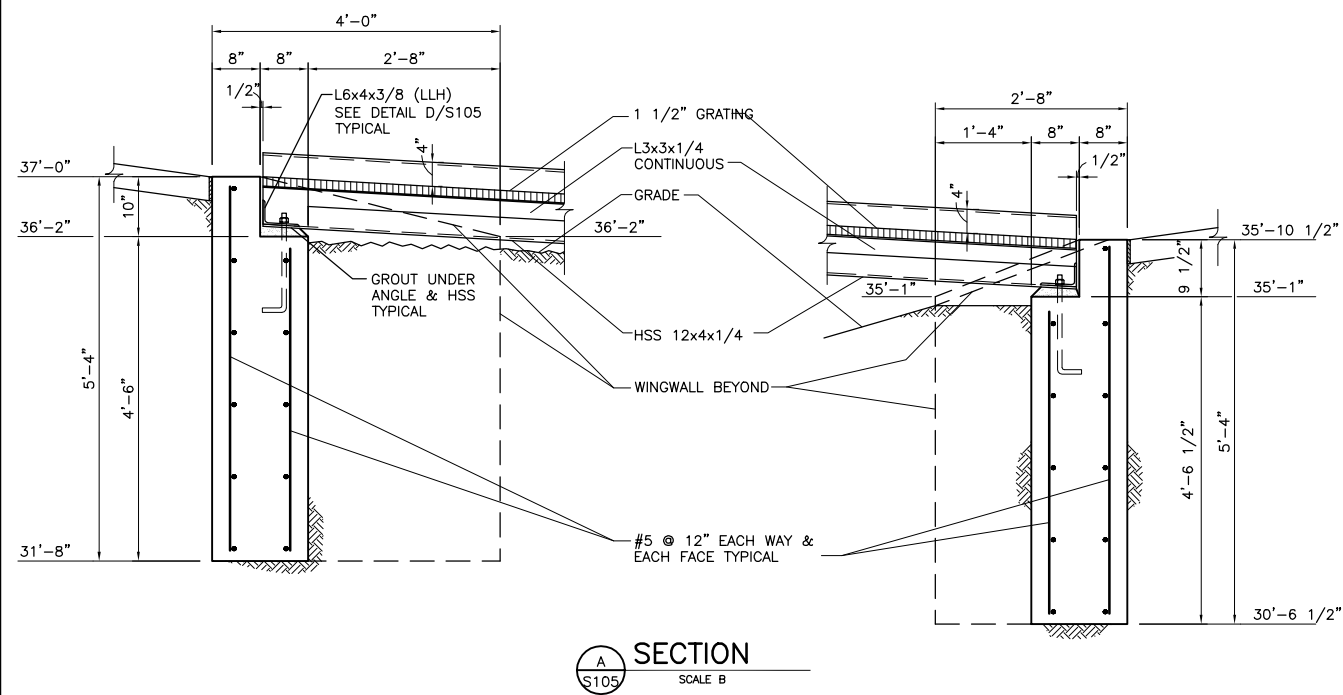
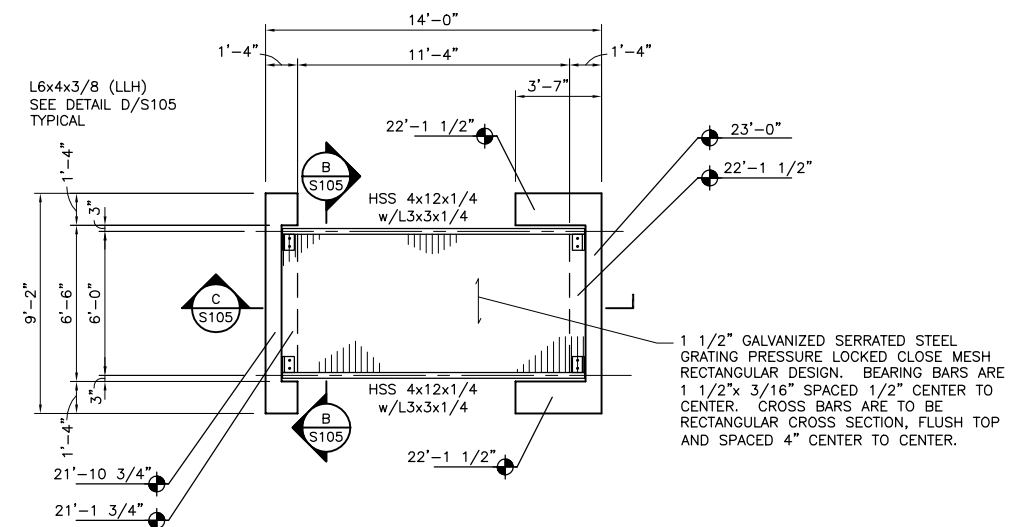
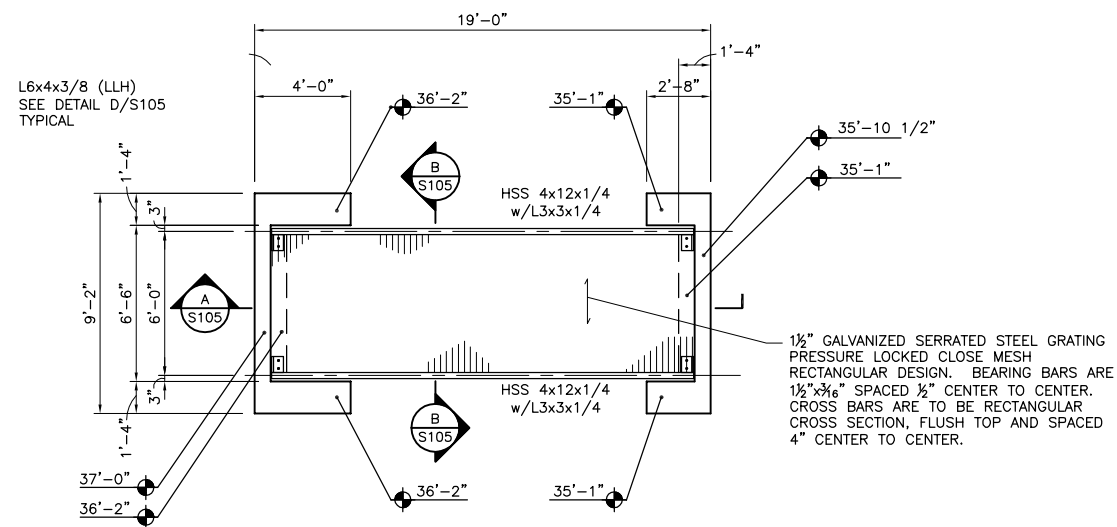
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OF

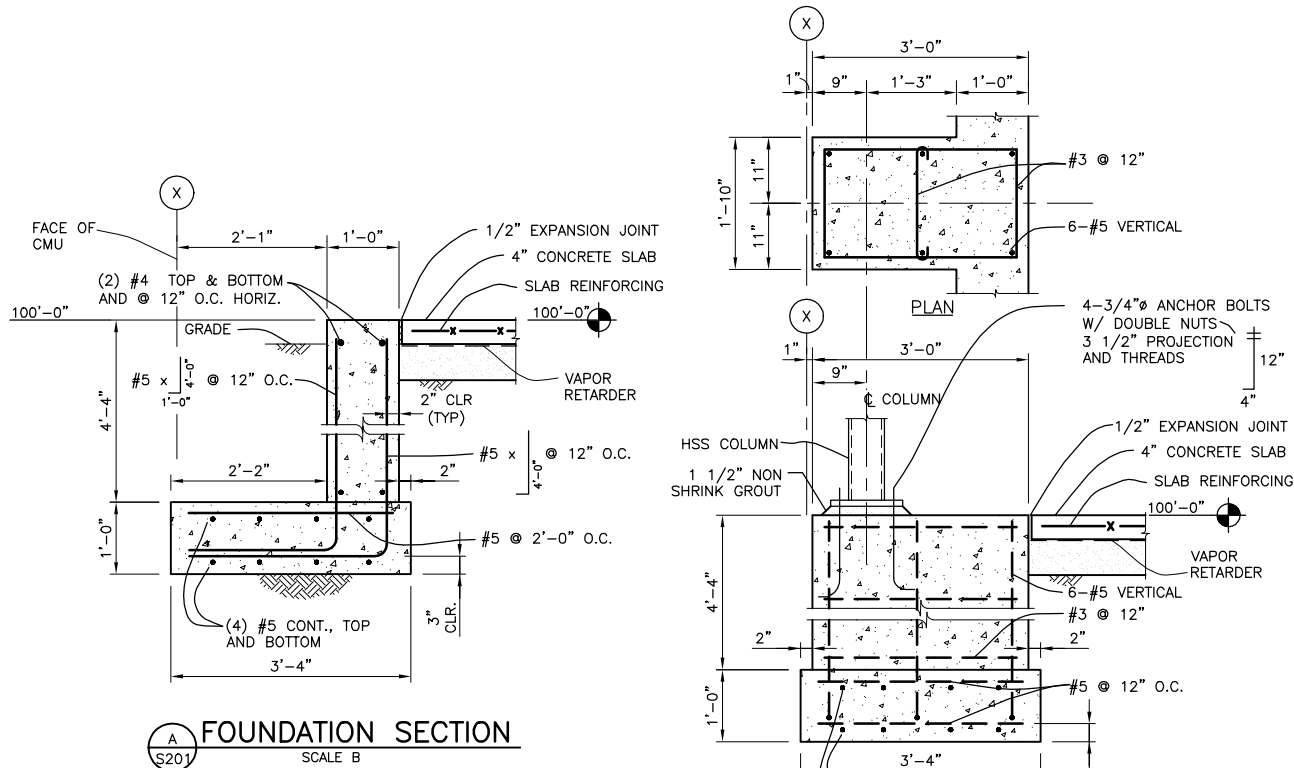
S102

PROJECT NO: P02124					
DESIGN OFFICE: LONCO, INC.					
DESIGNED BY: JAS					
DRAWN BY: EAB					
CHK'D BY: JAS					
APPROVED BY:					
DATE: 2-28-03	MARK	DESCRIPTION	DATE	APPROVED	RAWI INS

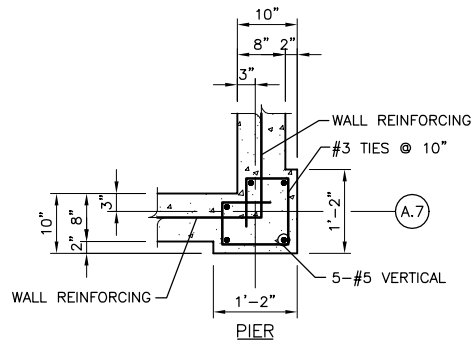
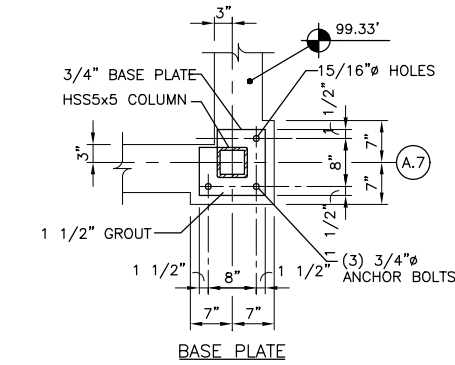




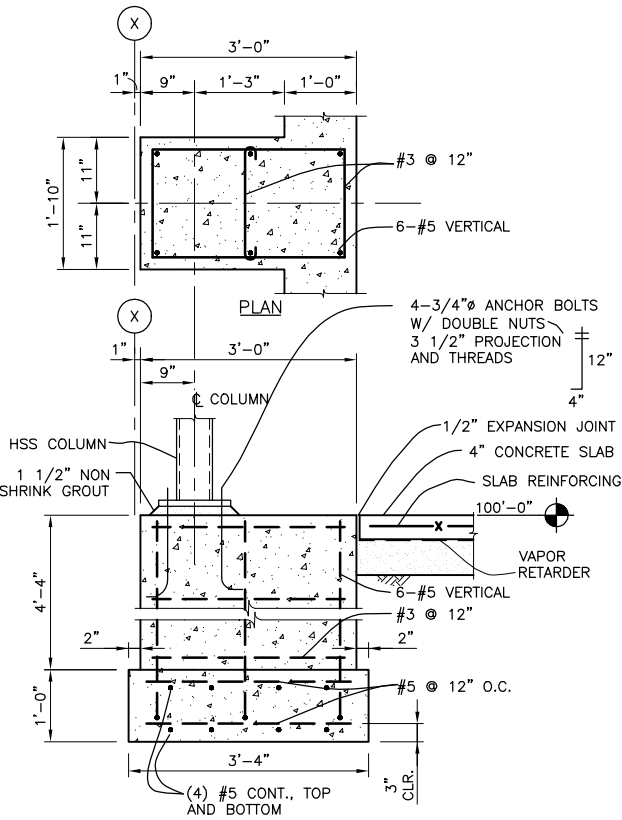
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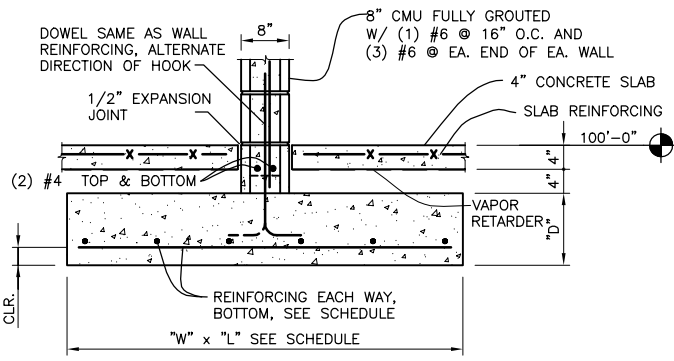
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S201 SCALE B



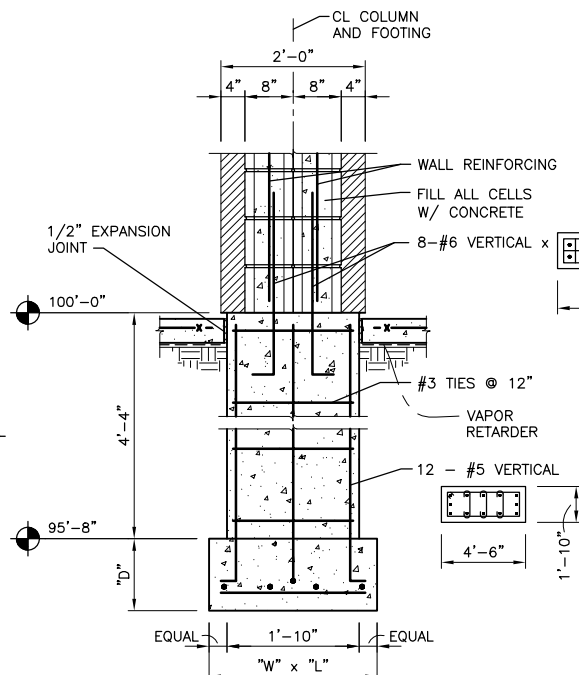
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S201 SCALE B



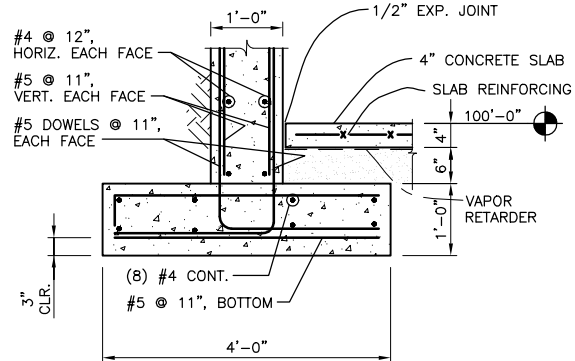
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S201 SCALE B



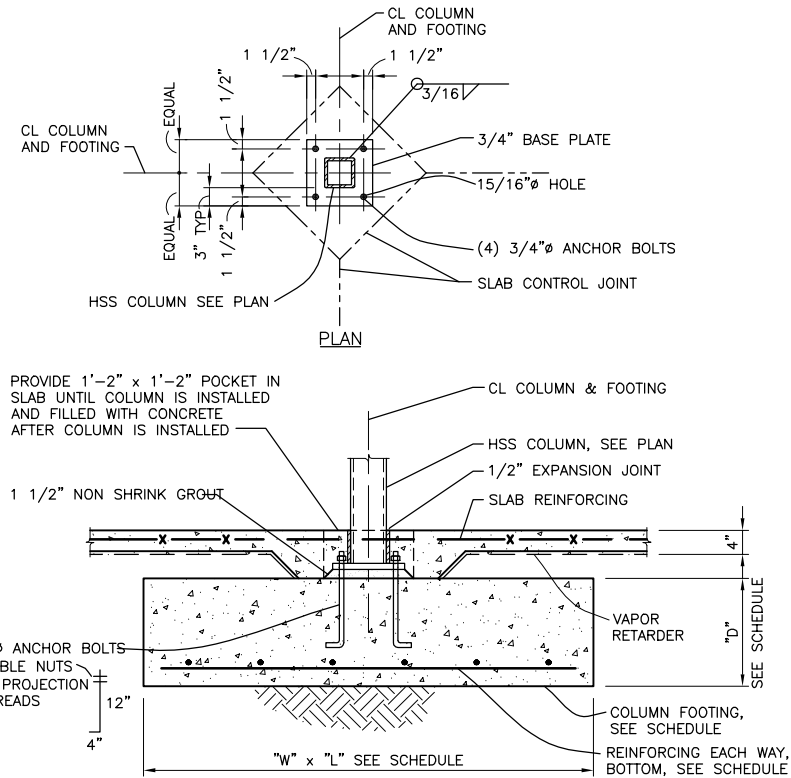
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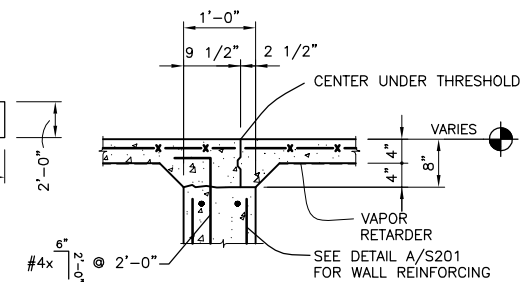
G FOUNDATION SECTION
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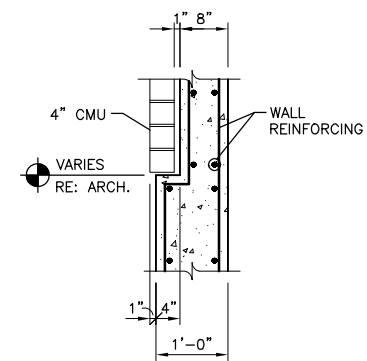
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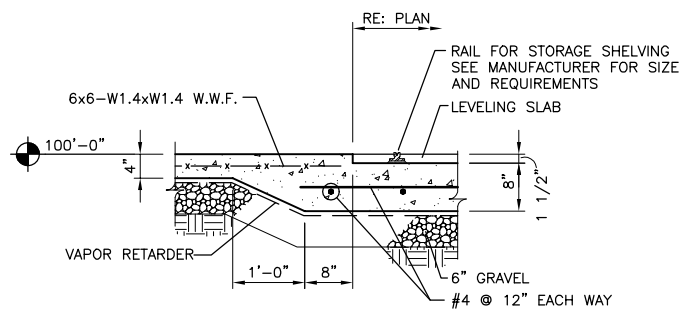
D INTERIOR COLUMN FOOTING DETAIL
S201 SCALE B



H FOUNDATION SECTION
S201 SCALE B

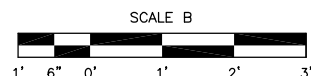


J FOUNDATION SECTION
S201 SCALE B



K FOUNDATION SECTION
S201 SCALE B

FOOTING SCHEDULE								
DESIGNATION	A	B	C	D	E	F	G	H
FOOTING SIZE "W"x"L"x"D"	9'-0"x9'-0"x2'-0"	8'-3"x8'-3"x2'-0"	7'-0"x7'-0"x1'-6"	6'-6"x6'-6"x1'-6"	5'-9"x5'-9"x1'-4"	5'-0"x21'-0"x1'-2"	5'-0"x12'-0"x1'-2"	2'-8"x5'-4"x1'-0"
FOOTING REINFORCING (EACH WAY BOTTOM)	11-#7	12-#6	10-#6	9-#6	9-#5	6-#6 LONG WAY 21-#6 SHORT WAY	6-#6 LONG WAY 12-#6 SHORT WAY	4-#4 LONG WAY 6-#4 SHORT WAY



FOR GENERAL NOTES SEE SHEET S301

ALWAYS THINK SAFETY



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

FOUNDATION DETAILS

RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

APPROVED

DATE

DESCRIPTION

MARK

PROJECT NO: P02124

DESIGN OFFICE: LONCO, INC.

DESIGNED BY: JAS

DRAWN BY: JOA

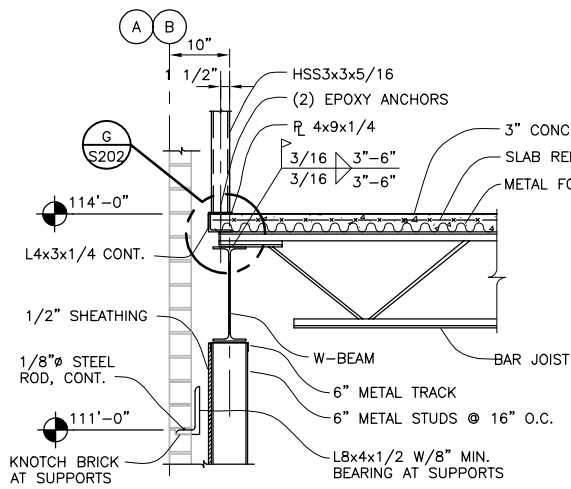
CHK'D BY: JAS

APPROVED BY:

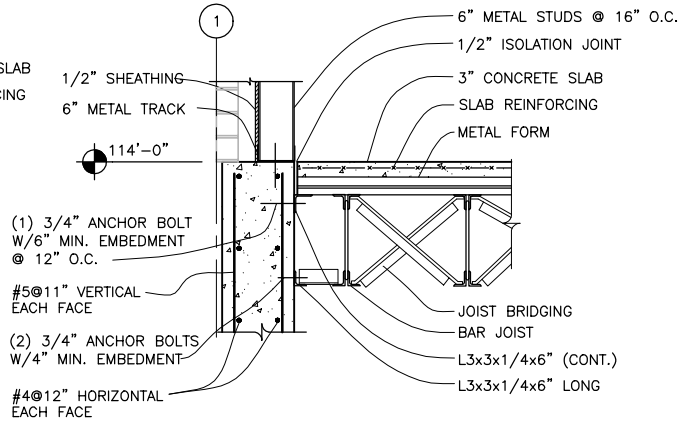
DATE: 2-28-03

S201
SHEET OF

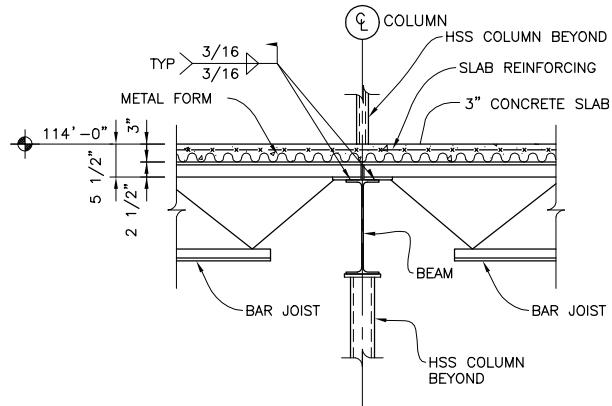
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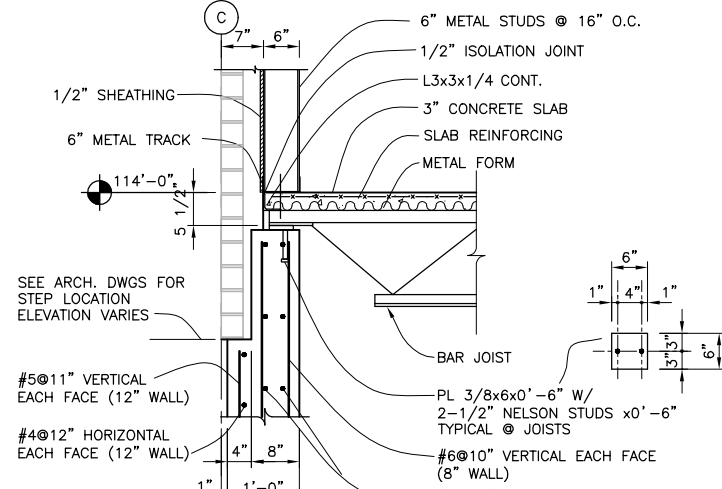
FLOOR FRAMING DETAIL
SCALE B



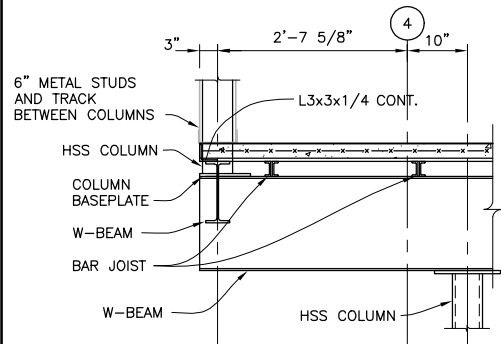
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SCALE B



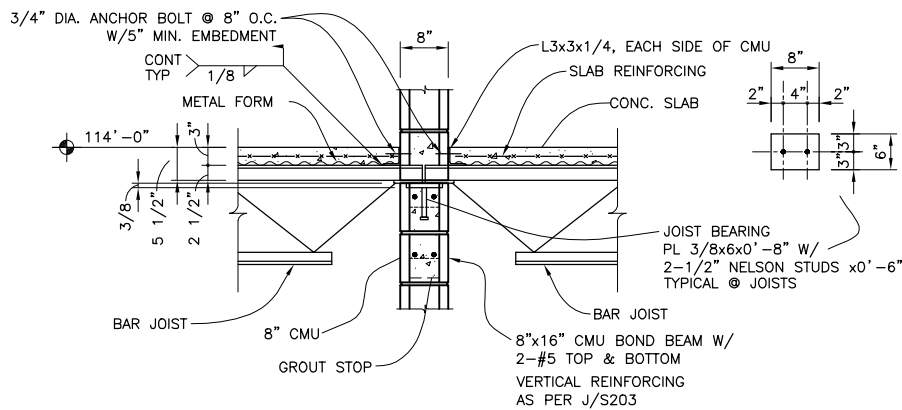
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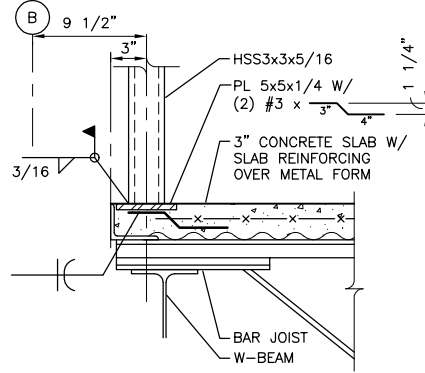
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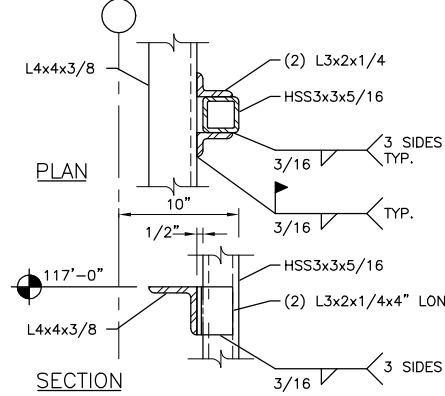
FLOOR FRAMING DETAIL
SCALE B



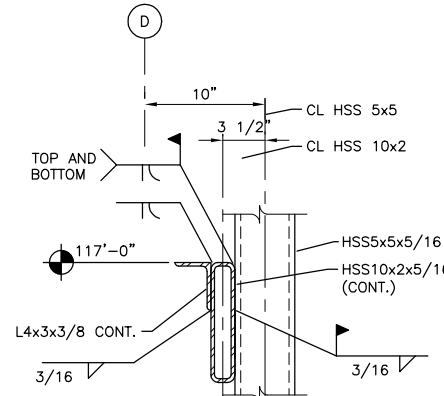
FLOOR FRAMING DETAIL
SCALE B



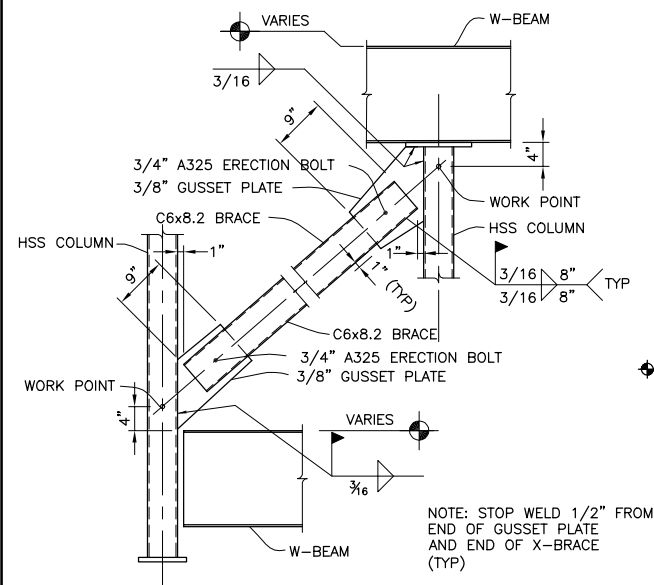
DETAIL
SCALE D



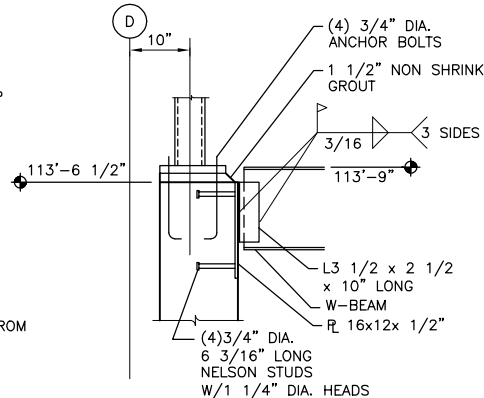
DETAIL
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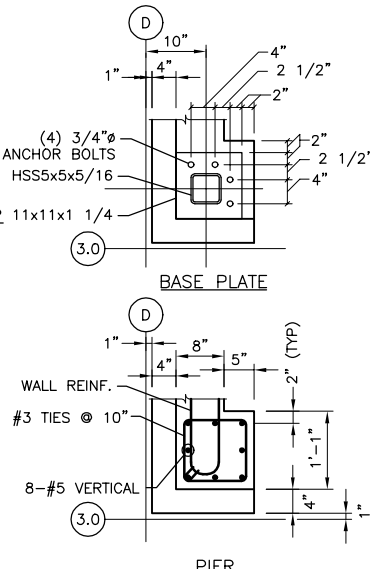
DETAIL
SCALE D



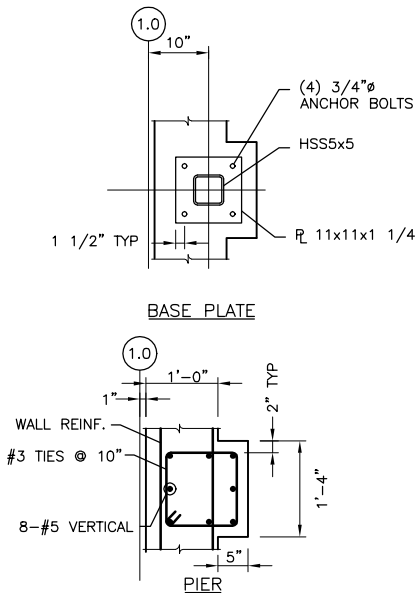
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N.T.S.



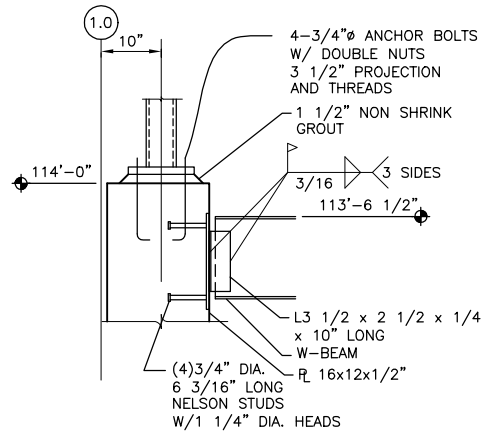
FRAMING DETAIL
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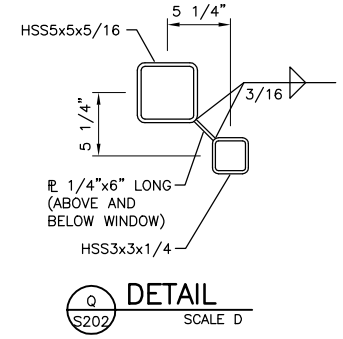
FRAMING DETAIL
SCALE B



FRAMING DETAIL
SCALE B



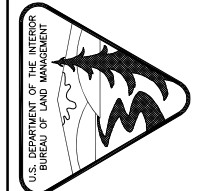
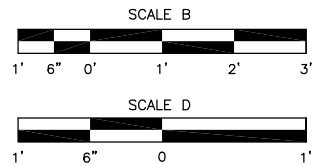
FRAMING DETAIL
SCALE B



DETAIL
SCALE D

FOR GENERAL NOTES SEE SHEET S301

ALWAYS THINK SAFETY



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

FLOOR FRAMING DETAILS

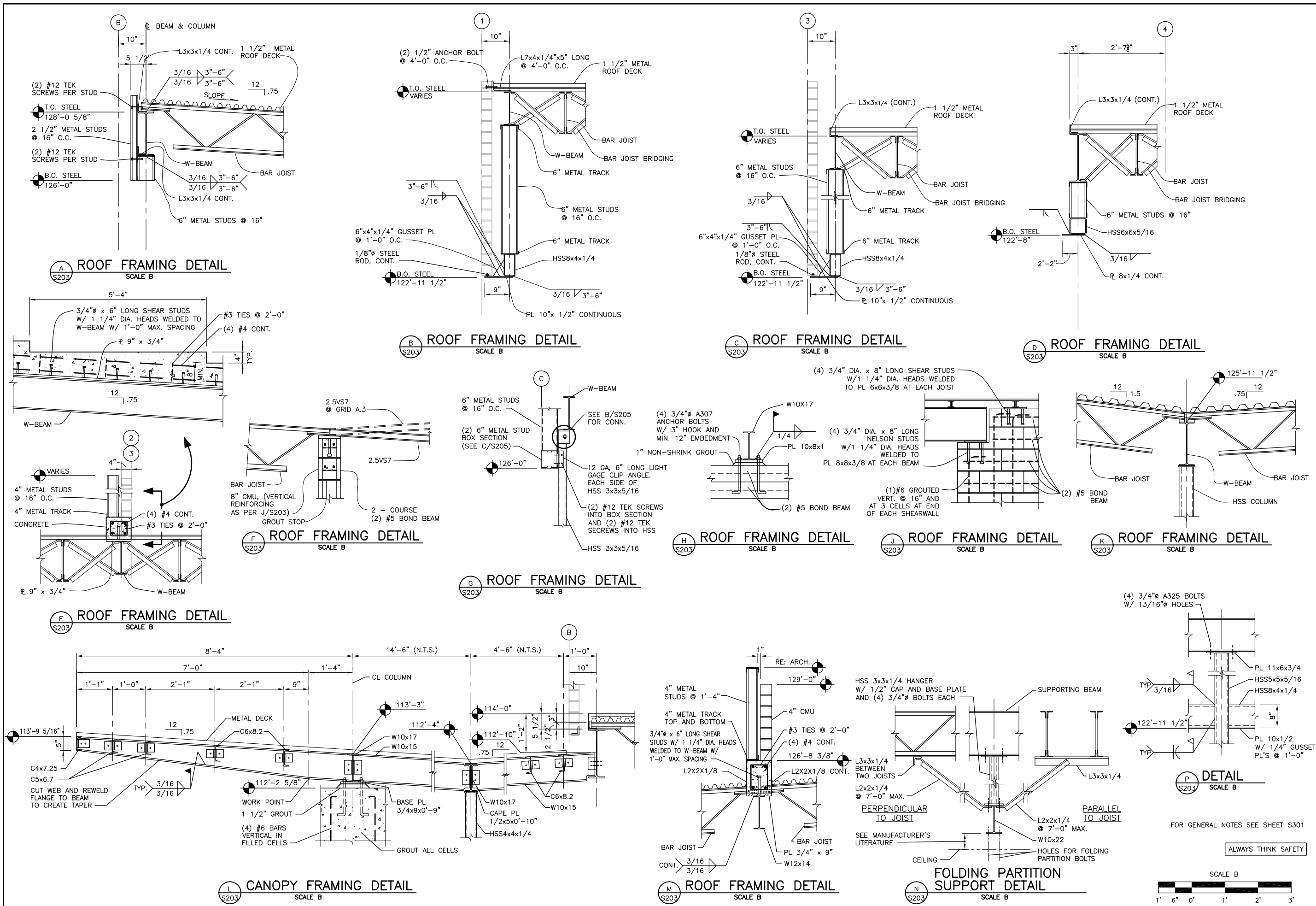
RAWLINS FIELD OFFICE BUILDING

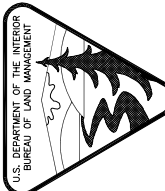
RAWLINS

MARK	DESCRIPTION	DATE	APPROVED

PROJECT NO: P02124	DESIGN OFFICE: LONCO, INC.
DESIGNED BY: JAS	DRAWN BY: BFB
CHK'D BY: JAS	APPROVED BY: JAS
DATE: 2-28-03	

F:\2002\p02124\drawings\S203 Roof Framg Dets.dwg - Current tab: Plot





U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

ROOF FRAMING DETAILS

RAWLINS FIELD OFFICE BUILDING

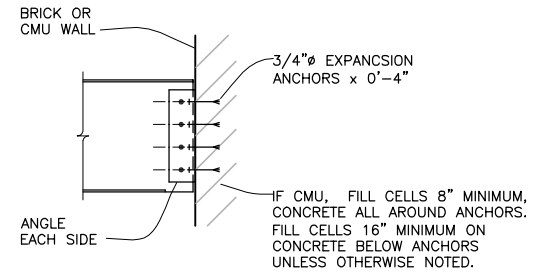
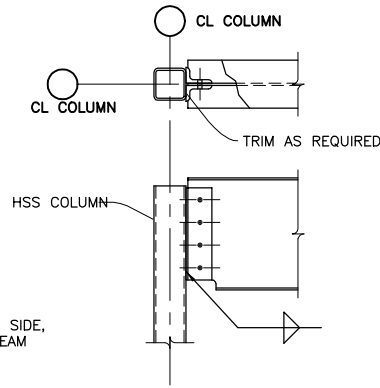
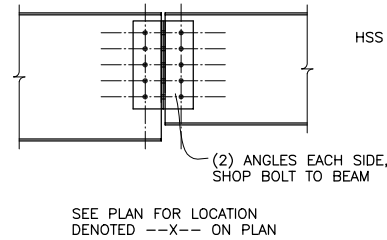
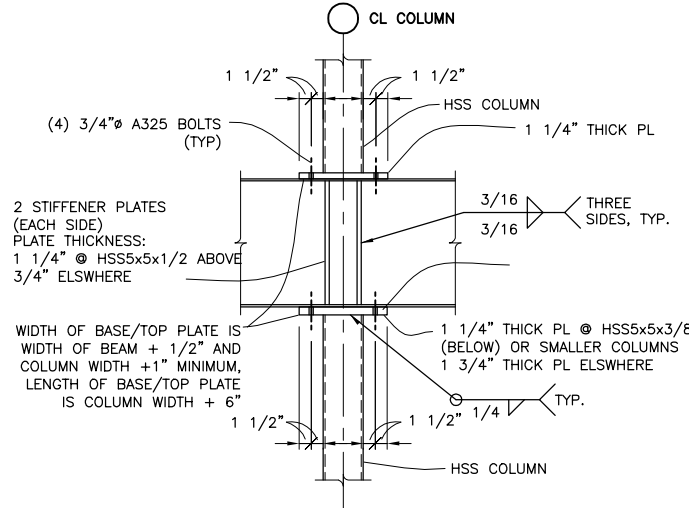
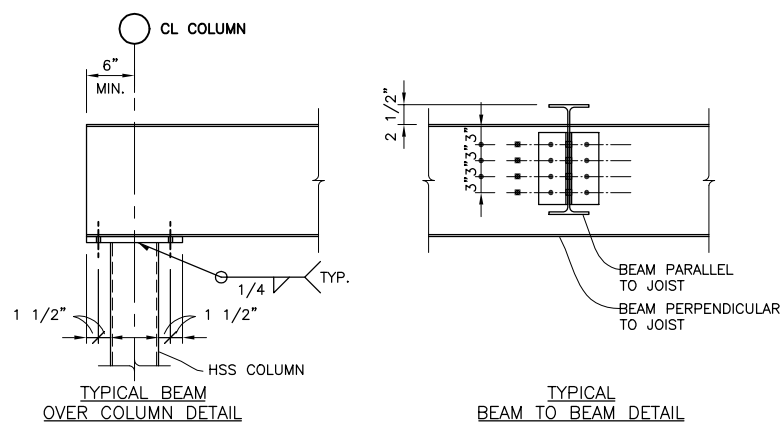
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P02124	LONCO, INC.	JAS	BFB	JAS		2-28-03

RAWLINS

DATE	APPROVED

SHEET **S203** OF

F:\2002\p02124\drawings\S204 Frmng Dets.dwg - Current tab: Plot



TYPICAL CONTINUOUS
BEAM TO COLUMN DETAIL

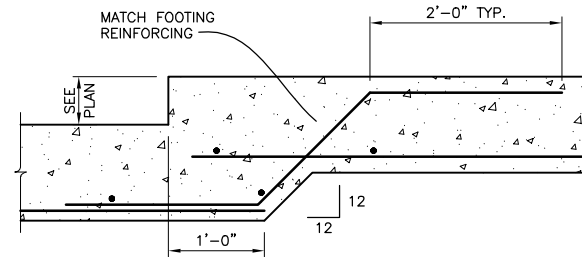
TYPICAL
BEAM SPLICE DETAIL

TYPICAL
BEAM TO COLUMN DETAIL

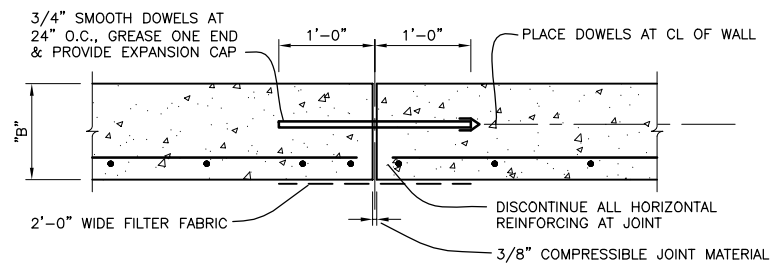
TYPICAL
BEAM TO WALL DETAIL

TYPICAL STEEL BEAM AND COLUMN CONNECTION DETAILS

NOTE:
ALL CONNECTIONS ARE TO BE EQUIVALENT TO AISC, EIGHTH EDITION, FRAMED BEAM
CONNECTIONS, TABLE 11-A WITH 5/16" THICK ANGLES AND MAXIMUM NUMBER OF 3/4" Ø
H.S. BOLTS AT 3" O.C. UNLESS OTHERWISE NOTED. BEARING TYPE CONNECTION WITH
THREADS IN SHEAR PLANE.

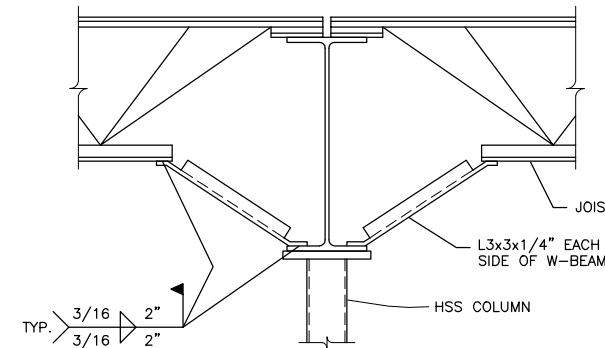


TYPICAL RETAINING WALL
EXPANSION JOINT PLAN DETAIL

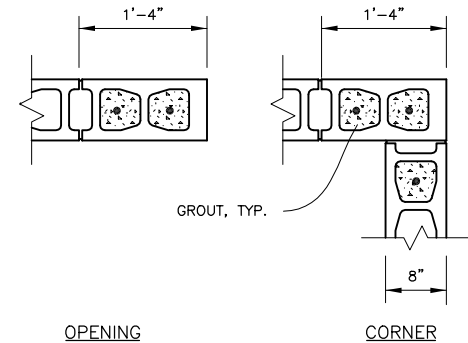


TYPICAL RETAINING WALL EXPANSION JOINT PLAN DETAIL

SPACE JOINTS IN WALL AT 20'-0" MAX., SEE PLAN.
JOINTS ARE NOT REQUIRED IN FOOTINGS.



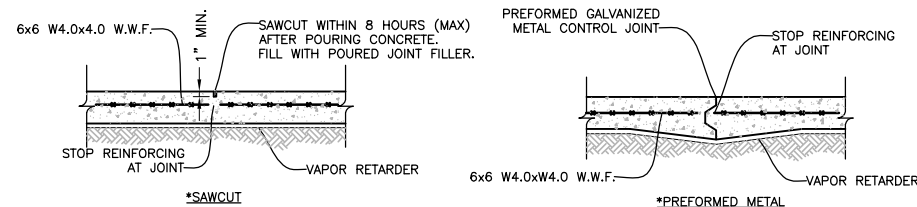
W-BEAM BRACING DETAIL



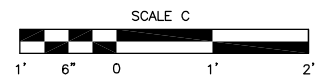
CMU NOTES:

- CELLS AROUND REINFORCING ARE TO BE FILLED WITH 3,000 PSI GROUT.
- PROVIDE THREE (3) FILLED CELLS OF TYPICAL WALL REINFORCING AT CORNERS, PROVIDE TWO (2) FILLED CELLS OF TYPICAL WALL REINFORCING AT EACH SIDE OF EACH OPENING.
- TYPICAL WALL REINFORCING UNLESS OTHERWISE NOTED:
 - REINFORCE 8" BLOCK WITH #6 VERTICAL AT 16" AND STANDARD (#9 WIRE) TRUSS TYPE WALL REINFORCING HORIZONTAL AT 16".
 - THE LAST (3) VERTICAL #6 BARS IN 8" BLOCK WALLS IS TO HOOK 90° INTO UPPERMOST BOND BEAM WITH 9" LEG.
 - GROUT STOP IS A FIBERGLASS MESH CONFORMING TO ASTM STANDARD D1668-73 TYPE 207.
 - SPLICE ALL MASONRY REINFORCING 48 BAR DIAMETERS MINIMUM.
 - EXTEND REINFORCING AROUND CORNERS OR PROVIDE CORNER BARS.

TYPICAL CMU WALL DETAILS



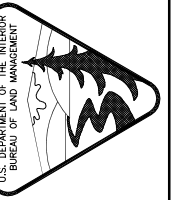
CONTROL JOINT DETAIL



RETAINING WALL SECTION

GENERAL CONCRETE RETAINING WALL SCHEDULE															
CONCRETE DIMENSIONS							REINFORCING								
"H"	"W"	"A"	"B"	"C"	"D"	"G"	"M"	"N"*	"P"*	"R"	"S"*	"U"	"X"	"Y"	"Z"
20'-0"	13'-0"	5'-0"	1'-4"	6'-8"	2'-2"	10"	#9 @ 6"	4'-0"	8'-0"	#6 @ 12"	19'-10"	#6 @ 6"	5'-10"	10 - #4	#5 @ 12"
16'-0"	9'-0"	3'-4"	1'-2"	3'-6"	1'-6"	6"	#7 @ 6"	4'-0"	8'-0"	#5 @ 18"	15'-10"	#5 @ 6"	4'-0"	7 - #4	#5 @ 12"
12'-0"	6'-0"	2'-0"	1'-0"	3'-0"	1'-2"		#4 @ 9"	4'-0"	7'-10"	#4 @ 18"	7'-10"	#4 @ 9"	1'-2"	4 - #4	#4 @ 12"
8'-0"	3'-6"	1'-0"	8"	1'-10"	1'-0"		#6 @ 9"	4'-0"	8'-0"	#5 @ 18"	11'-10"	#5 @ 9"	2'-6"	5 - #4	#4 @ 12"

* PROVIDE HOOK TO DEVELOP BAR LENGTH
** LENGTH VARIES WHEN TOP OF WALL SLOPES



UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

TYPICAL DETAILS

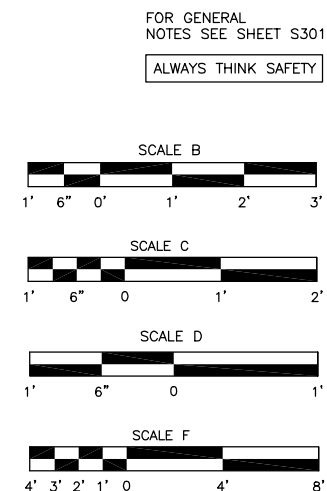
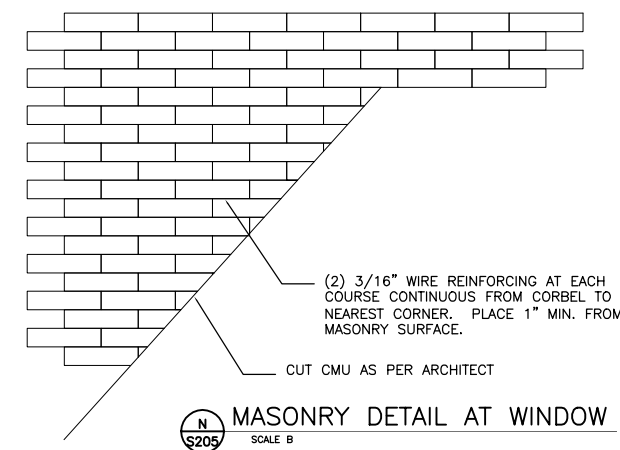
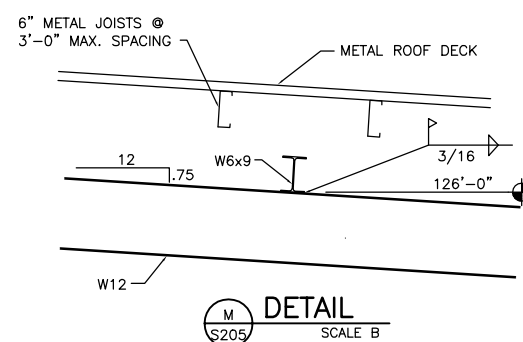
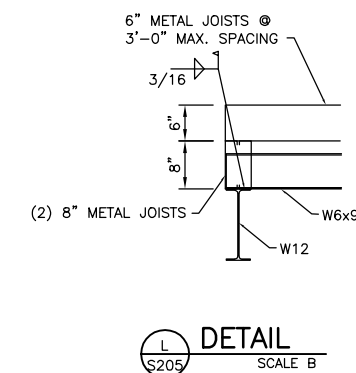
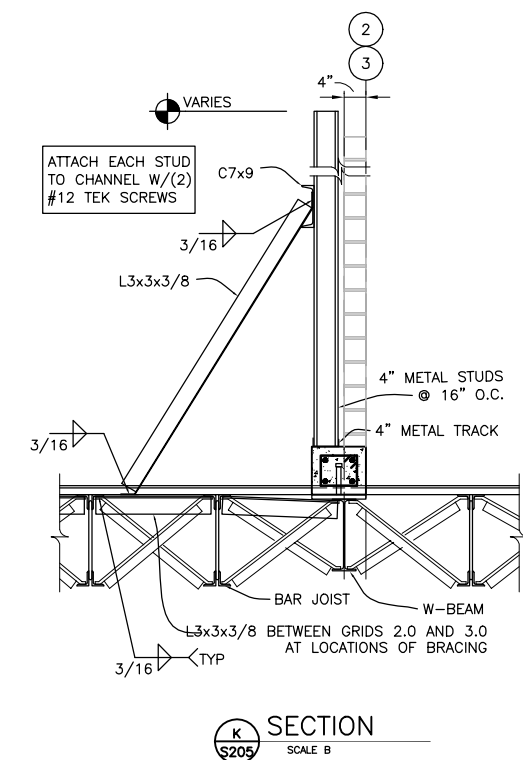
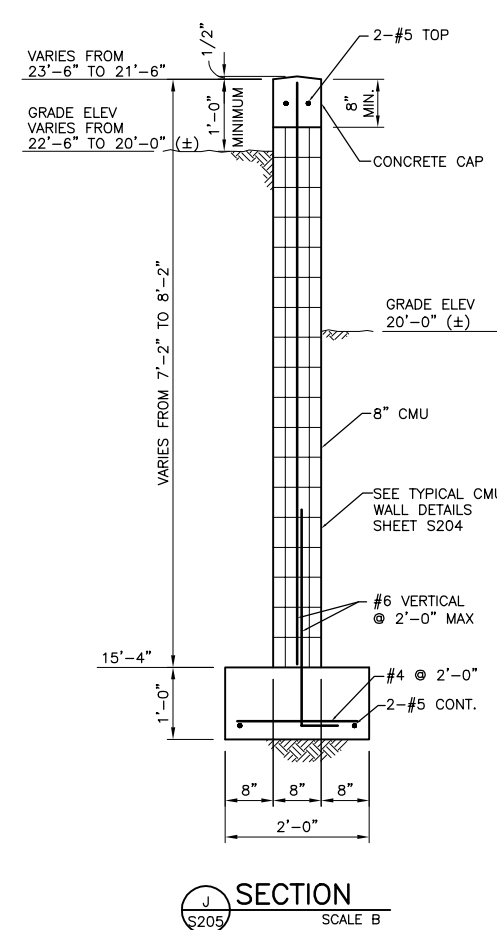
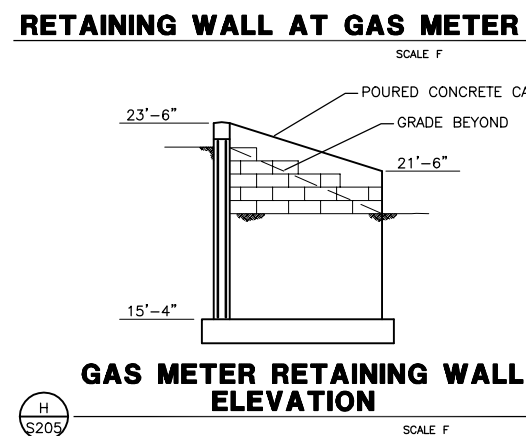
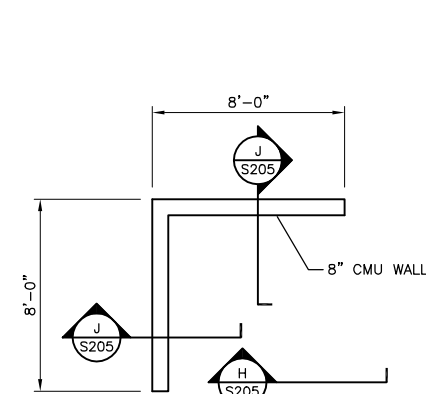
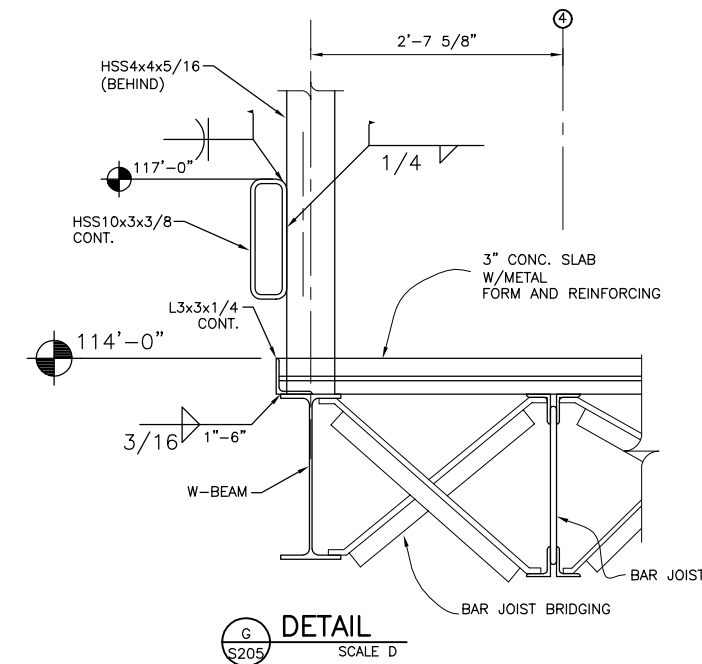
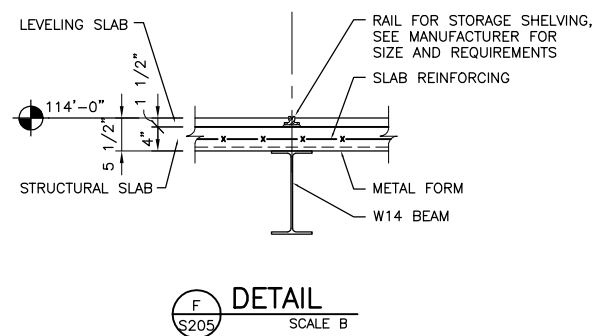
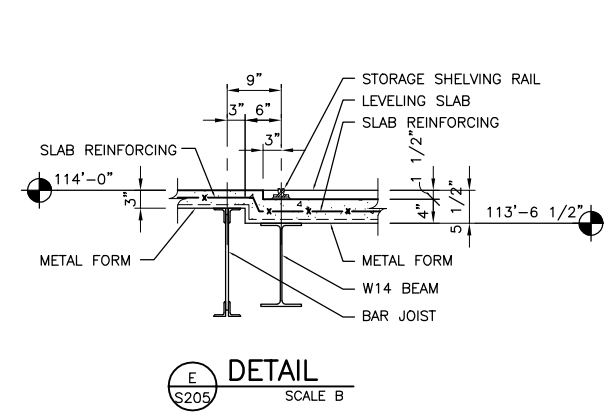
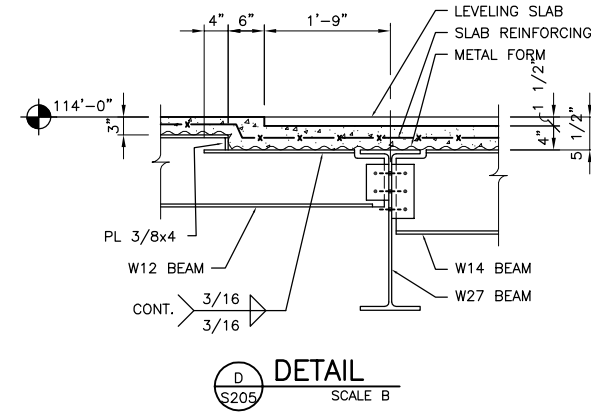
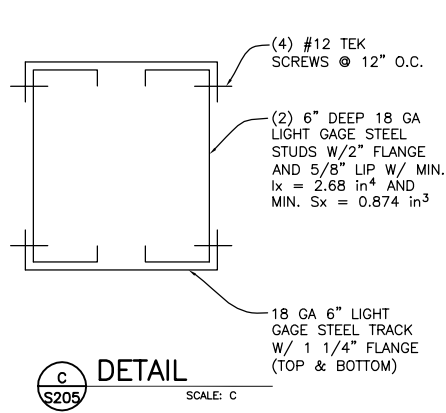
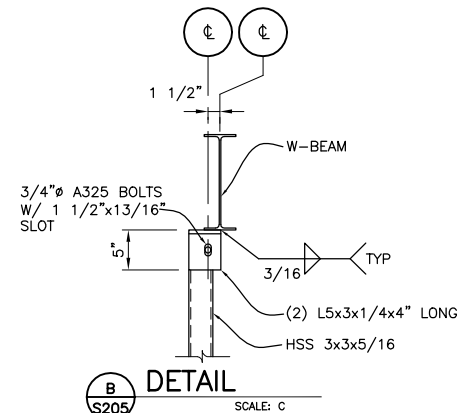
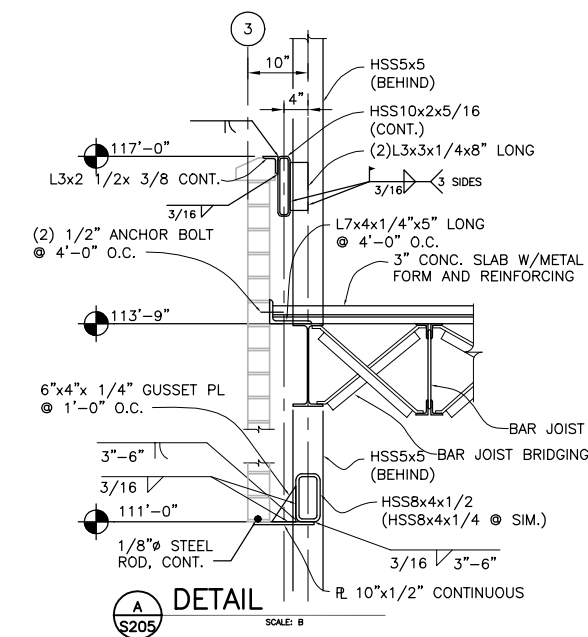
RAWLINS FIELD OFFICE BUILDING

WYOMING

RAWLINS

PROJECT NO:	DESIGN OFFICE:	DESIGNED BY:	DRAWN BY:	CHK'D BY:	APPROVED BY:	DATE:	DESCRIPTION:	MARK:
P02124	LONCO, INC.	JAS	JOA	JAS		2-28-03		

S204
SHEET OF



Technical drawing of a roof edge detail. The drawing shows a cross-section of a steel column (HSS5x5) and its connection to a roof structure. The column is labeled "HSS5x5 COLUMN BEYOND" and "B.O. STEEL 129'-0\"". A gusset plate, labeled "6\"x4\"x1/4\" GUSSET PL (LLV) @ 1'-0\" O.C.", is welded to the column. The gusset is connected to a "W12x14" beam. The roof structure includes a "1 1/2\" METAL ROOF DECK" and a "BAR JOIST". The roof deck is shown with a "SLOPE" indicated by a triangle with a vertical side of "12" and a horizontal side of ".75". The connection between the gusset and the beam is detailed with welds: "3/16\" 3'-6\"" and "3/16\" 10\"x1 1/2\" CONTINUOUS". A dimension of "10\" is shown for the distance from the column face to the start of the continuous weld. A circular detail callout "D" is located at the top of the column.

Technical drawing of a wall section showing a window opening. The drawing includes dimensions and labels for various components:

- Horizontal dimension: 1'-11 7/8"
- Vertical dimension: 103'-0"
- Window opening: 103'-0"
- Window frame: L4x4x1/4
- Window opening: (2) #12 TEK SCREWS PER STUD
- Window opening: 2 1/2" METAL STUDS
- Wall: L3
- Wall: 1/2x2 1/2x1/4 (EMBEDDED IN MORTAR JOINT)
- Callout B: Circular symbol at top left
- Callout D: Circular symbol at top right with S301

LOADS

SCHEDULE

MAXIMUM OPENING	SIZE OF ANGLE PER 4" MASONRY	SIZE OF ANGLE AT 8" CMU	MINIMUM BEARING
4'-0"	3 1/2 x 2 1/2 x 1/4	(2) 4 x 3 1/2 x 1/4	8"
7'-0"	3 1/2 x 2 1/2 x 1/4	(2) 4 x 3 1/2 x 3/8	8"
7'-9"	4 x 3 1/2 x 1/4	(2) 5 x 3 1/2 x 3/8	8"
8'-2"	5 x 3 1/2 x 5/16	(2) 6 x 3 1/2 x 5/16	8"
10'-0"	5 x 3 1/2 x 3/8		

8" MAS.

14" CAVITY WALL

Diagram E shows the exterior wall and roofline. The L3 angle is shown embedded in the mortar joint of the masonry, with a dimension of 1'-0" MIN. The L4 angle is shown attached to the exterior wall, with a dimension of 1'-11 7/8" from the vertical line to the center of the angle. The roofline is shown with a 3/16 slope.

3. LIVE LOADS USED IN DESIGN:

- A. FOOTING 20 PSF
- B. WIND 110 MPH, WITH USE FACTOR OF 1.0 (PER ASCE7)
- C. SEISMIC ZONE 0

4. MATERIALS ARE TO CONFORM TO THE FOLLOWING:

- A. CONCRETE 3000 PSI AT 28 DAYS
(MAXIMUM AGGREGATE SIZE $\frac{3}{4}$ ")
- B. REINFORCING BARS ASTM A615, GRADE 60
- C. REINFORCING MESH (WWF) ASTM A185
- D. STRUCTURAL STEEL ASTM A588, Fy=50 KSI
- E. STRUCTURAL HSS TUBES ASTM 500, GRADE B, Fy=46 KSI
- F. STRUCTURAL BOLTS ASTM A325

5. LIGHT GAGE STEEL FRAMING SHALL HAVE 33 KSI MINIMUM Fy.

- A. FOR 6" WALL STUDS UP TO 13'-0" IN LENGTH, 18 GA. STEEL WITH 2" FLANGE AND 5/8" LAP WITH MINIMUM $I_x = 2.683 \text{ in}^2$ AND MINIMUM $S_x = 0.874 \text{ in}^3$ SHALL BE USED.
- B. FOR 6" WALL STUDS LONGER THAN 13'-0", 12 GA. STEEL WITH 2" FLANGE AND 5/8" LAP WITH MINIMUM $I_x = 5.656 \text{ in}^2$ AND MINIMUM $S_x = 1.885 \text{ in}^3$ SHALL BE USED.
- C. FOR 4" WALL STUDS, 18 GA. STEEL WITH 1 1/2" FLANGE AND 3/4" LAP WITH MINIMUM $I_x = 0.777 \text{ in}^2$ AND MINIMUM $S_x = 0.388 \text{ in}^3$ SHALL BE USED.
- D. FOR 3 1/2" WALL STUDS, 18 GA. STEEL WITH 1 1/2" FLANGE AND 3/4" LAP WITH MINIMUM $I_x = 0.568 \text{ in}^2$ AND MINIMUM $S_x = 0.325 \text{ in}^3$ SHALL BE USED.
- E. FOR 2 1/2" WALL STUDS, 18 GA. STEEL WITH 1 1/2" FLANGE AND 3/4" LAP WITH MINIMUM $I_x = 0.261 \text{ in}^2$ AND MINIMUM $S_x = 0.209 \text{ in}^3$ SHALL BE USED.

6. ALL CONCRETE WORK IS TO CONFORM TO ACI 318.

7. LAP REINFORCING STEEL A MINIMUM OF 36 BAR DIAMETERS. MAKE ALL BARS CONTINUOUS AROUND CORNERS.

8. PROVIDE ALL ACCESSORIES NECESSARY TO SUPPORT REINFORCING AT POSITIONS SHOWN ON DRAWINGS.

9. FOUNDATION HAVE BEEN DESIGNED ACCORDING TO GEI CONSULTANTS GEOTECHNICAL REPORT #02182, DATED JULY 2002 AND AMENDED AUGUST 5, 2002.

